IVI3: Information Visualization

05 - Visualization Techniques | WS 2026/27

FH-Prof. Dr. Mandy Keck

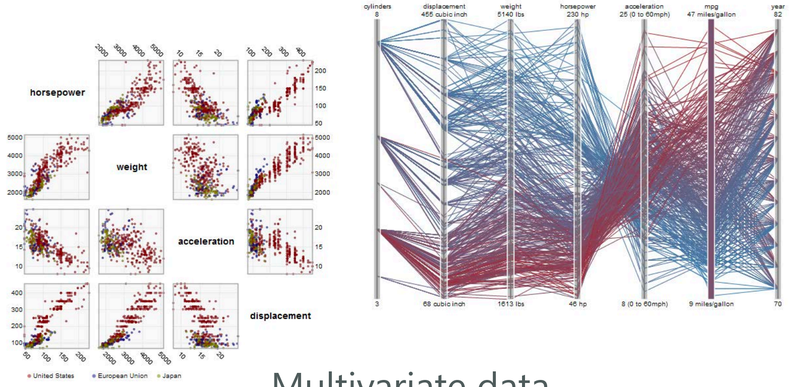
Professor of User Experience and Interaction Design

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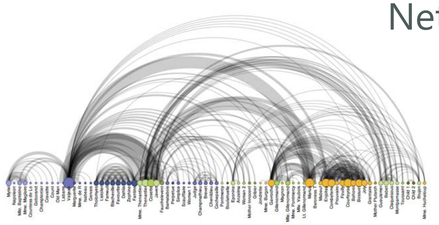
mandy.keck@fh-hagenberg.at

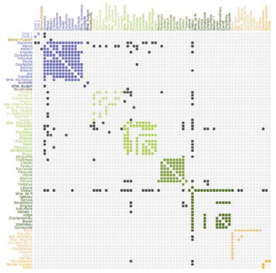
A BRIEF TOUR THROUGH THE VISUALIZATION ZOO

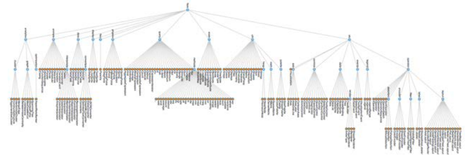


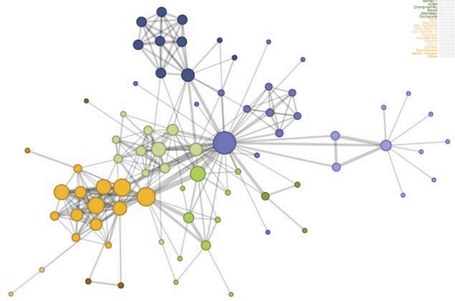
Multivariate data

Network data

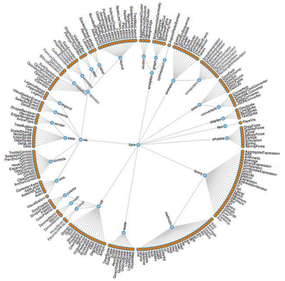


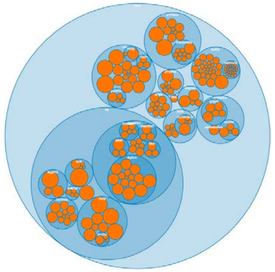




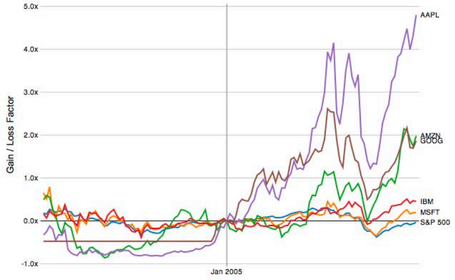


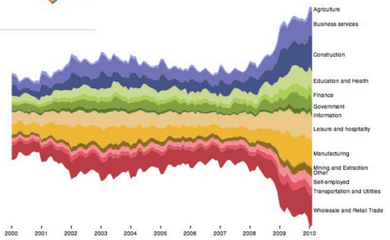
WITH WHOM





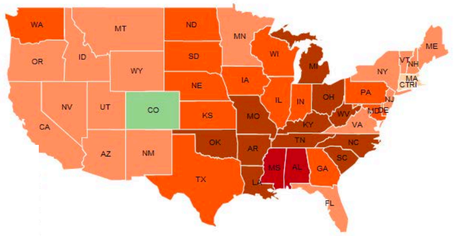
Hierarchies

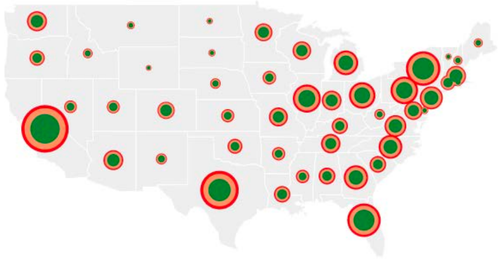




Temporal data

WHEN

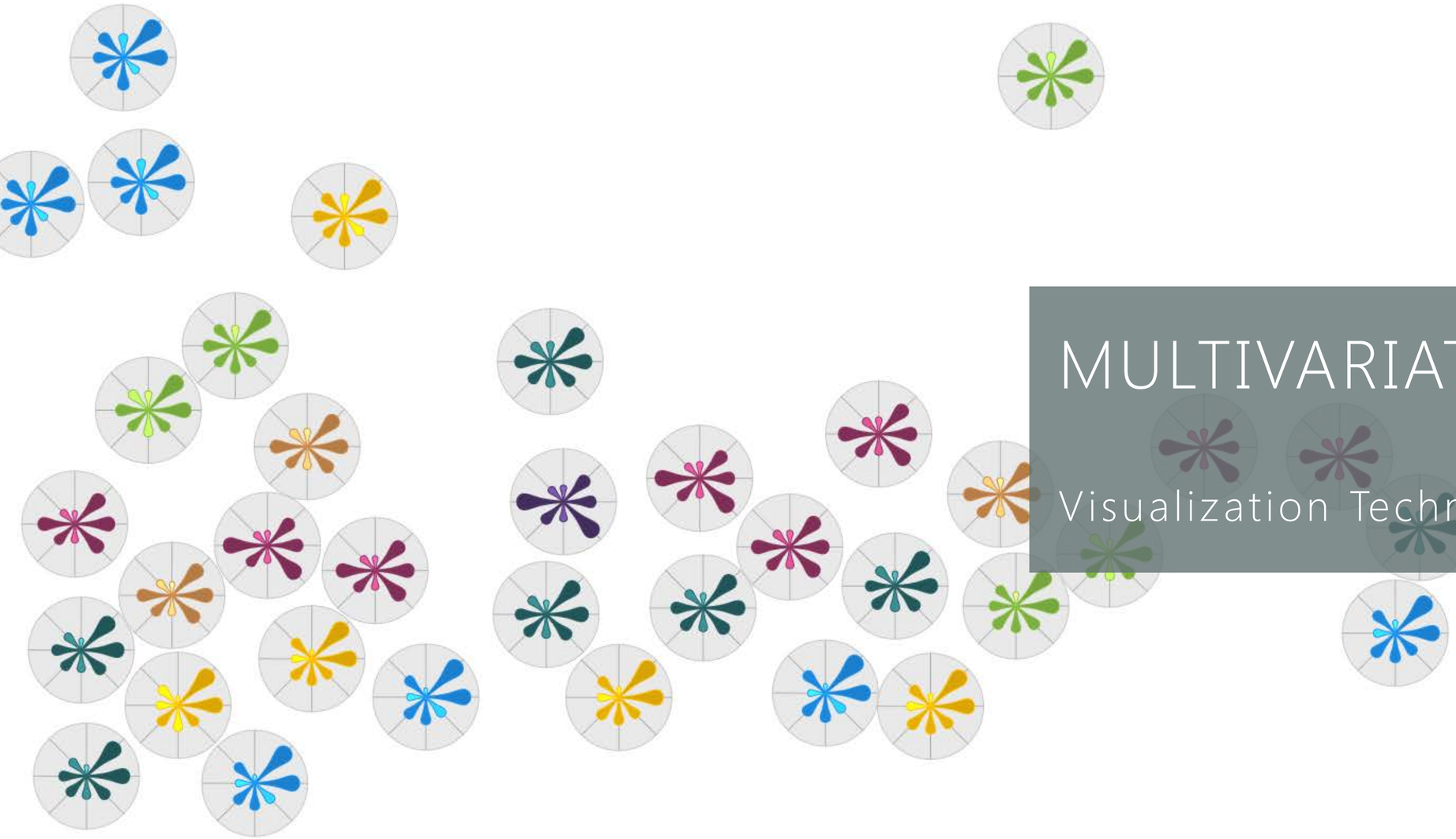




Spatial data

WHERE

J. Heer, M. Bostock, & V Ogievetsky (2010), A Tour Through the Visualization Zoo, Comm of the ACM, 53(6): 59-67



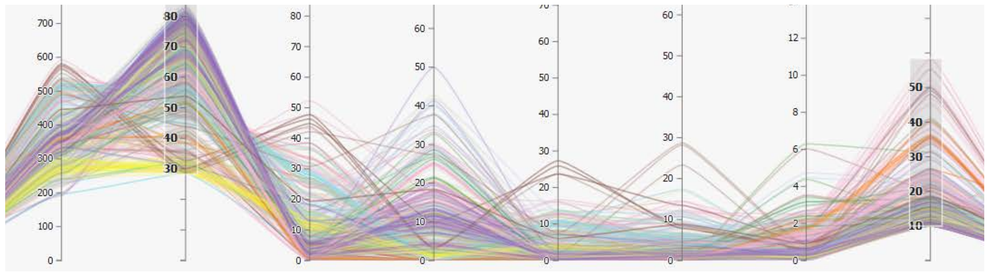
MULTIVARIATE DATASETS

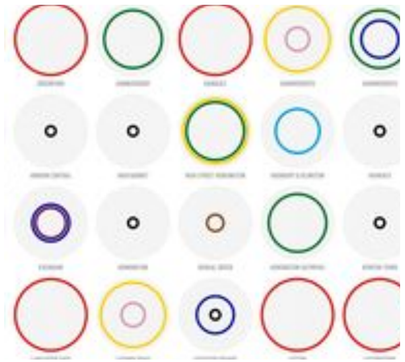
Visualization Techniques

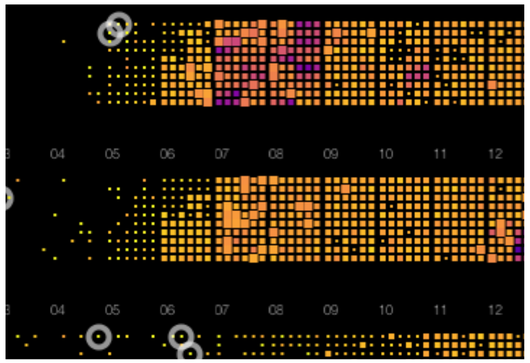
MULTIVARIATE DATA

Visualization techniques:

- Table-based visualizations
- Axes-based visualizations
  - Combined bivariate visualizations
  - Polyline-based visualizations
- Glyph-based visualizations
- Pixel-based visualizations
- Nested visualizations







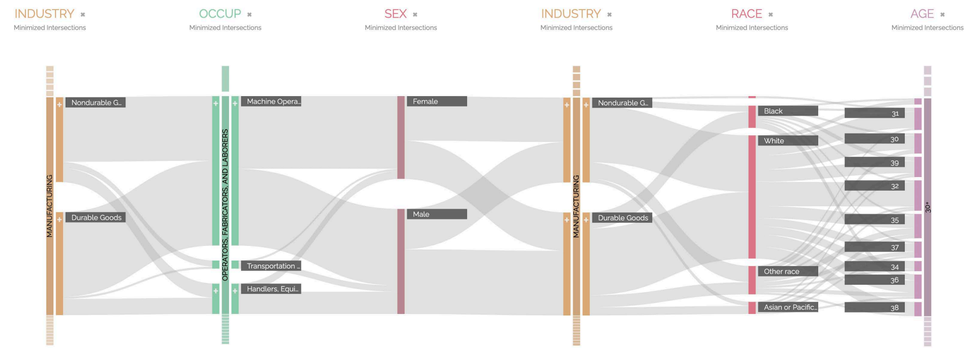
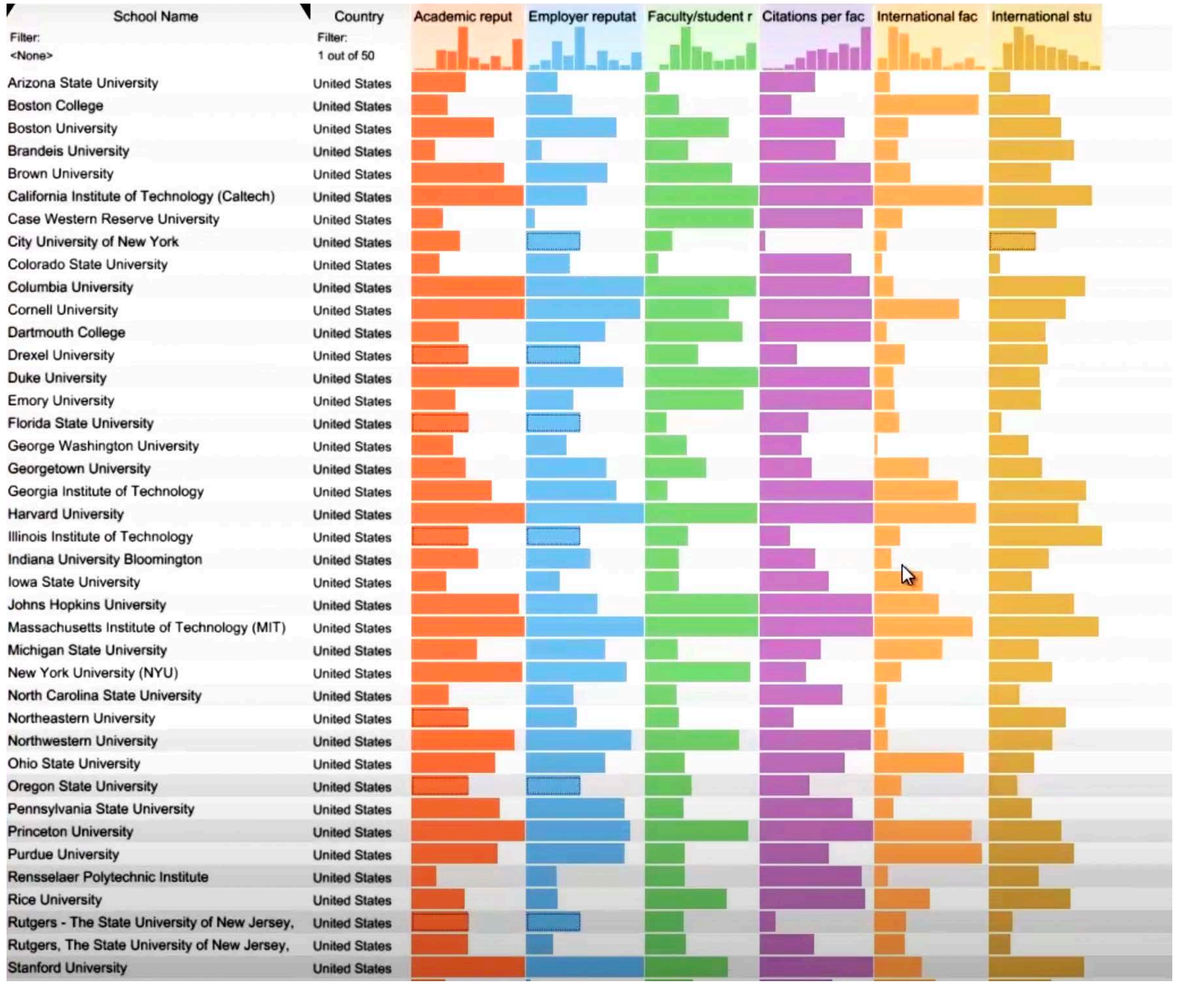


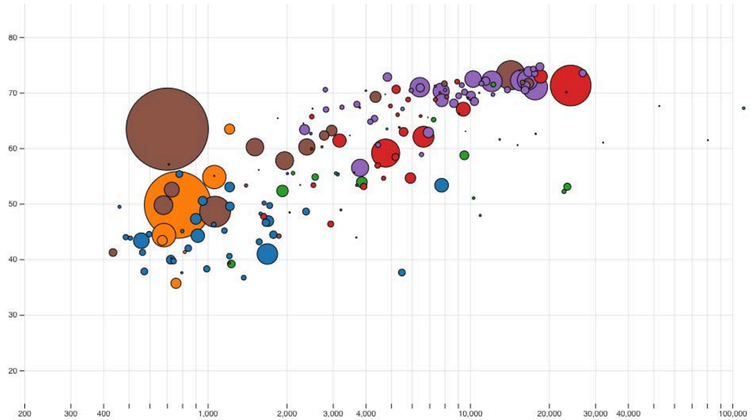
TABLE-BASED VISUALIZATIONS

- retains the tabular layout of spreadsheets, but replace the textual representation of data values by a visual representation
- a visual representation makes interpretation of the data much easier and also requires less display space

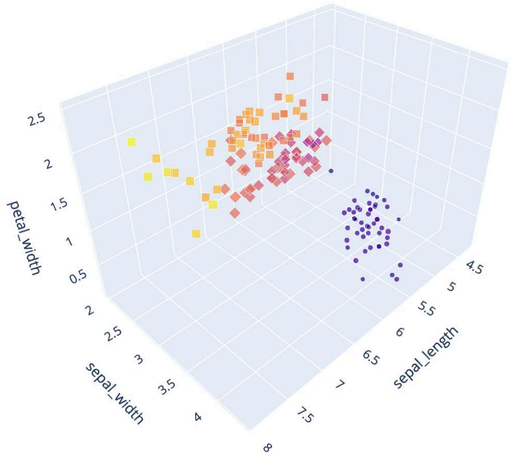


LineUp: table-based visualization with bar charts

AXES-BASED VISUALIZATIONS



2-dimensional



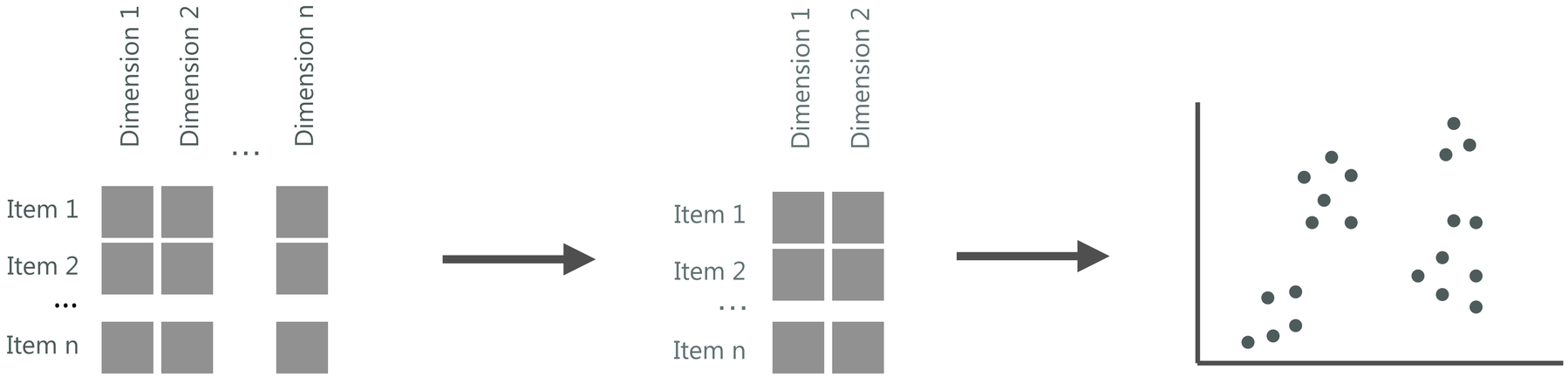
3-dimensional

?

More dimensions?

DIMENSIONALITY REDUCTION

- projects similarity of high-dimensional data elements to proximity in low-dimensional space



High-dimensional Data

Two-dimensional Data

Two-dimensional Plot

- advantage: visual clustering
- disadvantage: information loss

Example in Google Colab: https://colab.research.google.com/drive/1TQPLkGbu6N9L-8OZrS_2rDUjuAkzvvPi

DIMENSIONALITY REDUCTION



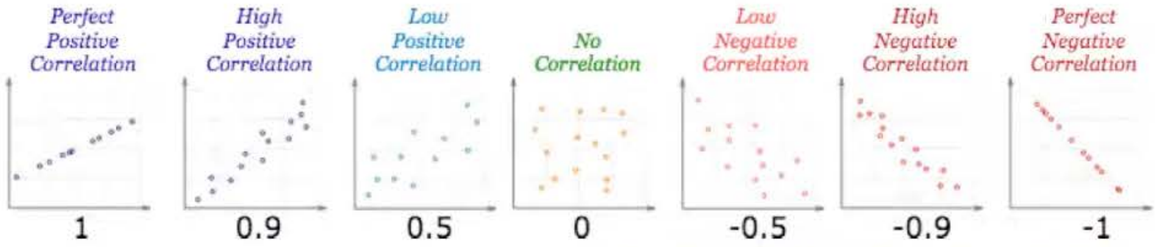
Google Arts & Culture: https://artsexperiments.withgoogle.com/tsnemap/

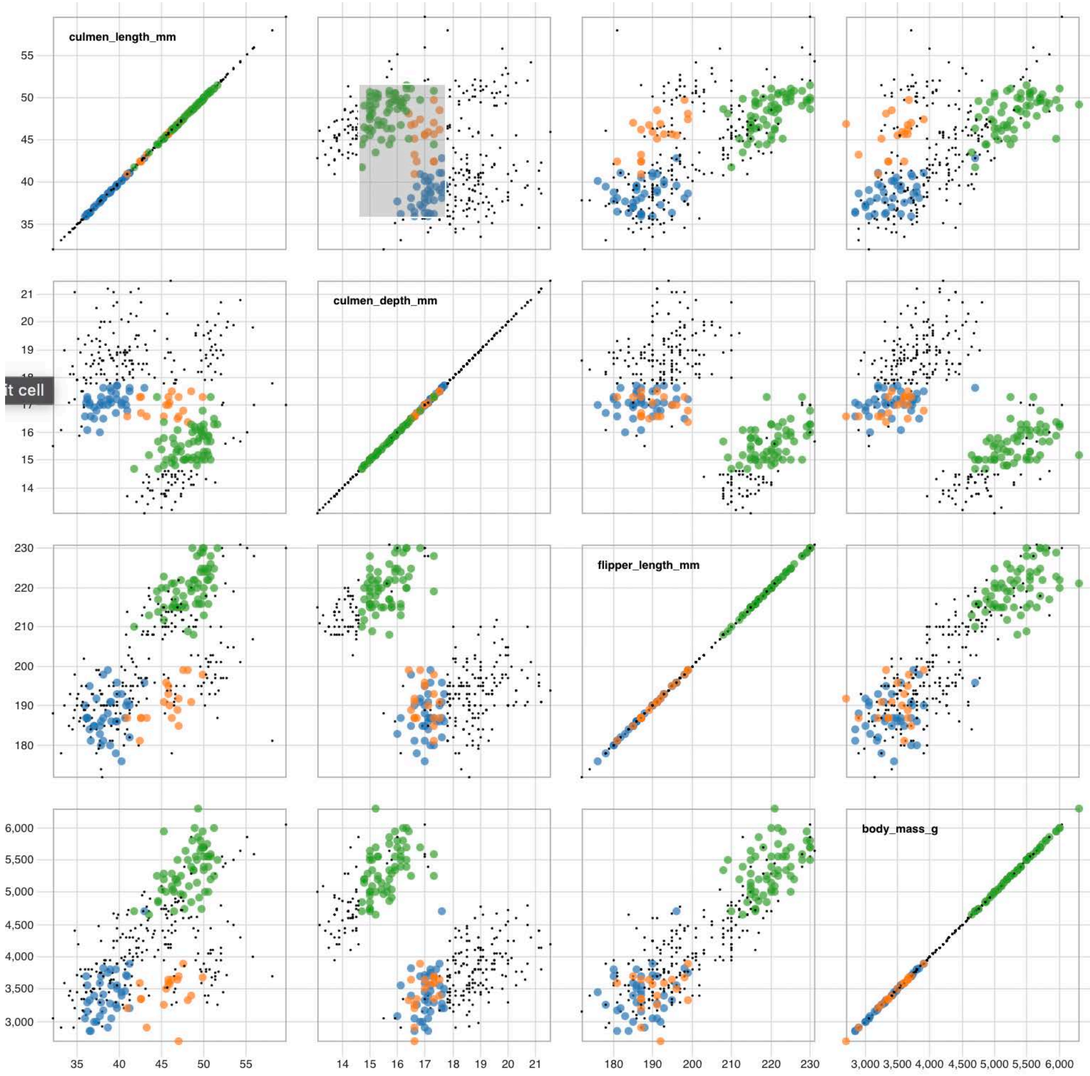
COMBINED BIVARIATE VISUALIZATIONS

- visualize multivariate data by combing several bivariate displays

Scatterplot Matrix (SPLOM):

- each scatter plot consists of two orthogonal axes that represent two attributes, and dots visualize data tuples with respect to the two data attributes
- is suited to communicate two-dimensional values distributions, bivariate correlations, clusters, and outliers

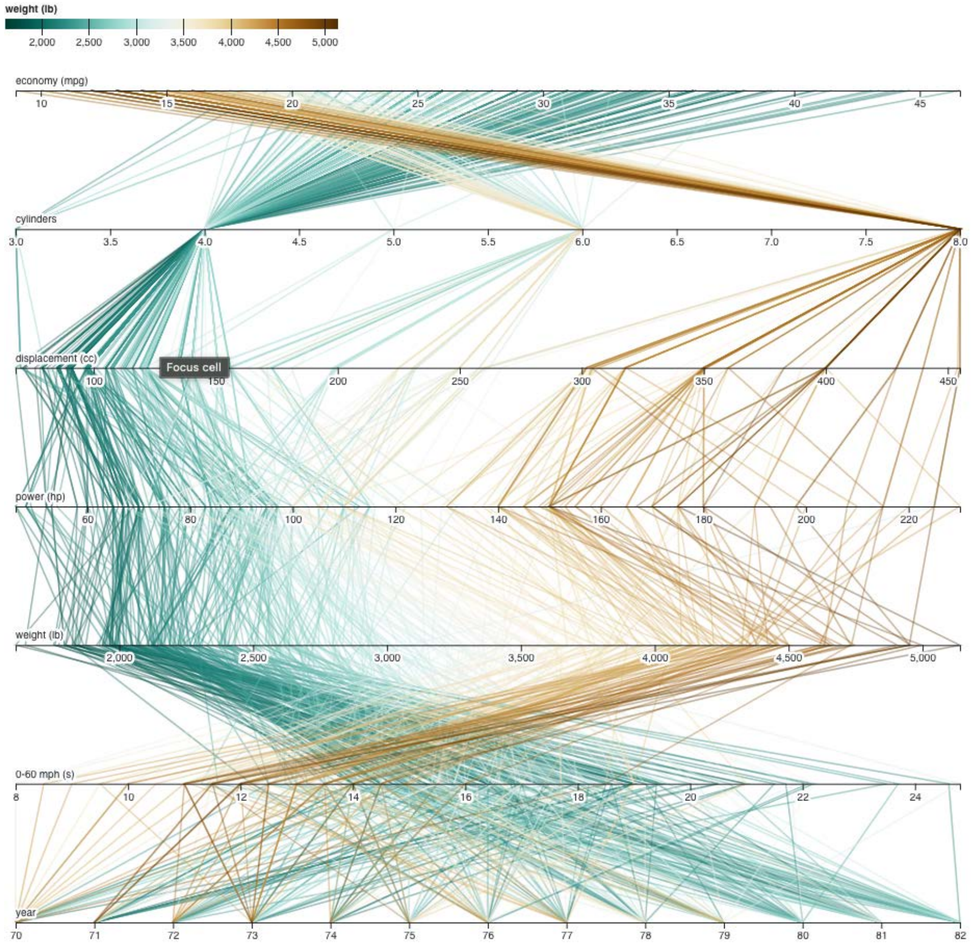




POLYLINE-BASED VISUALIZATION

Parallel Coordinates:

- approach for visualizing many quantitative attributes at once using spatial position
- shows data elements as poly lines across axes
- Limitations: visible patterns only between neighboring axis pairs
- usual solution: reorderable axes, interactive exploration

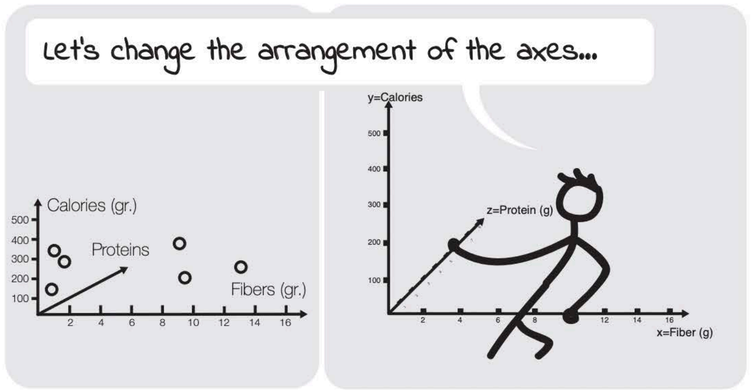


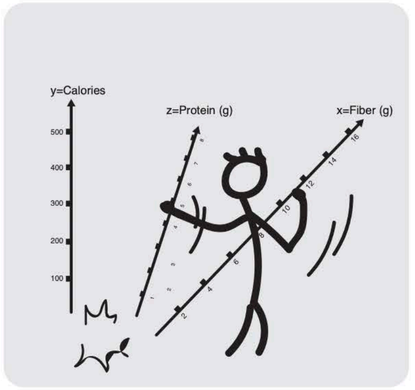
https://observablehq.com/@d3/brushable-parallel-coordinates

A. Inselberg: Parallel Coordinates: Visual Multidimensional Geometry and Its Applications, Springer, 2009

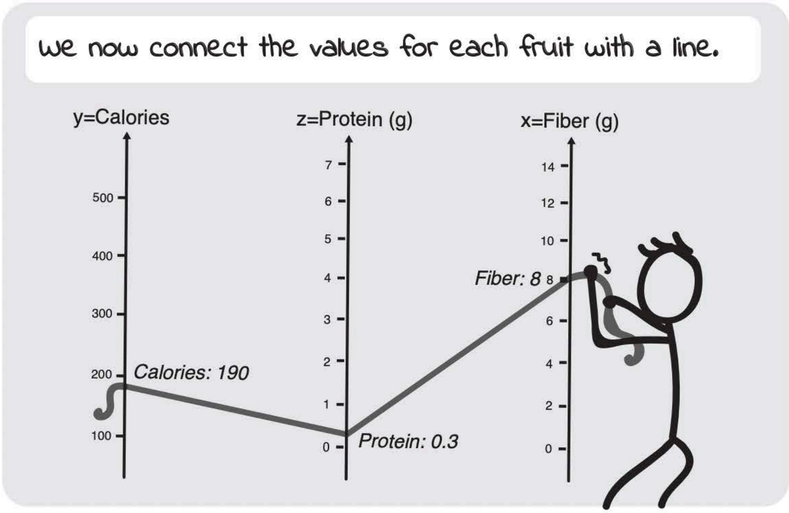
PARALLEL COORDINATES

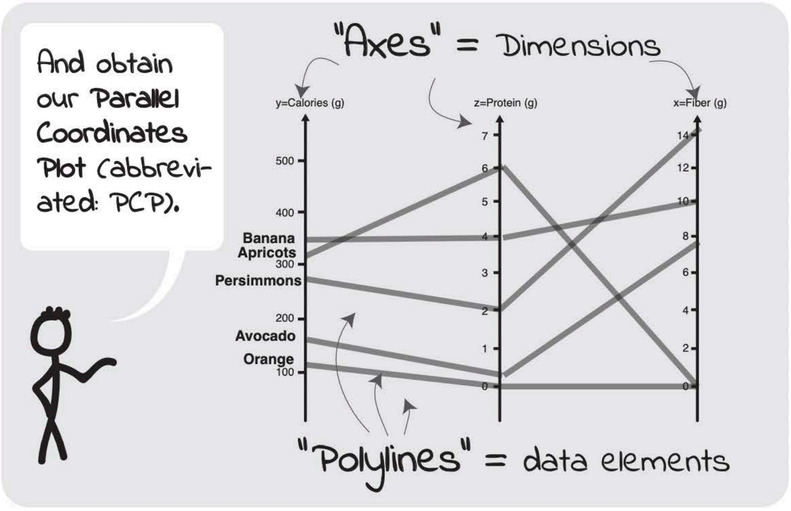
Construction of Parallel Coordinates:





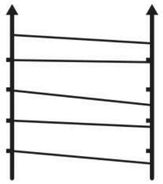




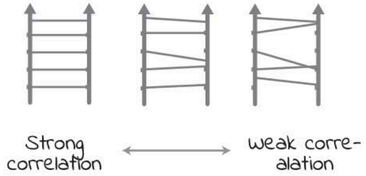


PARALLEL COORDINATES

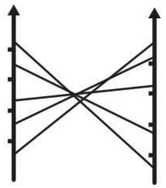
Positive Correlation = Parallel Lines



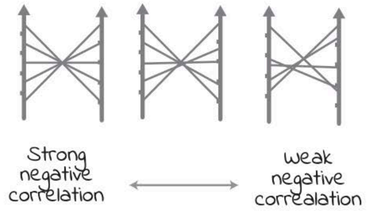
A correlation is visible through rather parallel polylines between two axes.



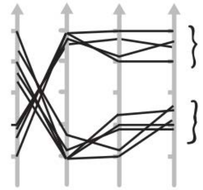
Negative Correlation = Crossing Lines



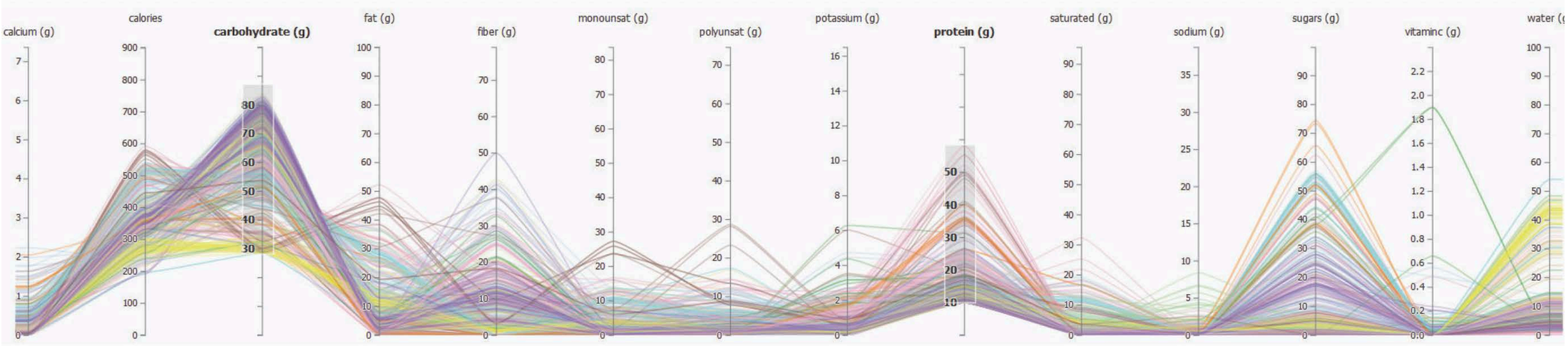
An inverse correlation is visible through lots of crossing polylines between two axes.



Grouped Lines = Cluster



Clusters are visible as polylines 'following' each other across several axes, resulting in bundles.



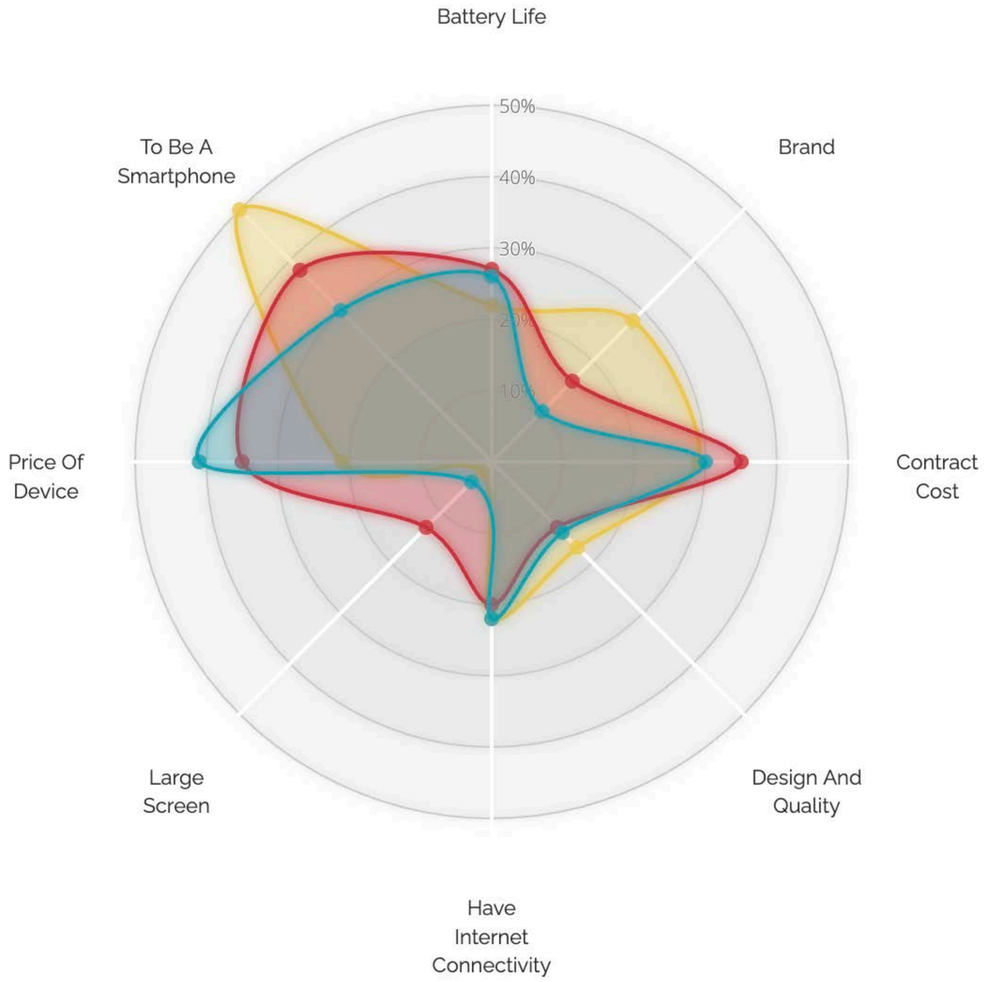
PC Cheat Sheet: https://visualizationcheatsheets.github.io -Nutrient Explorer: http://bl.ocks.org/syntagmatic/raw/3150059/

RADAR CHARTS

- use polar coordinates
- one dimension is measured as an angle from a starting line and the other is measured as a distance from a center point

- angle channel is less accurately perceived than rectilinear spatial position

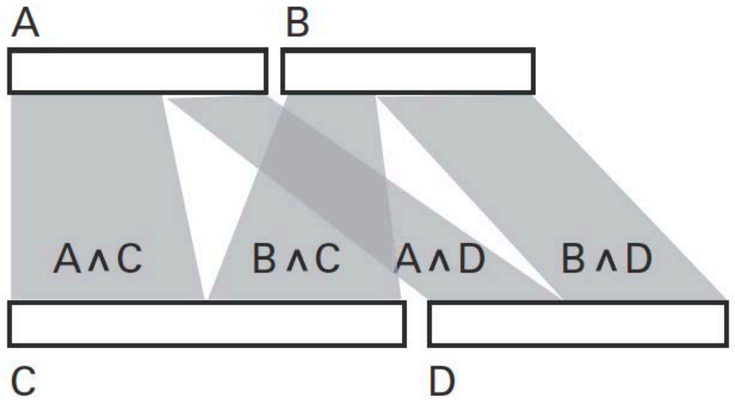
+ radial layouts may be more effective in showing the periodicity of patterns

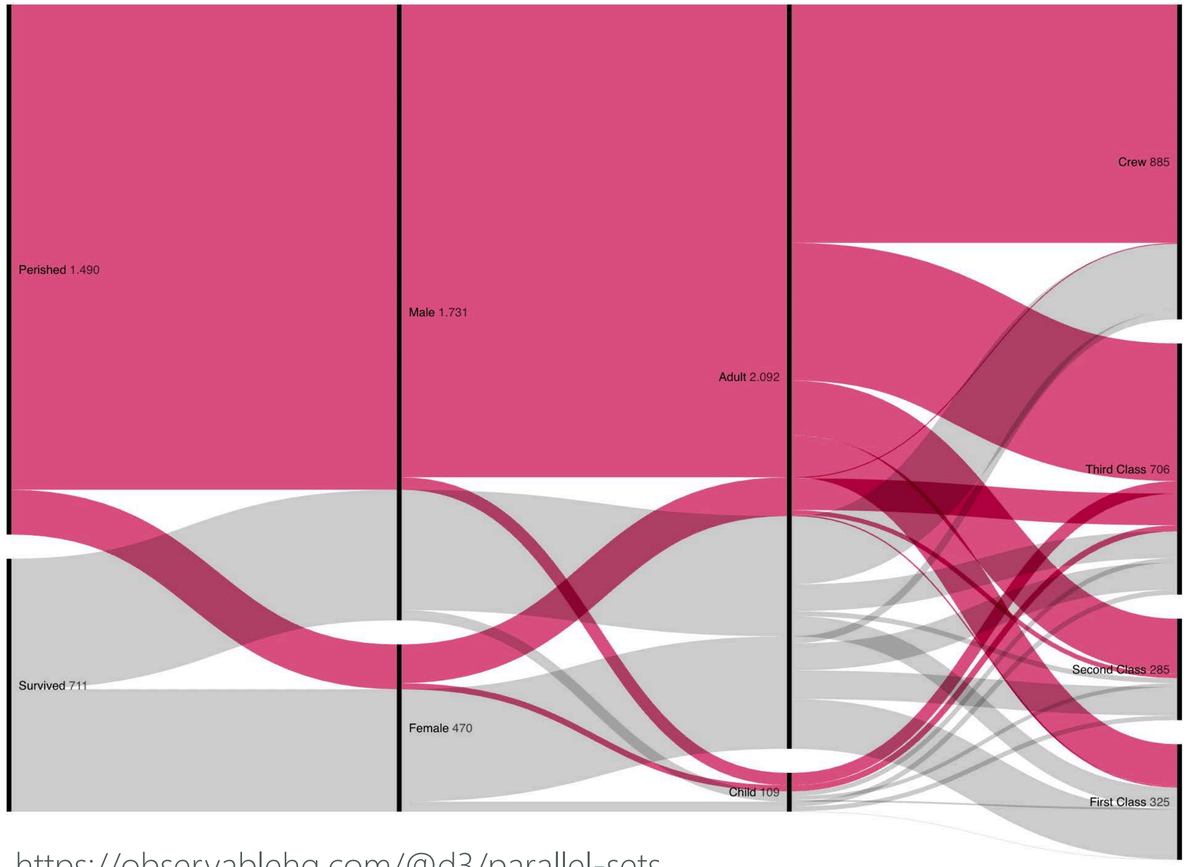


https://www.visualcinnamon.com/2015/10/different-look-d3-radar-chart/ - https://observablehq.com/@observablehq/plot-radar-chart

PARALLEL SETS » ALLUVIAL DIAGRAM

- interactive visualization for exploring categorical data
- technique extends the parallel coordinates by representing the set of categories along each axis, while scaling the categories according to their corresponding frequencies

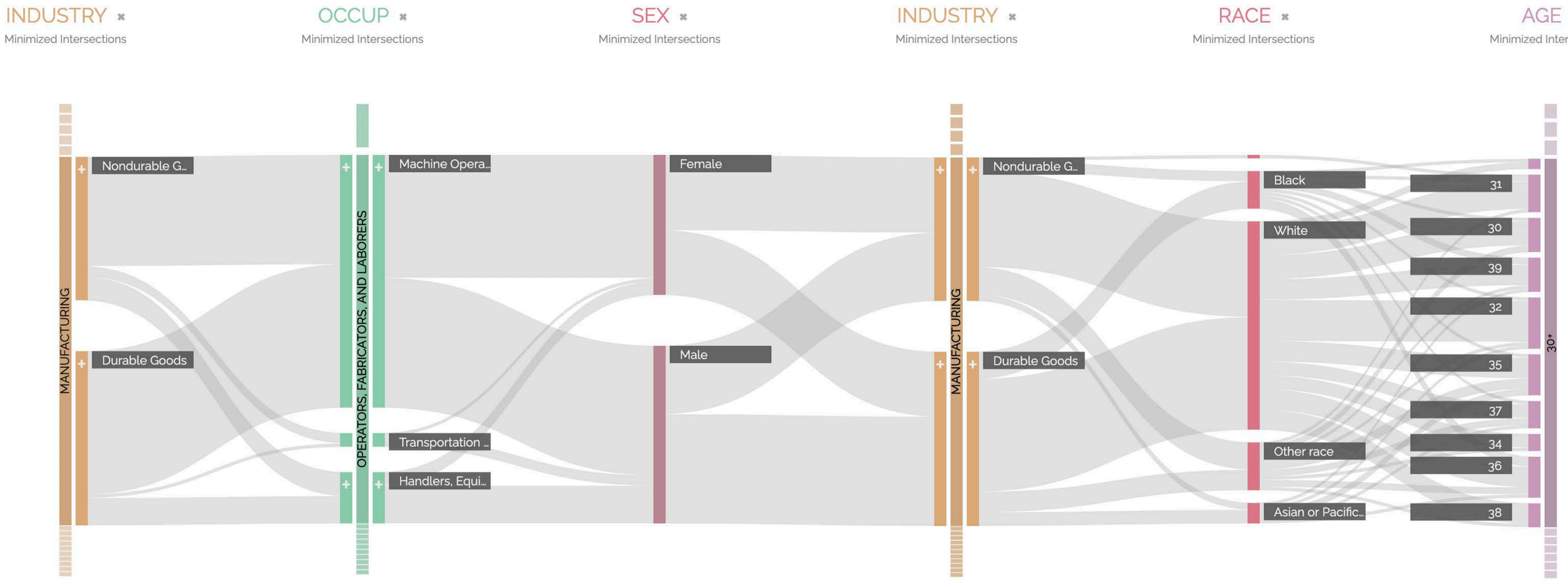




https://observablehq.com/@d3/parallel-sets

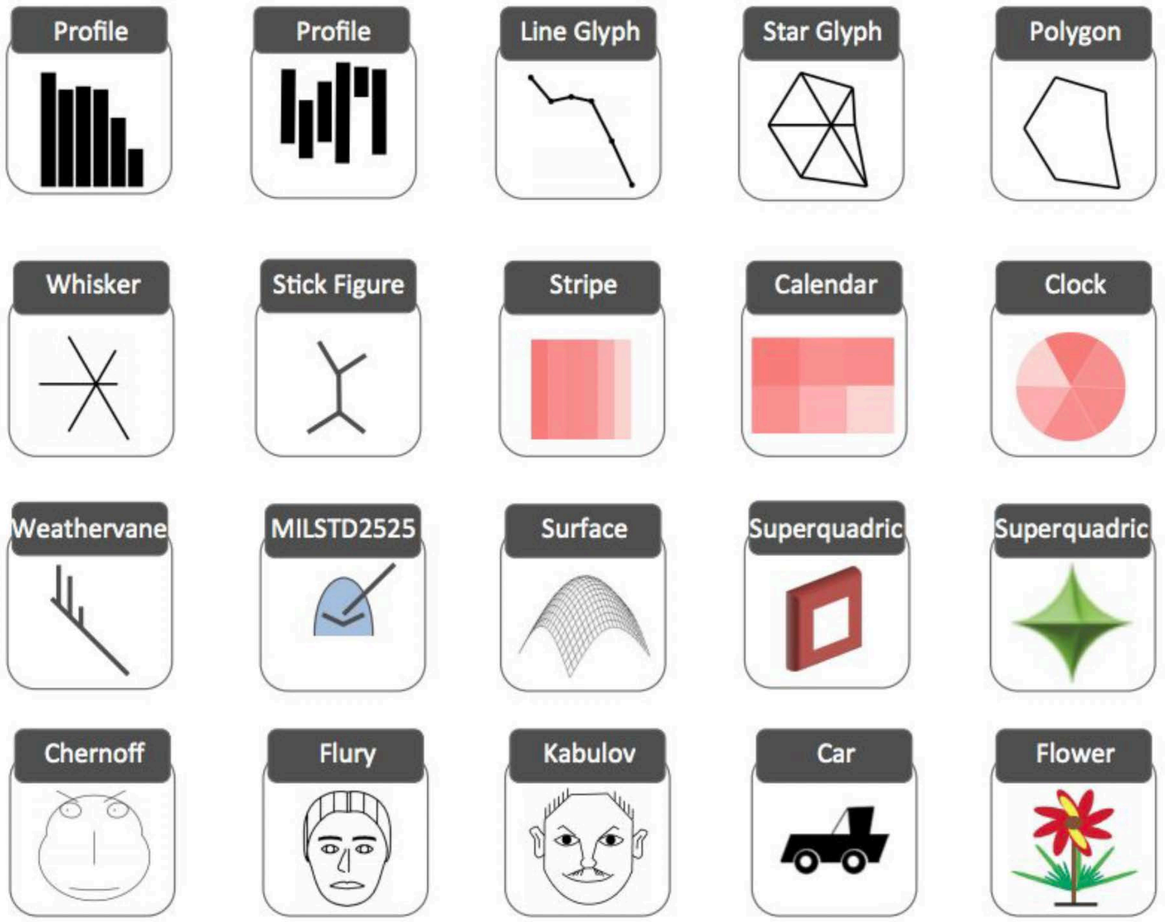
Kosara R, Bendix F, Hauser H. Parallel sets: interactive exploration and visual analysis of categorical data. IEEE Trans Vis Comput Graph. 2006

PARALLEL HIERARCHIES



GLYPHS

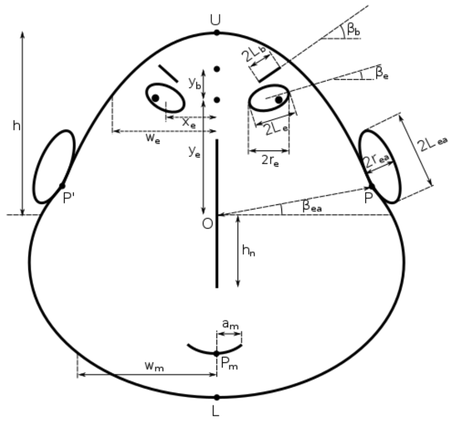
- single data points are encoded individually by assigning their dimensions to one or more marks and their visual variable
- allow the identification of patterns involving several dimensions
- they can be placed in different layout strategies

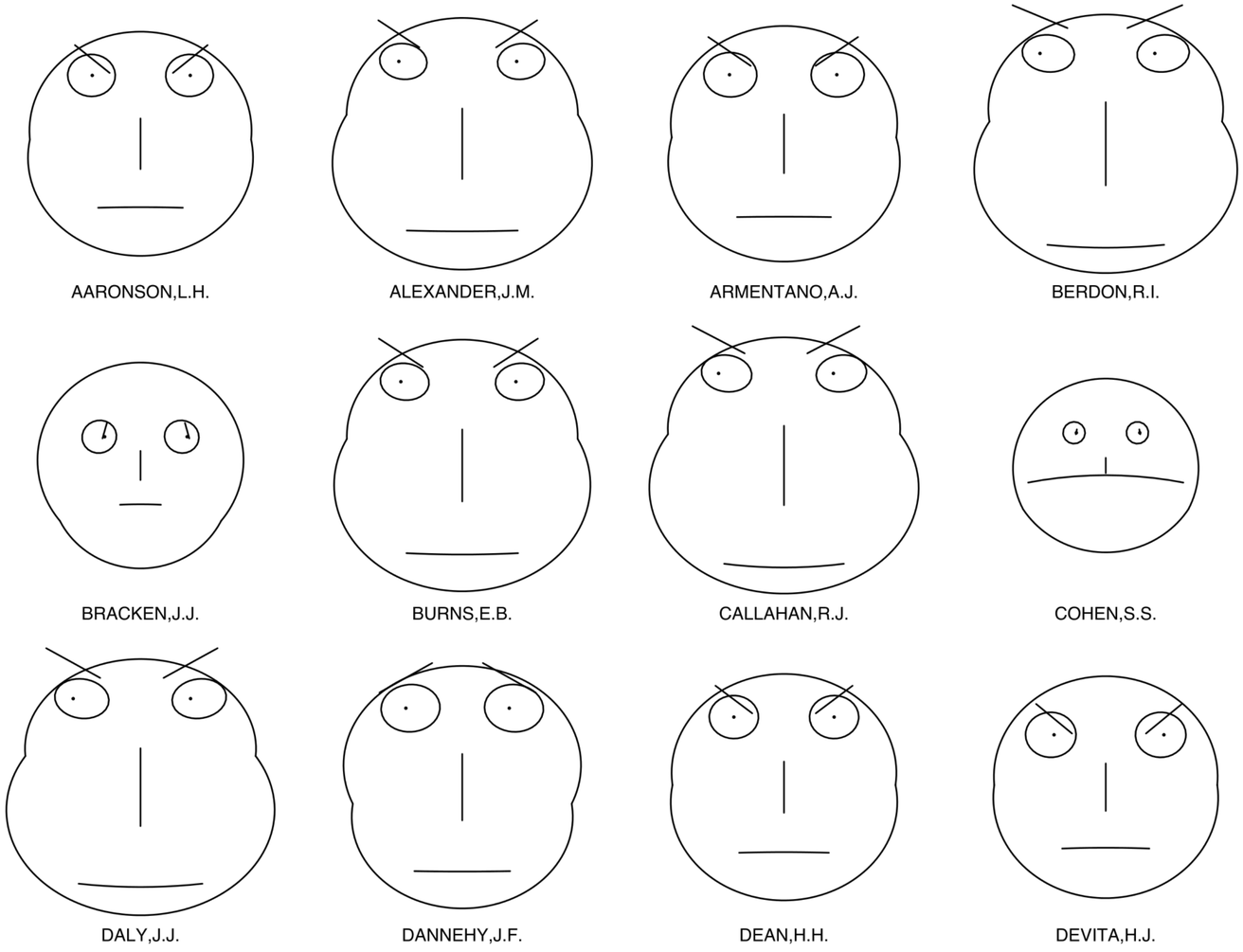


Fuchs, Johannes, et al. "Evaluation of alternative glyph designs for time series data in a small multiple setting." Proceedings of the SIGCHI 2013

CHERNOFF FACES

- encode data by varying the features of a human face, such as the expression of the mouth or the eyes or the shape and size of the nose
- **One-to-One-Mapping:** each data attribute is mapped to a different visual attribute





Chernoff, H. (1973). The use of faces to represent points in k-dimensional space graphically. Journal of the American Statistical Association

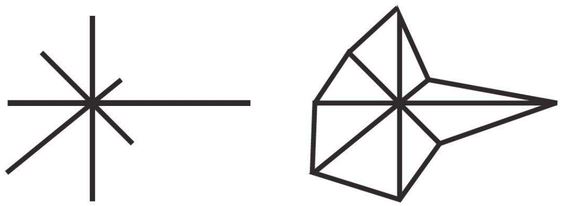
GLYPHBOARD




GLYPHBOARD
learning_time
success
credits
secondoldday
membershipduration
gender
score
ageContent
Colored Feature: Cluster
8.253 items
DATA
GLYPHS
FILTER
CONTEXT
Dataset
cmm_lernsituationen_...
Dataset
cmm_lernsituationen_...
Version
29102018
Version
29102018
Position Algorithm
pca
Position Algorithm
tsne
Feature Context
standard context
Feature Context
standard context

STAR PLOT

- **Star & Whisker Plot:** length of each line represents one quantitative attribute
- **Star Plot:** shape of star is emphasized by contour line
- **Many-to-One-Mapping:** several data attributes are mapped to the same visual attribute (length)





Keck, M. et al.: Towards Glyph-based Visualizations for Big Data Clustering, VINCI conference 2017

FLOWER GLYPH

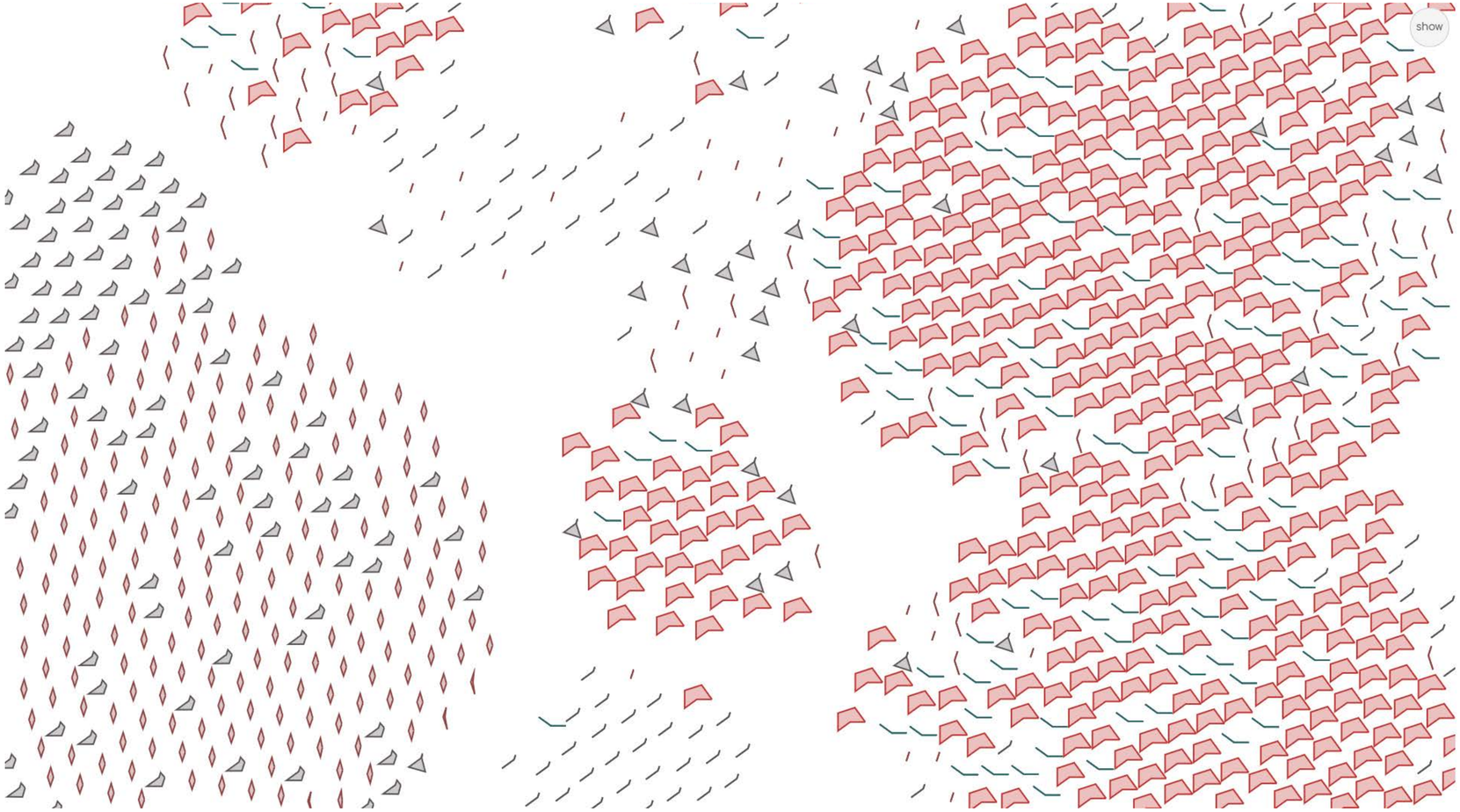
- length of each flower petal represents one quantitative attribute
- categorical attribute is encoded by color



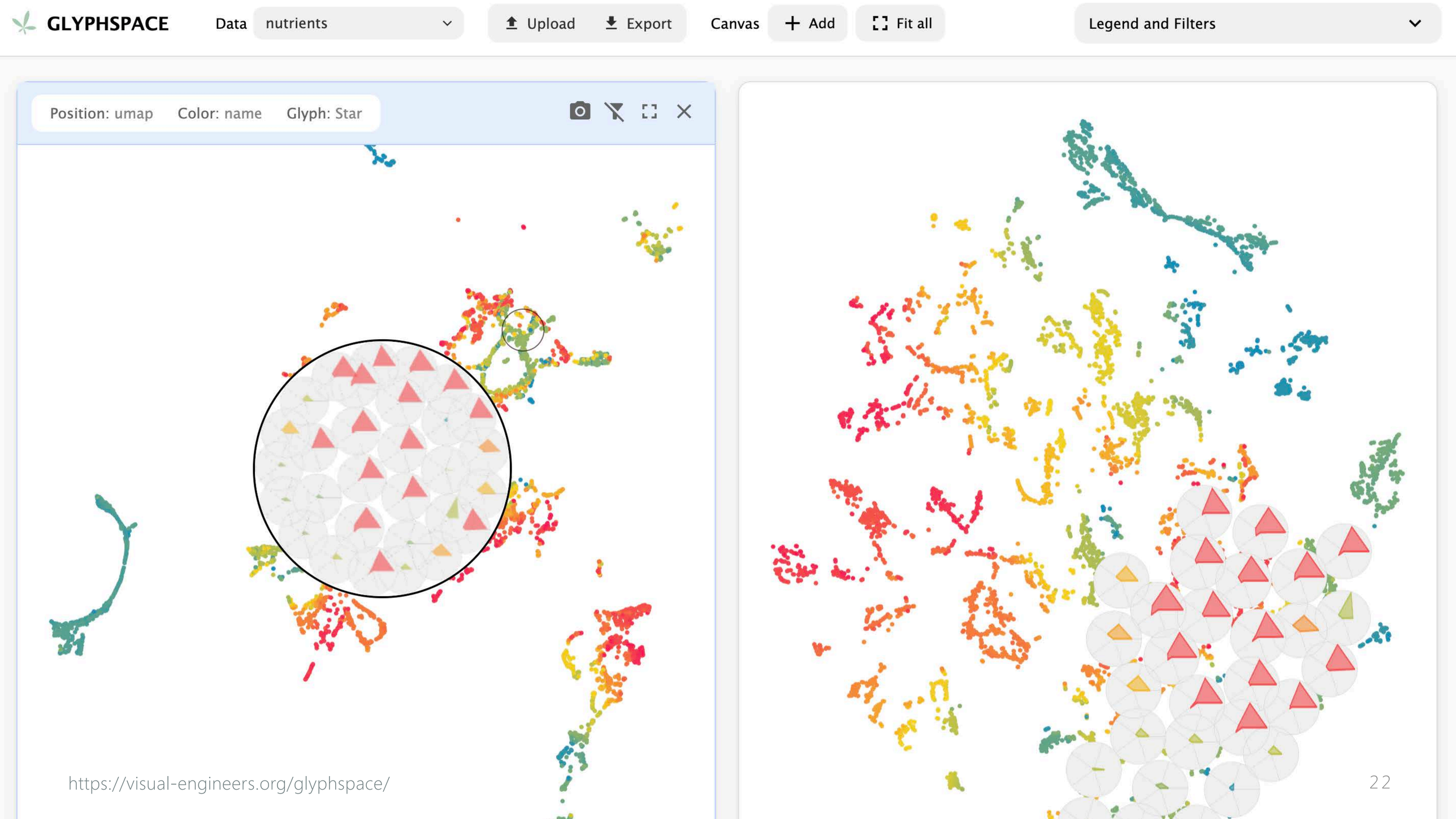
Keck, M. et al.: Towards Glyph-based Visualizations for Big Data Clustering, VINCI conference 2017

GLYPH PLACEMENT

- placing glyphs on grids is useful to compare values
- but glyphs can be placed more flexibly according to data criteria
- here: dimensionality reduction plot

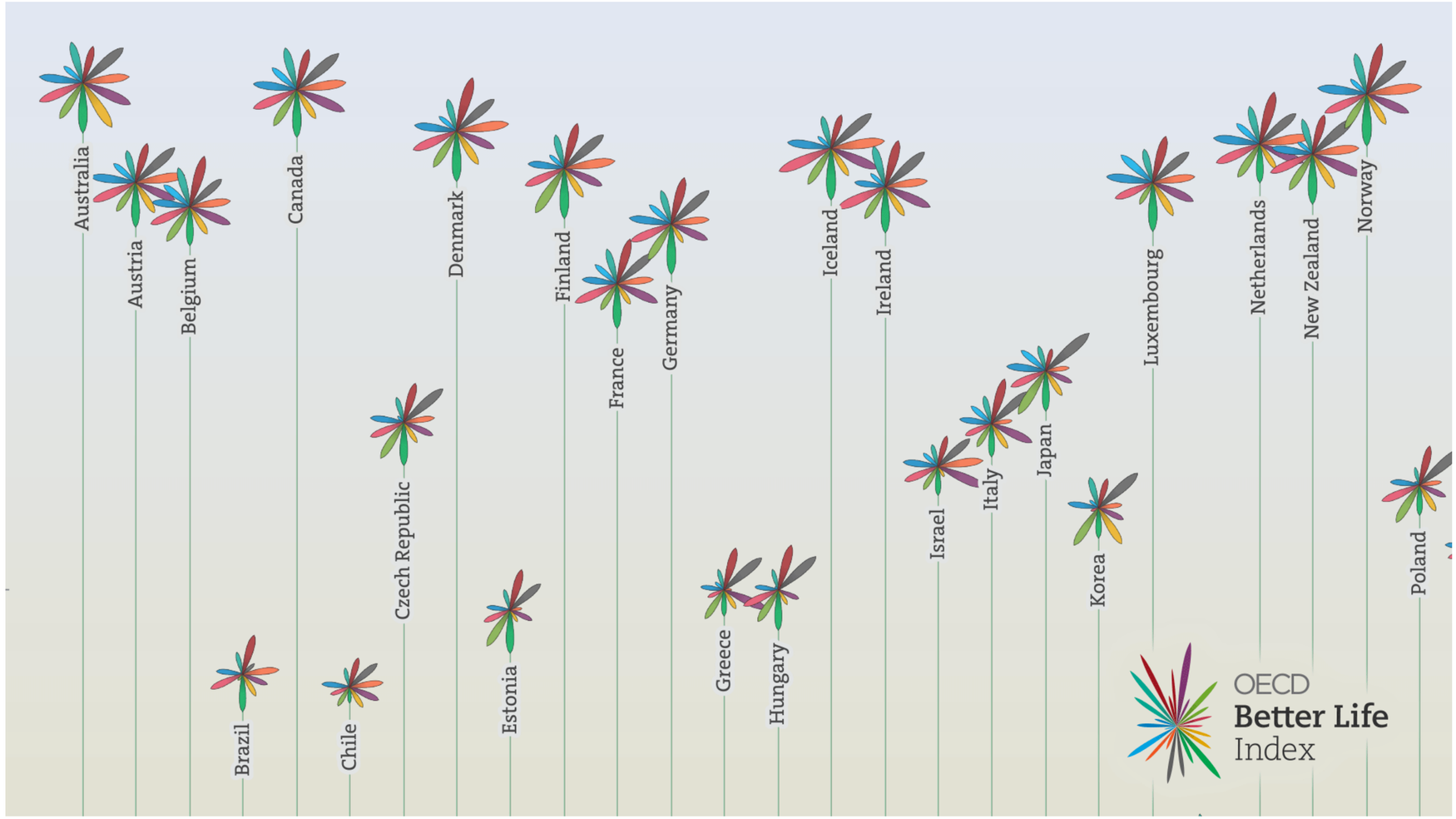


Keck, M. et al.: Towards Glyph-based Visualizations for Big Data Clustering, VINCI conference 2017


GLYPHSPACE
Data
nutrients
Upload
Export
Canvas
Add
Fit all
Legend and Filters
Position: umap
Color: name
Glyph: Star
https://visual-engineers.org/glyphspace/

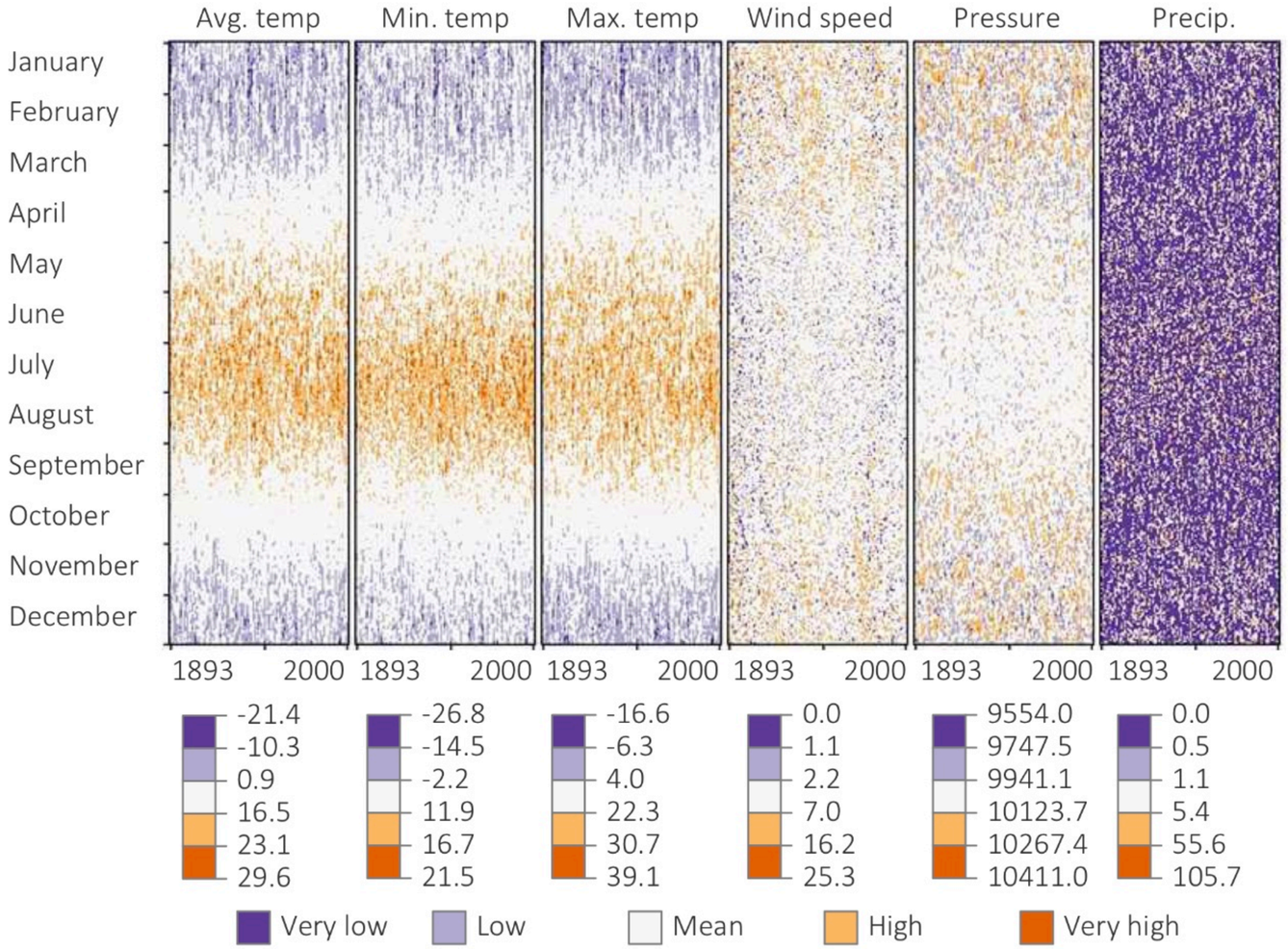
GLYPH PLACEMENT

- Y-axis encodes overall index



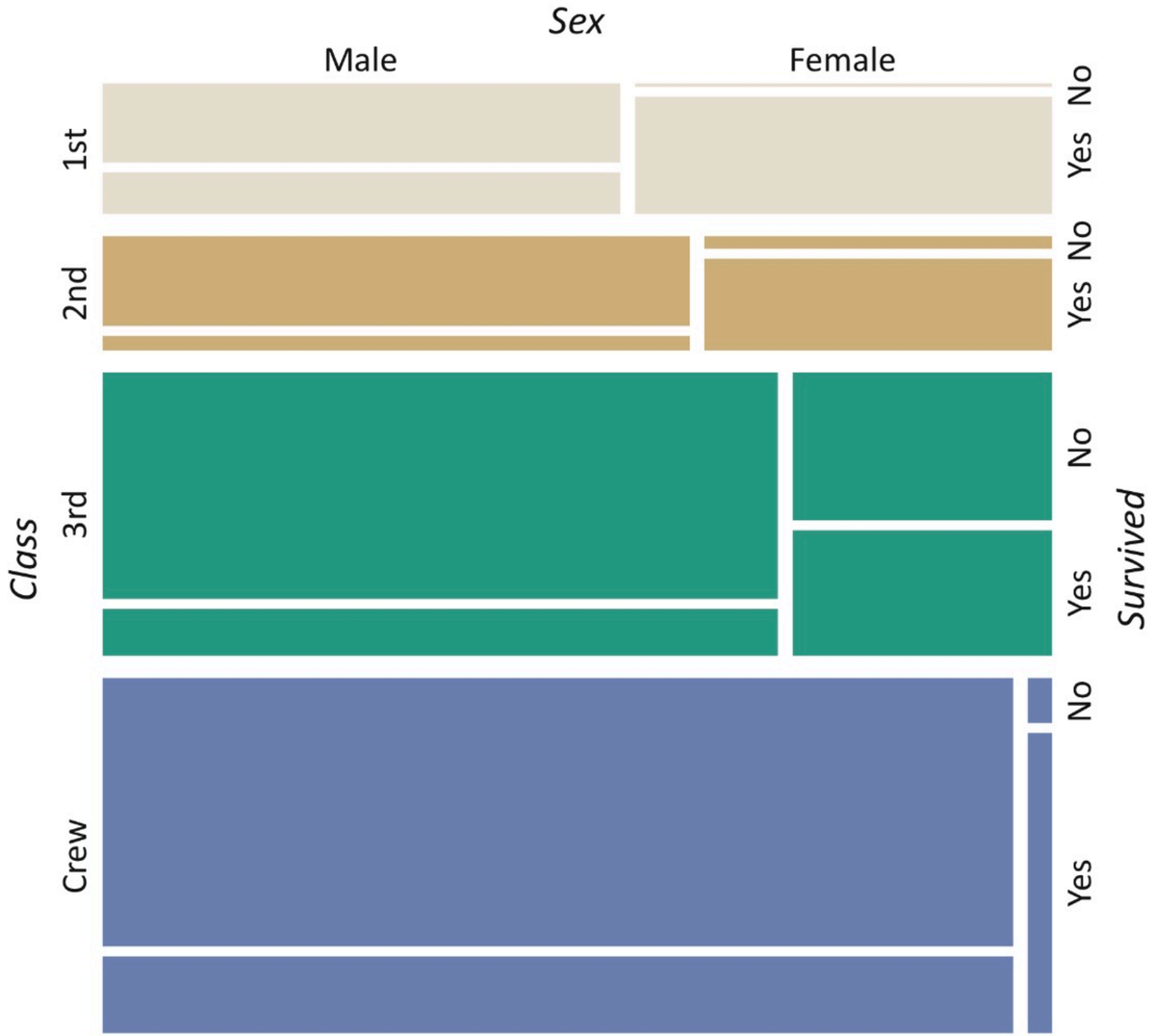
PIXEL-BASED VISUALIZATIONS

- each pixel visualizes exactly one day value by its pixel color
- very space-efficient encoding, even very large datasets can be visualized
- details cannot be discerned easily

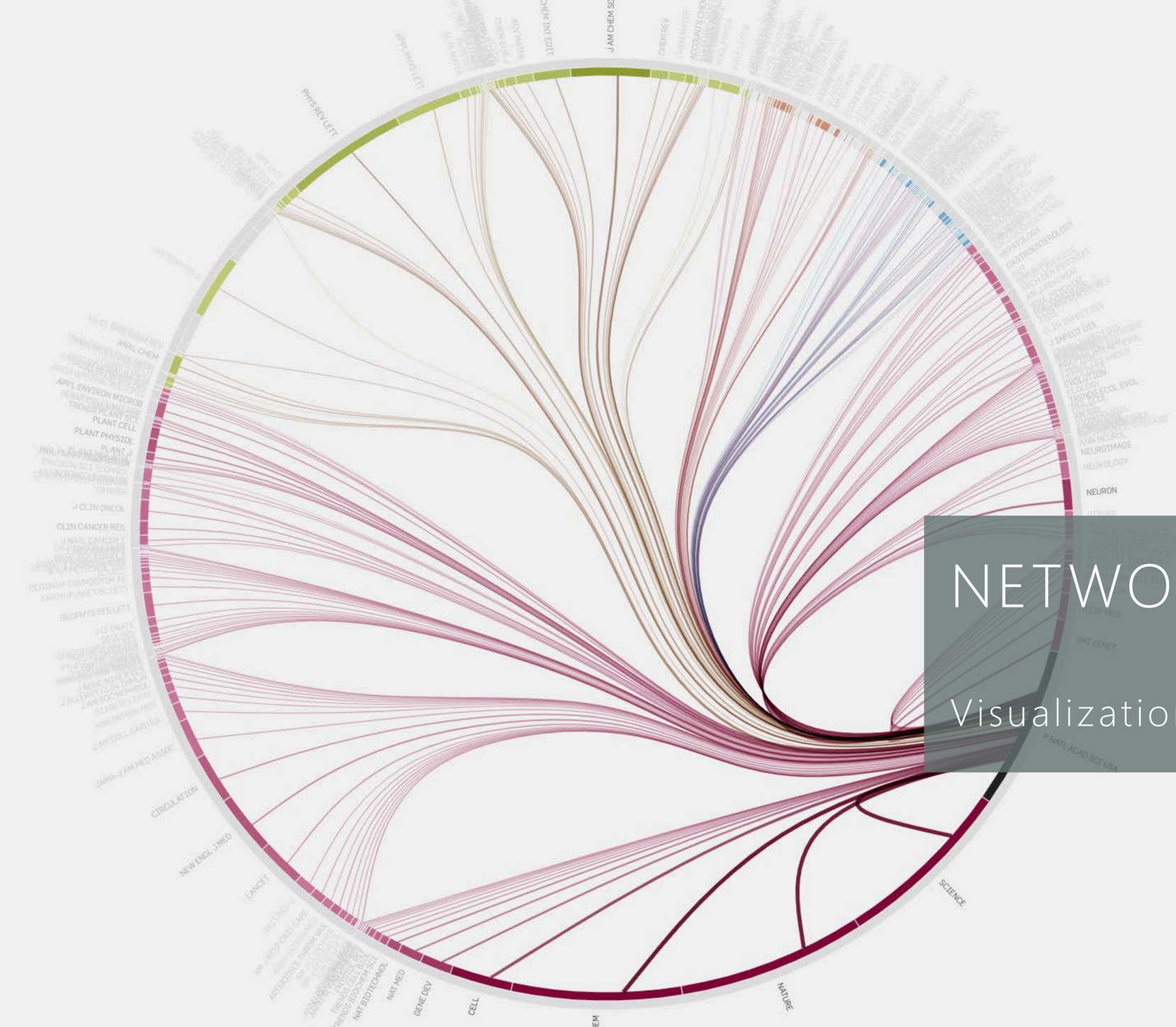


NESTED VISUALIZATIONS

- is based on a recursive embedding of attribute subsets
- resulting visual representations are very well suited to communicate frequencies of the different value combinations in the data set



Mosaic plot: shows an example with the distribution of surviving passengers of the Titanic with respect to class and sex



NETWORK

Visualization Techniques

RELATIONAL DATA

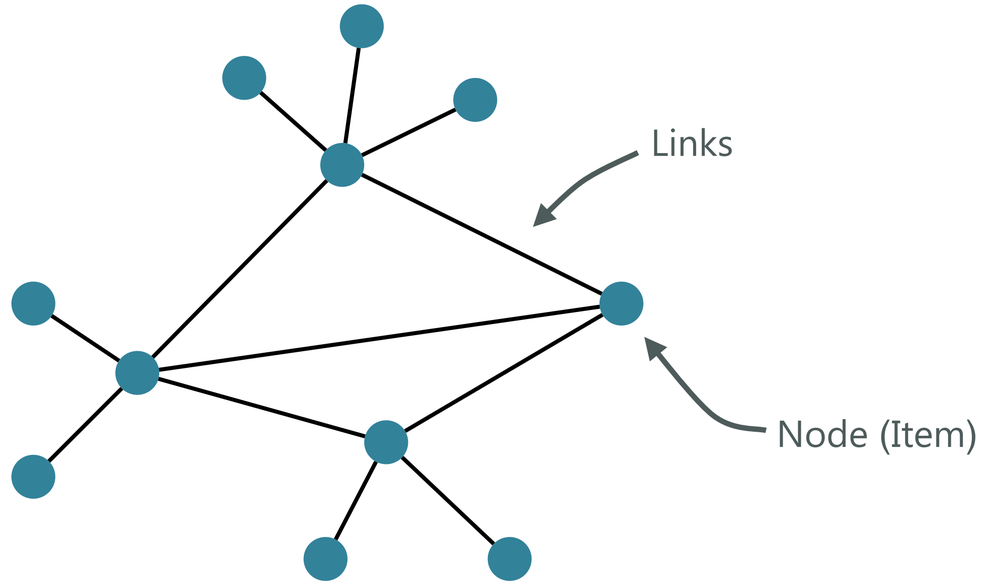
Networks are collections of nodes and links

Label	Value
A	127
B	96
C	65

Nodes

N1	N2	Value
A	B	4
A	D	3
B	C	1

Links



Nodes:

Degree: Number of links incident to a node

Label: Name of the represented entity

Attributes: additional information

Links:

Weight: strength of the connection between two nodes

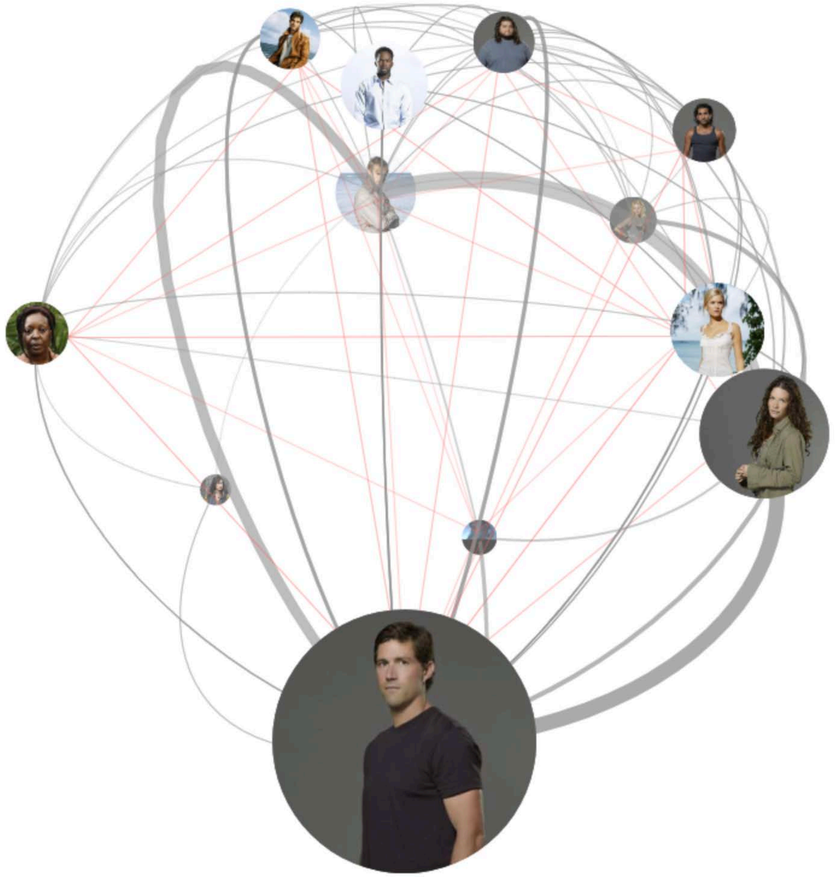
Direction: in case of directed graphs with origin destination

Path:

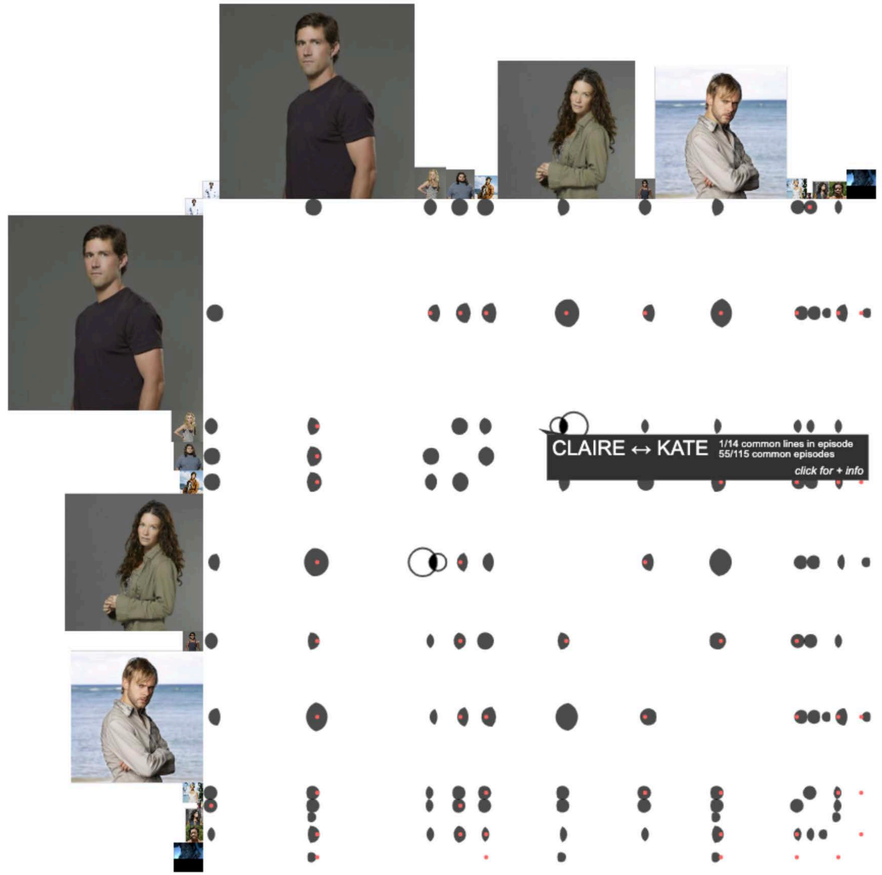
sequence of nodes and links

Cycle: if first and last node are the same

TYPES OF REPRESENTATION



Node-Link

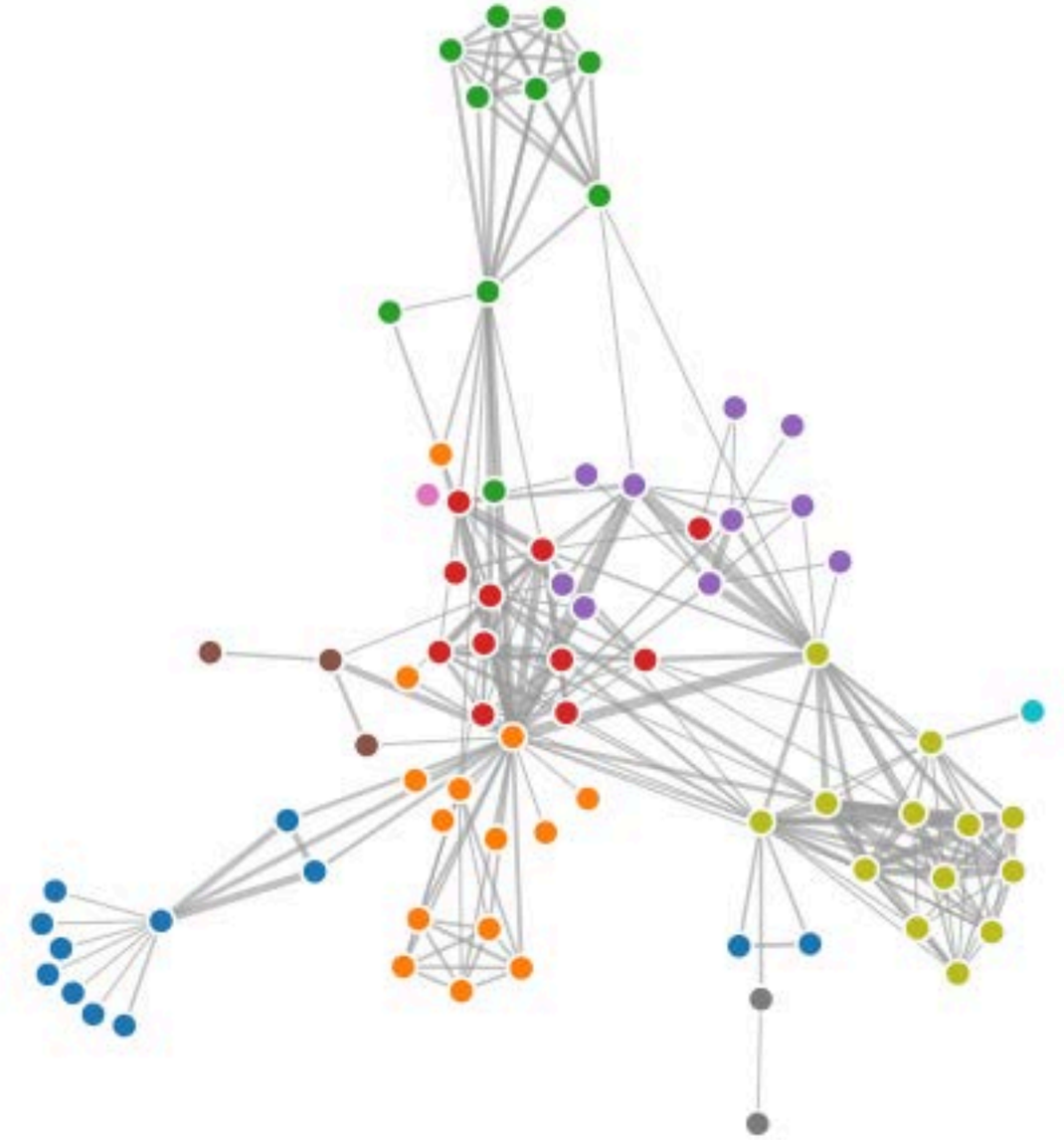


Matrix

http://intuitionanalytics.com/other/lostalgic/

FORCE-DIRECTED LAYOUT

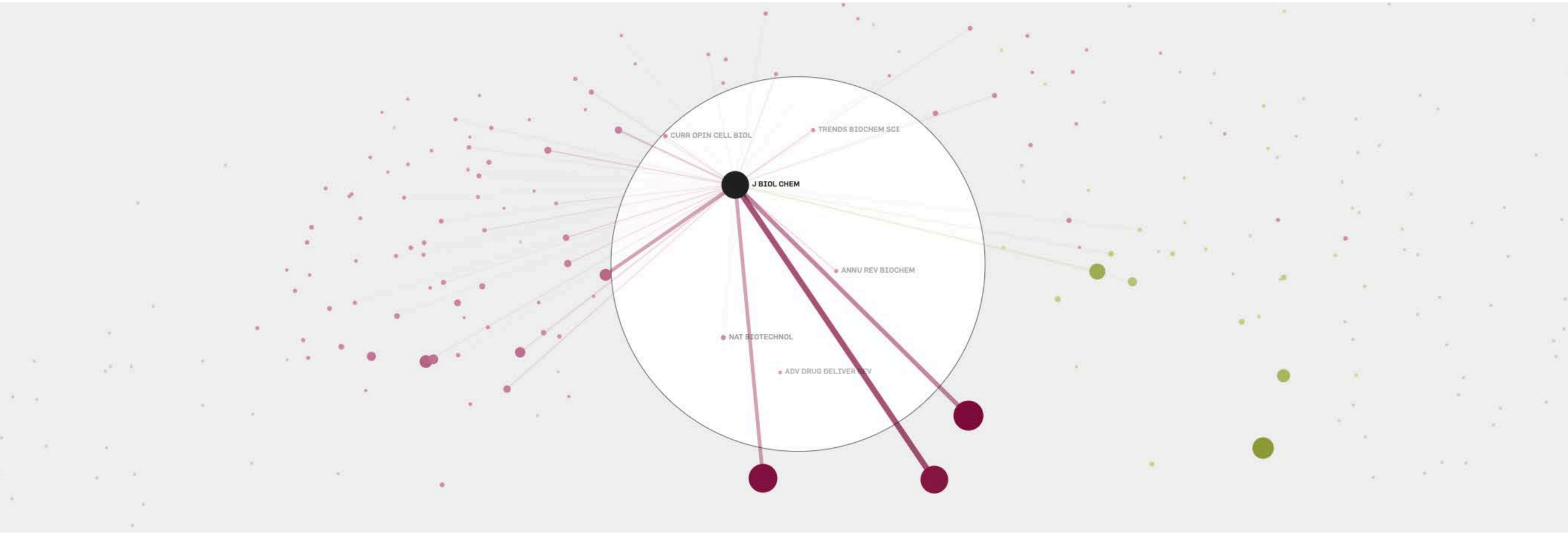
- nodes use a force-directed placement and links serve as springs that pull nodes together
- most widely used technique in node-link diagrams
- main goal: visualize the structure of the network
- **non-deterministic:** they look different each time the algorithm is run



https://observablehq.com/@d3/force-directed-graph

POLAR LAYOUT

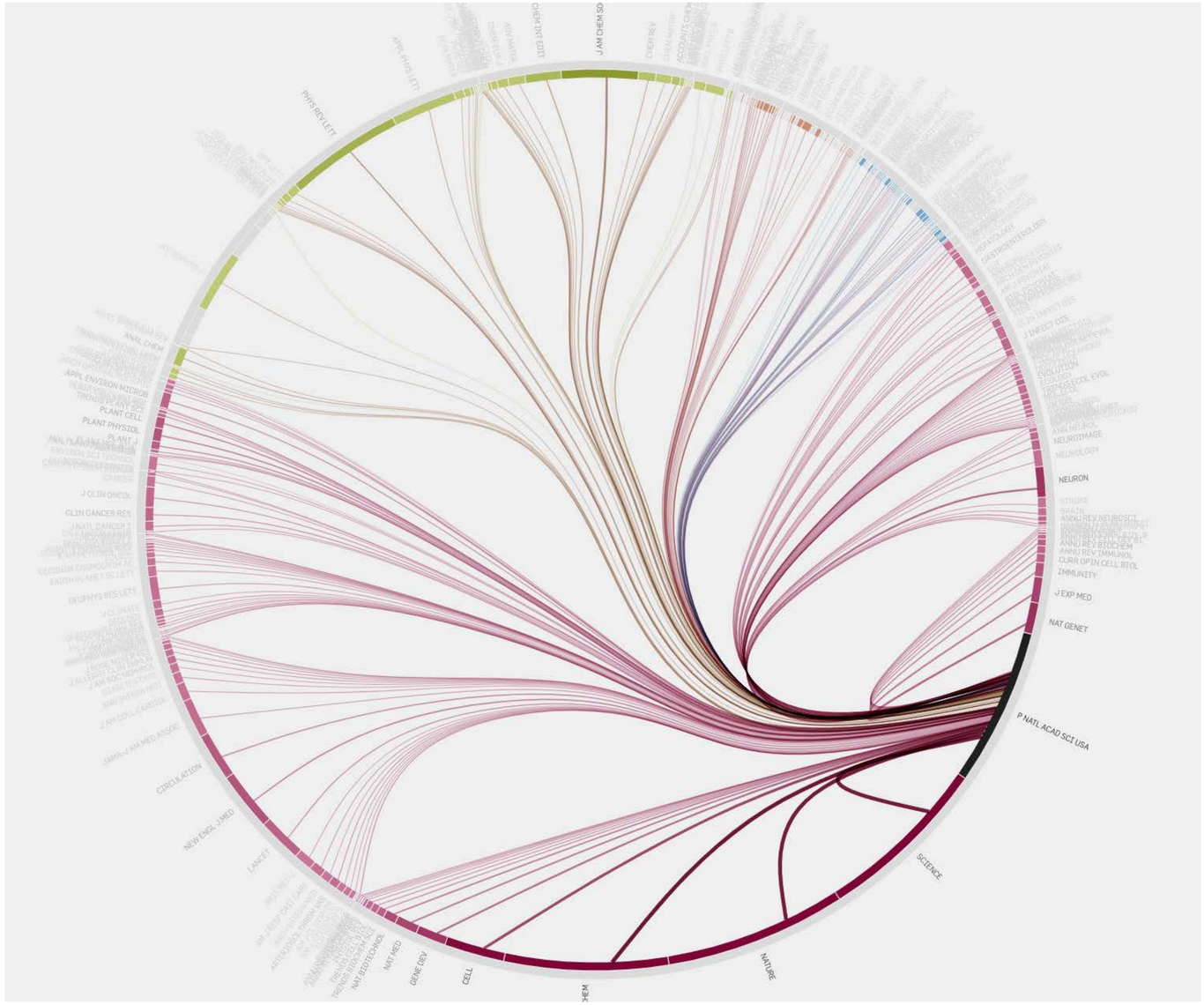
- nodes are organized around a central node



http://well-formed.eigenfactor.org/map.html

CIRCULAR LAYOUT

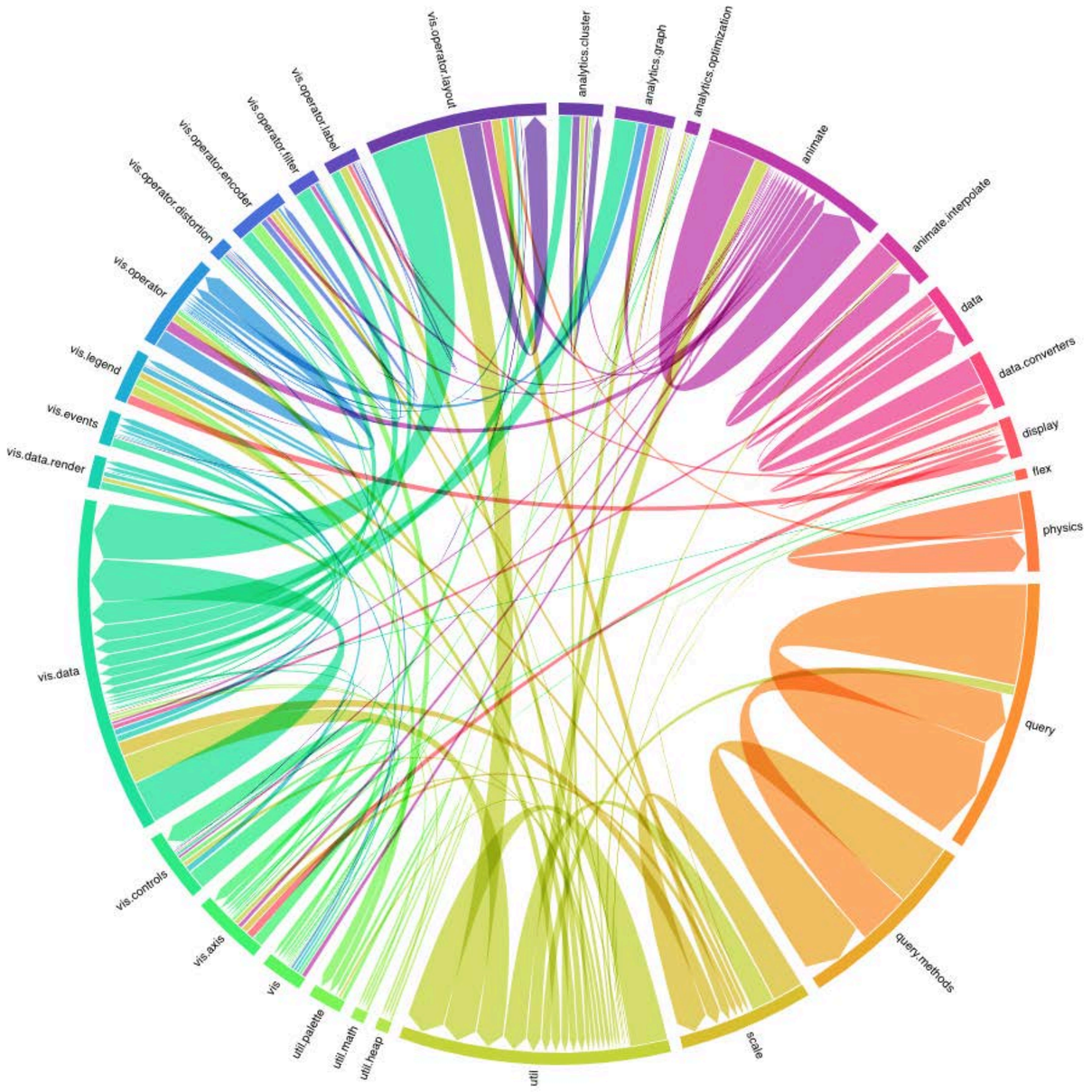
- nodes are located around the circumference of the circle and connected by links
- **Edge Bundling:** to reduce the clutter links with the same direction are bundled
- since the layout is fixed, nodes can be grouped in meaningful categories



http://well-formed.eigenfactor.org/radial.html

CHORD DIAGRAM

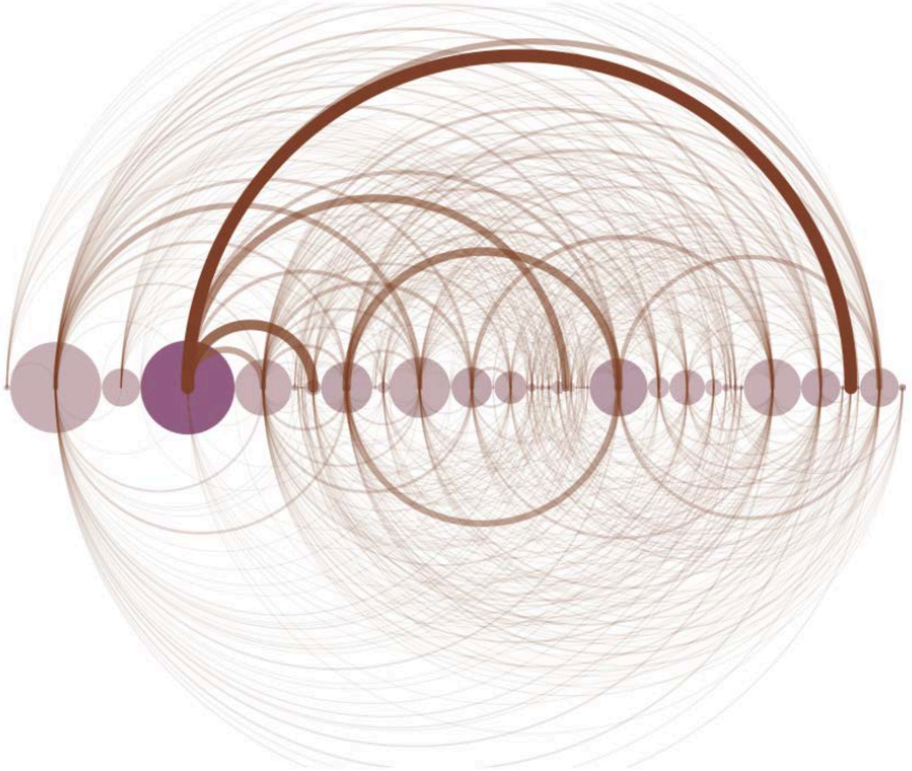
- shows relationships between nodes arrayed in a circle
- thickness of the arcs (often also differentiated by color or transparency) represent the degree of the connection between the different groups

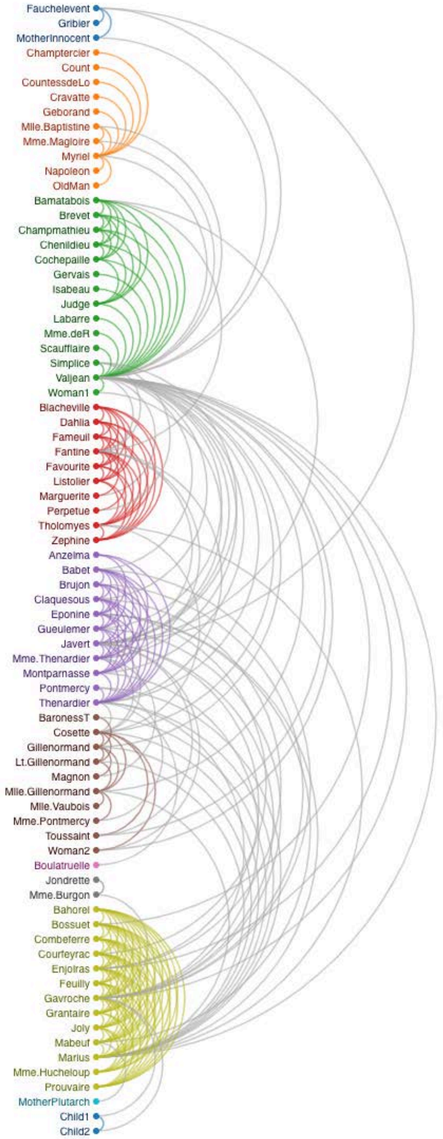


https://observablehq.com/@d3/chord-dependency-diagram

ARC CHART

- restricted node-link layout: layout nodes along line
- focus on ordering of nodes

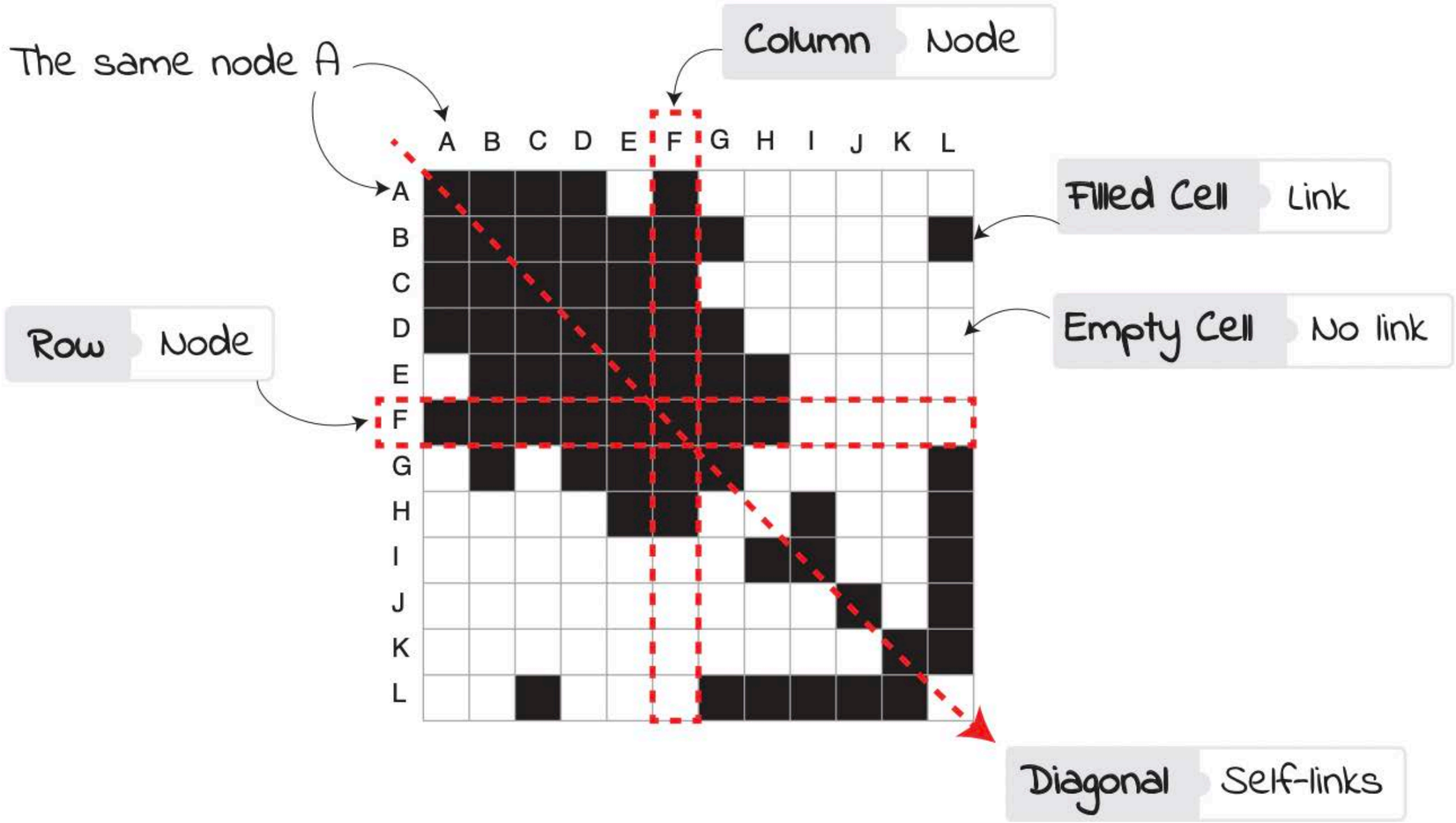




http://datavis.dekstop.de/irc_arcs/ - https://observablehq.com/@d3/arc-diagram

ADJACENCY MATRIX

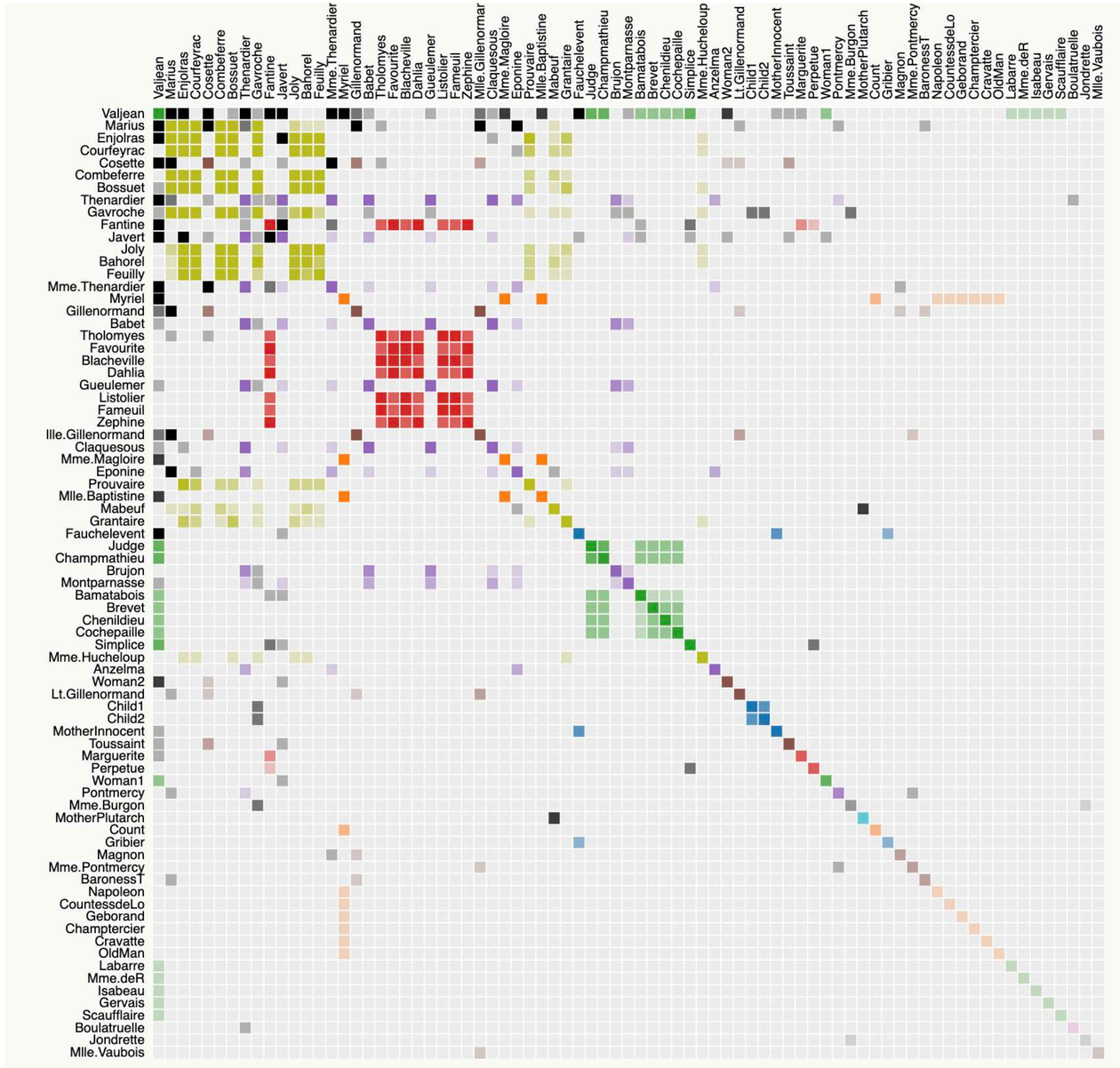
- matrix view with nodes listed along the horizontal and vertical axes
- links between two nodes are indicated by coloring an area mark in the matrix cell
- **advantages:** nodes are fully visible, no line crossings, less clutter, no hairball effect
- **disadvantages:** less familiar, scalability, reordering often required



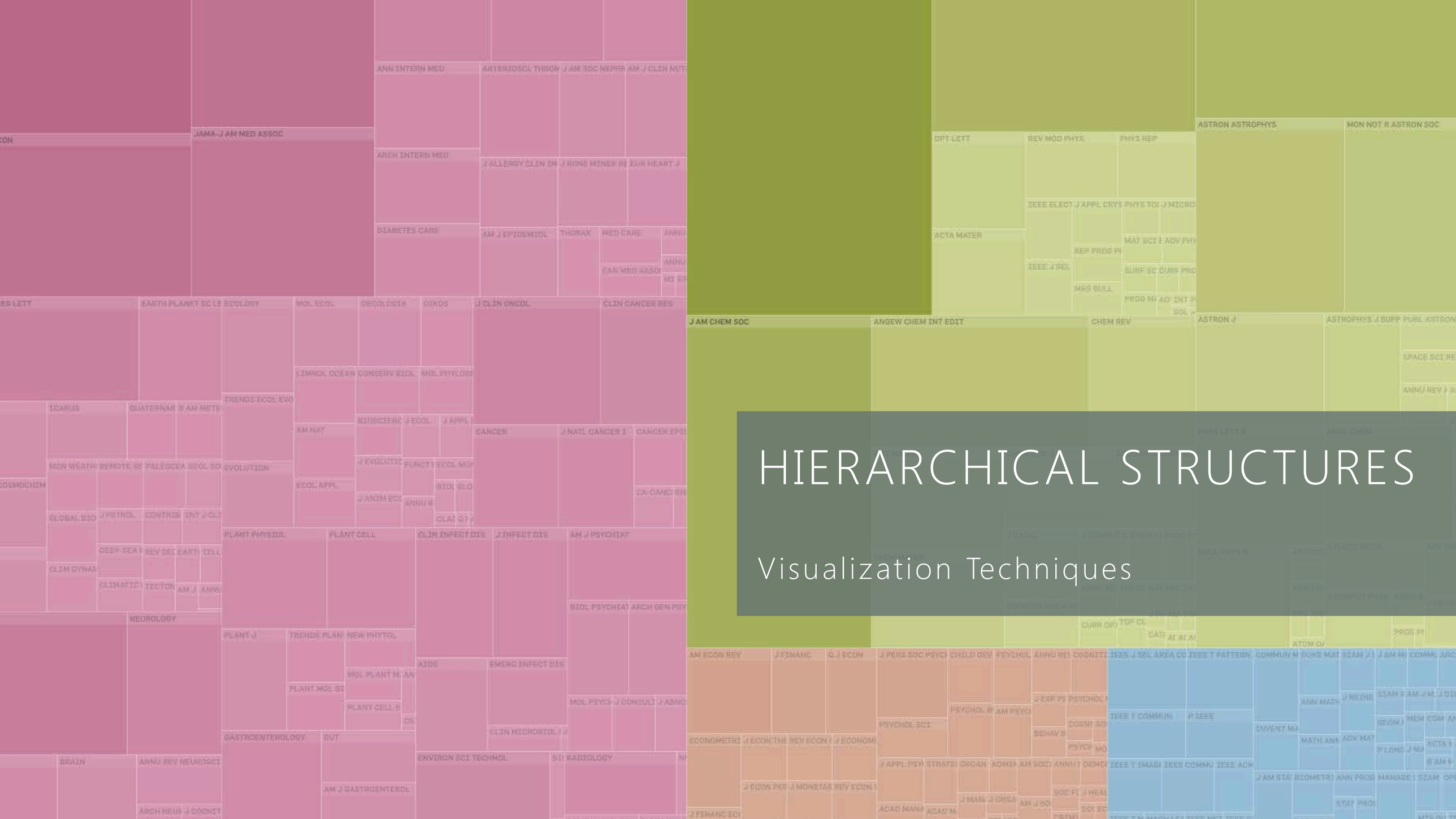
https://visualizationcheatsheets.github.io/matrix.html

ADJACENCY MATRIX

- **undirected networks:** links are symmetric
- **directed networks:** full square of the matrix has meaning, because links can be asymmetric



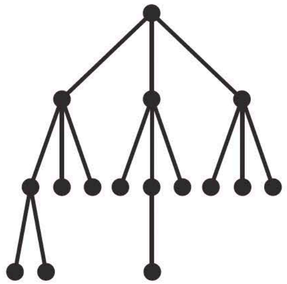
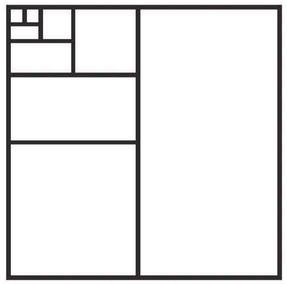
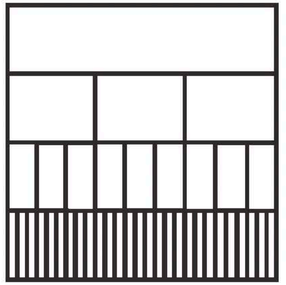
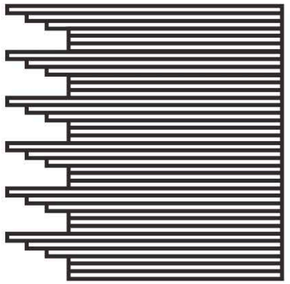
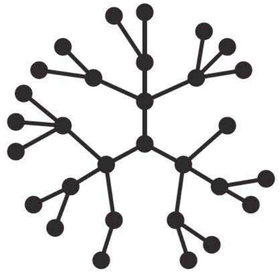
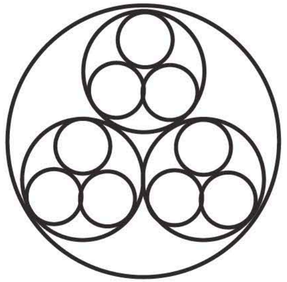
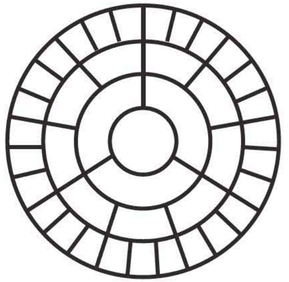
https://bost.ocks.org/mike/miserables/ - https://observablehq.com/@bstaats/matrix-diagram



HIERARCHICAL STRUCTURES

Visualization Techniques

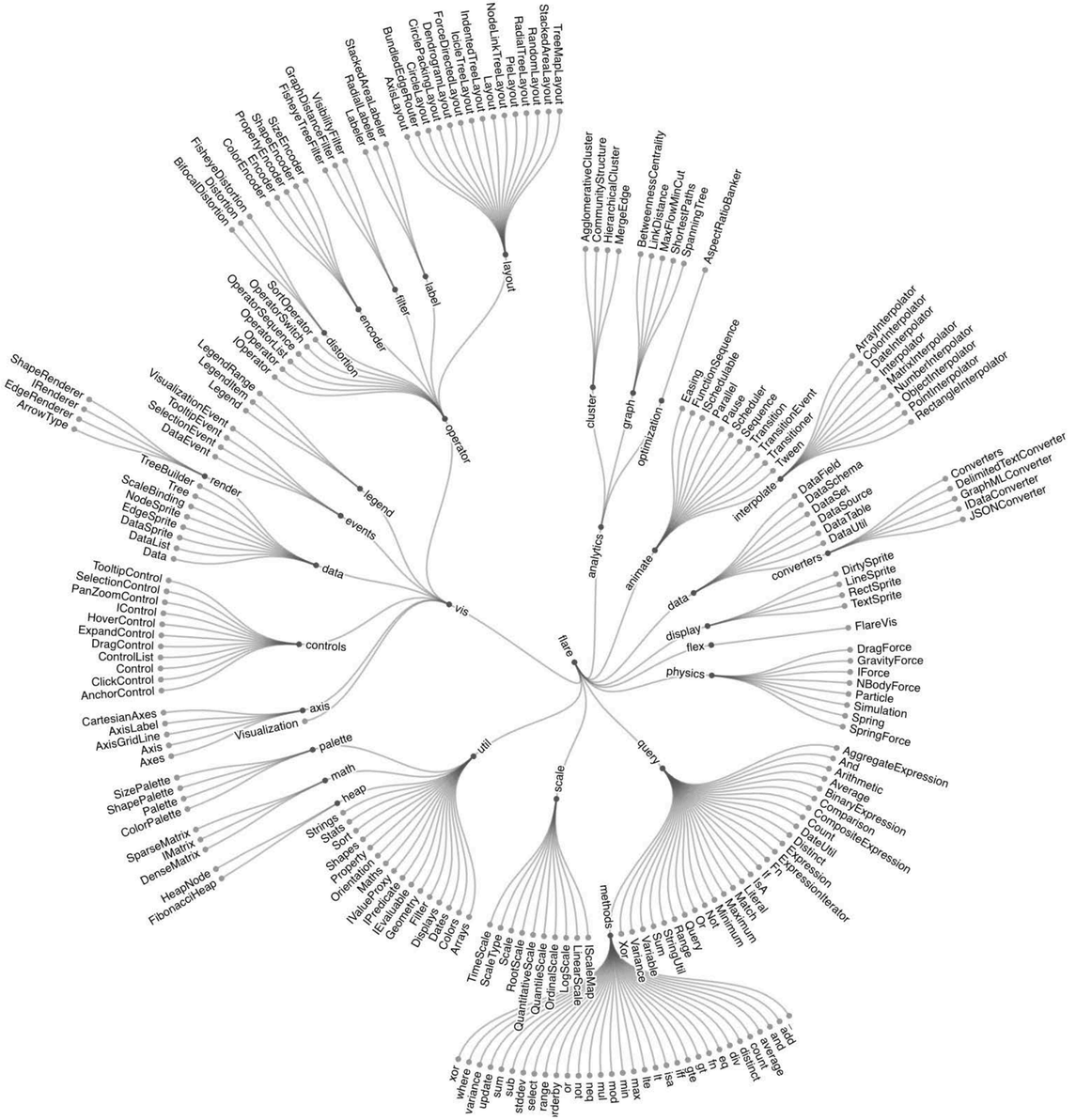
HIERARCHICAL STRUCTURES

	linked	nested	partitioned	indented
rectilinear	Node-Link 	Treemap 	Icicle Tree 	Indented Outline 
radial	Radial Node-Link 	Nested Circles 	Sunburst 	

I. Meirelles: Design for Information, Rockport Publishers, 2013 / McGuffin & Robert 2010

NODE-LINK DIAGRAM

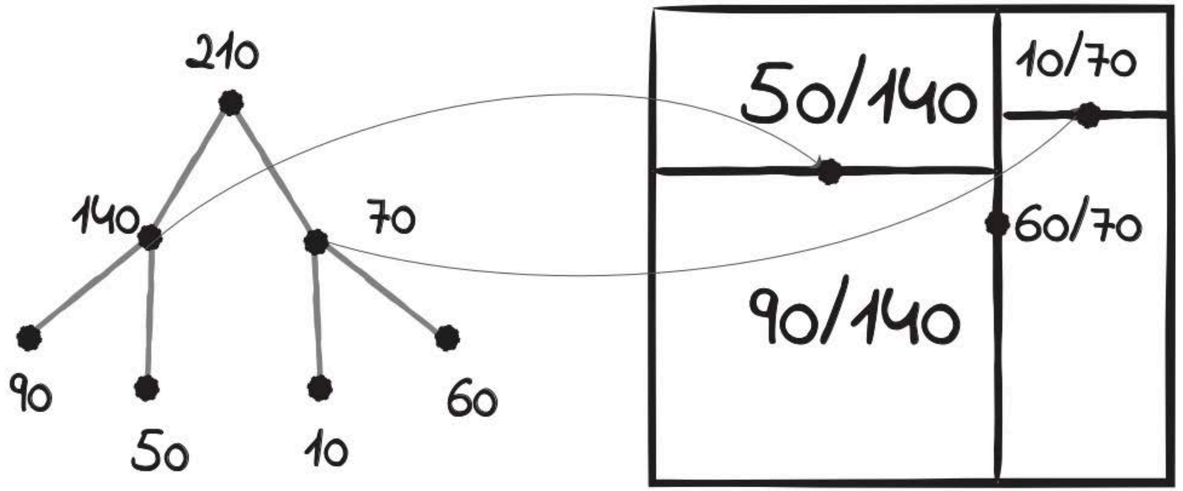
- effective for tasks focused on topological structure, such as tracing paths through the tree

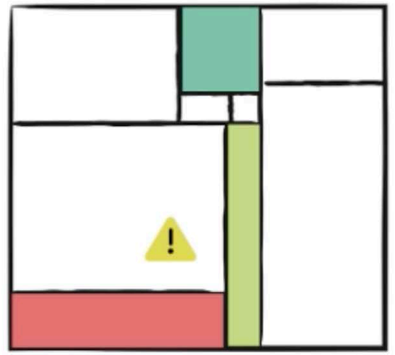


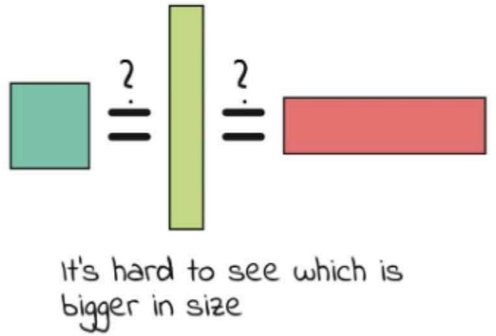
https://observablehq.com/@d3/radial-tidy-tree

TREEMAP

- **Nested layout:** hierarchical relationshIps are shown with containment rather than connection
- are not as effective as node-link for tasks focused on topological structure
- but they shine for tasks that pertain to understand attribute values at the leaves of a tree
- are very effective for spotting the outliers of very large attribute values, e.g. large files



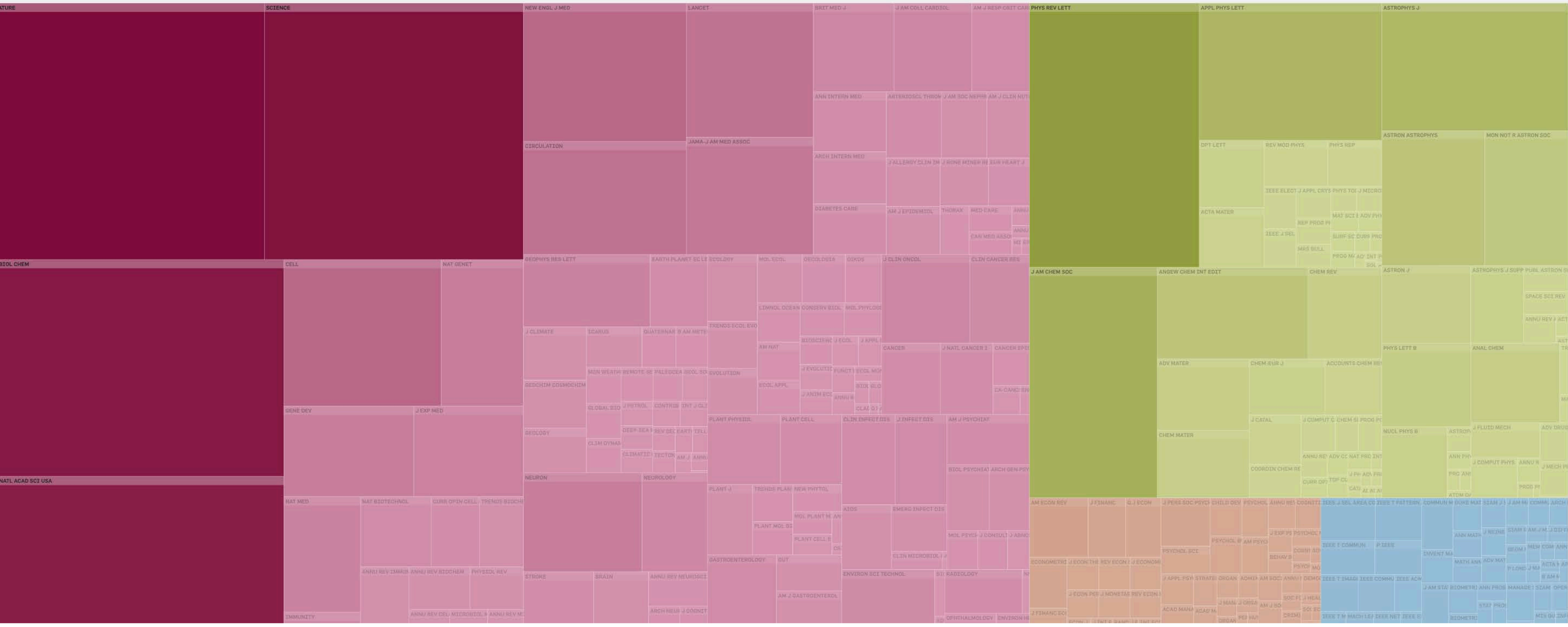




Hard due to different aspect ratios of rectangles

https://visualizationcheatsheets.github.io/treemap.html

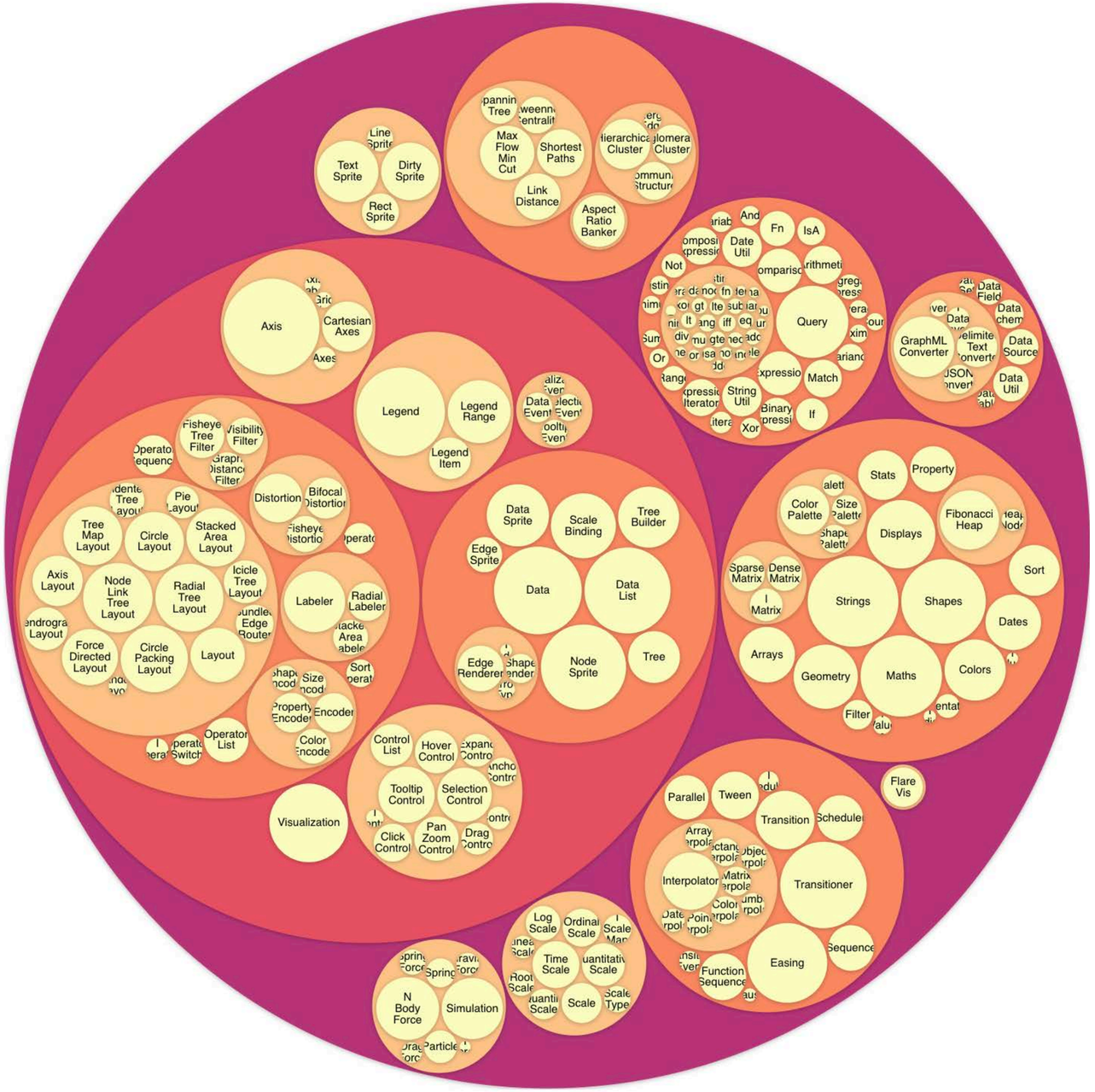
TREEMAP



http://well-formed.eigenfactor.org/treemap.html

NESTED CIRCLES

- although nested circles do not use space as efficiently as a treemap, the “wasted” space better reveals the hierarchical structure

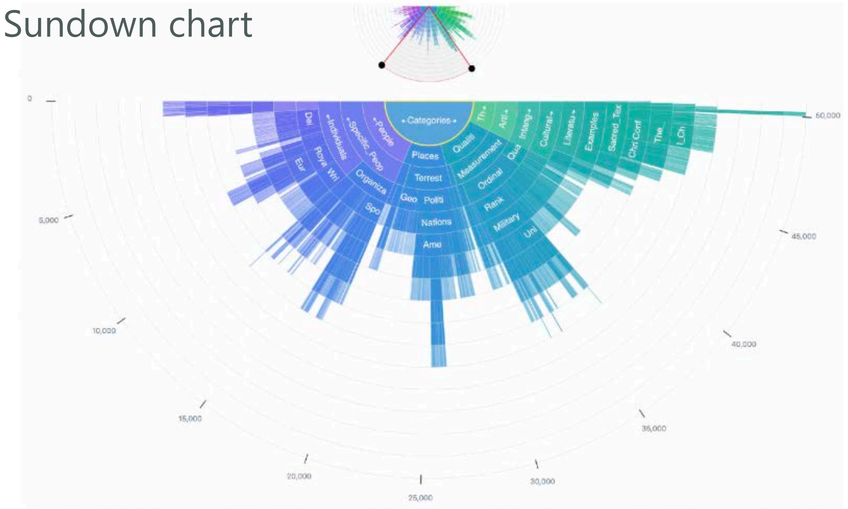


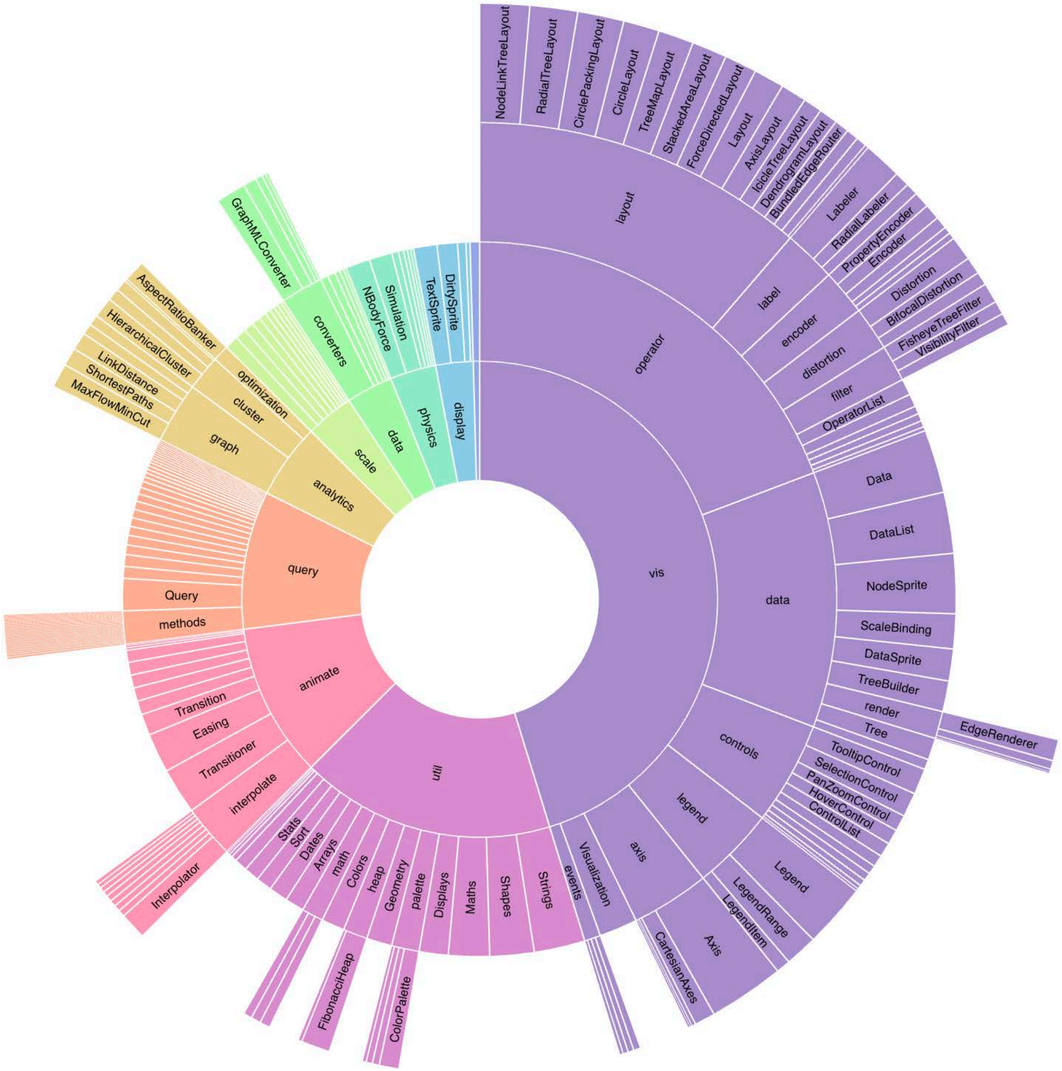
https://observablehq.com/@d3/circle-packing

SUNBURST

- show hierarchal structure
- central ring shows the top level (root)
- each ring in a subset corresponds to a different level in the hierarchy

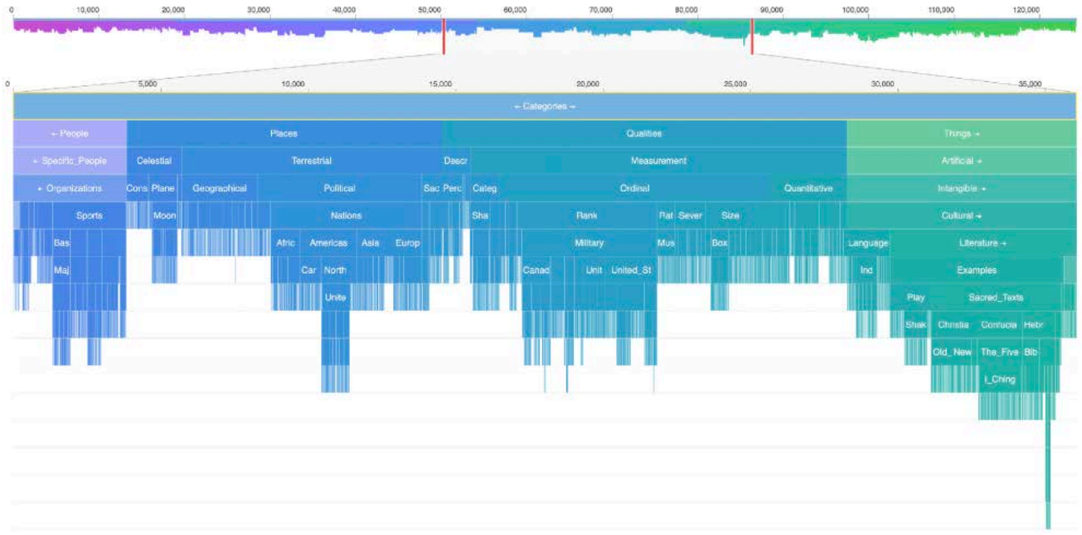
Sundown chart

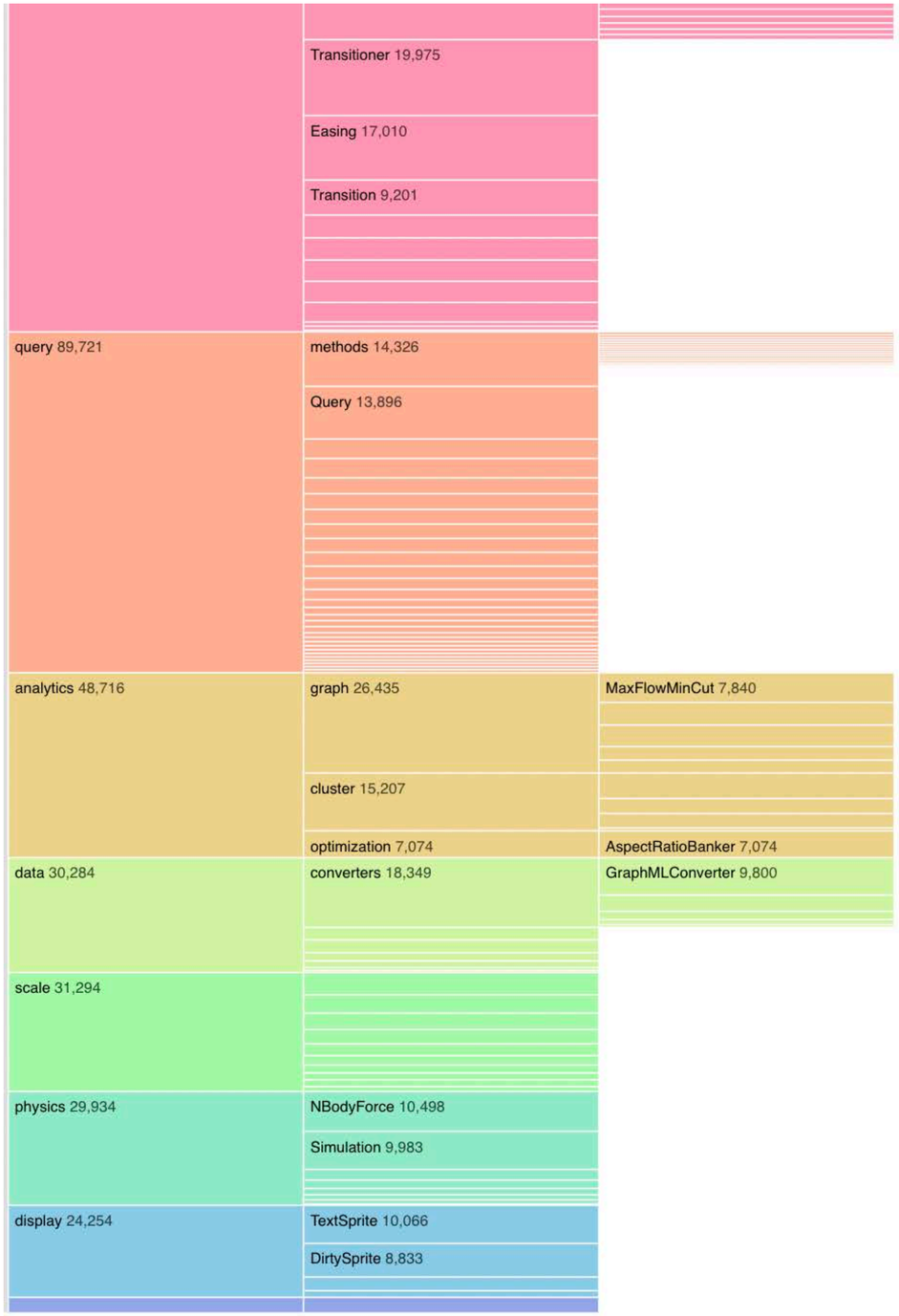




ICICLE PLOTS

- cartesian equivalent to a sunburst
- shows the cumulative values of subtrees
- **Advantage:** hierarchy is easier to perceive, fixed height of hierarchy level, only width varies
- **Disadvantage:** use space less efficiently, leaves can get very small





https://observablehq.com/@d3/icicle



SPATIAL DATA

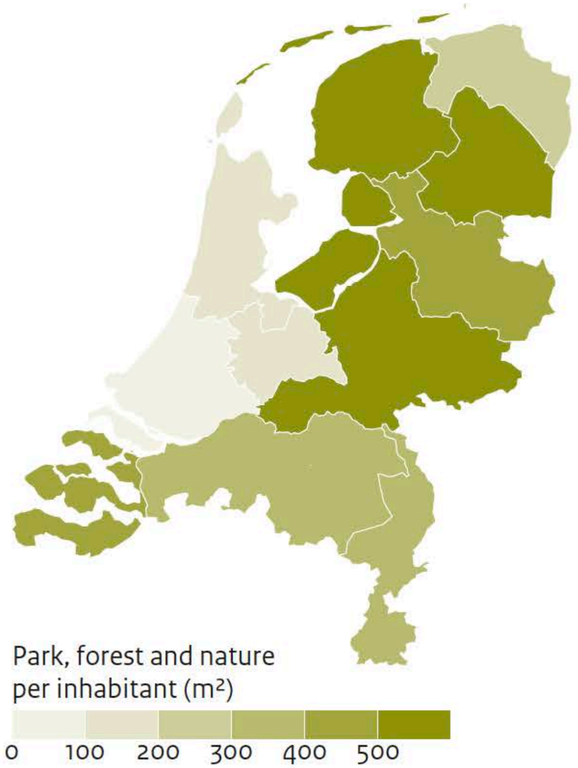
Visualization Techniques

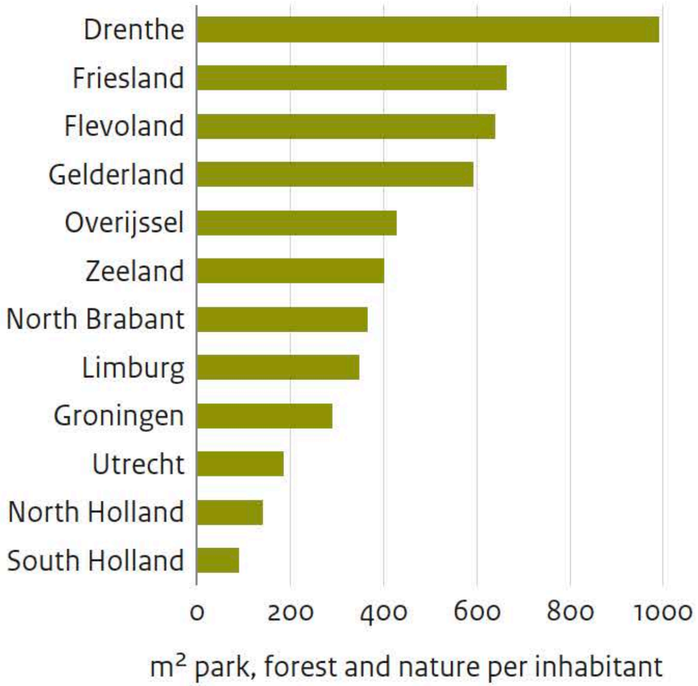
(GEO-)SPATIAL DATA

- maps give an impression of the geographical distribution of data
- disadvantage: less suitable for delivering exact amounts

Maps are not always the ***best*** or ***only*** solution!

Useful when the question is inherently spatial and when maps help to find information

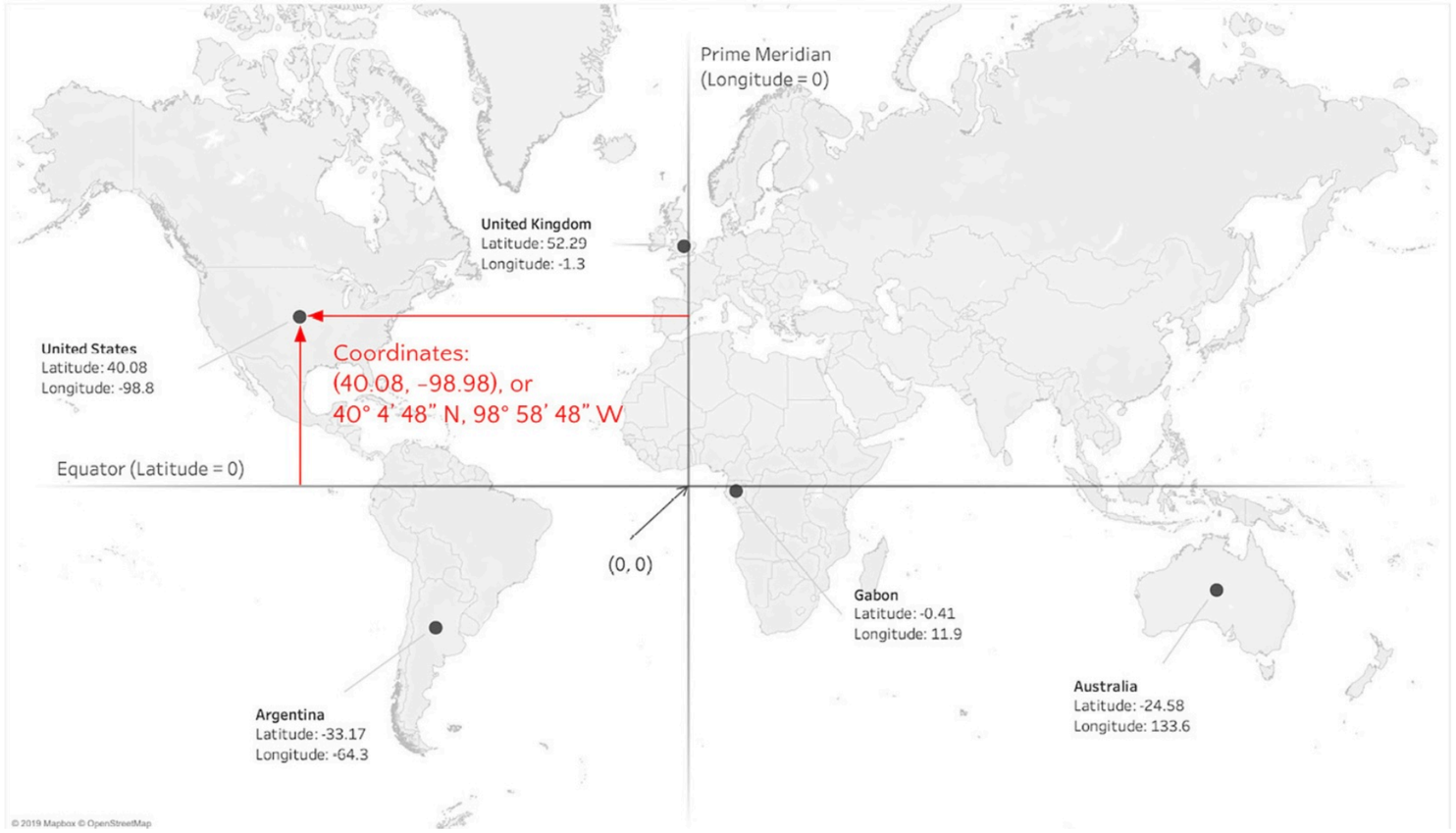


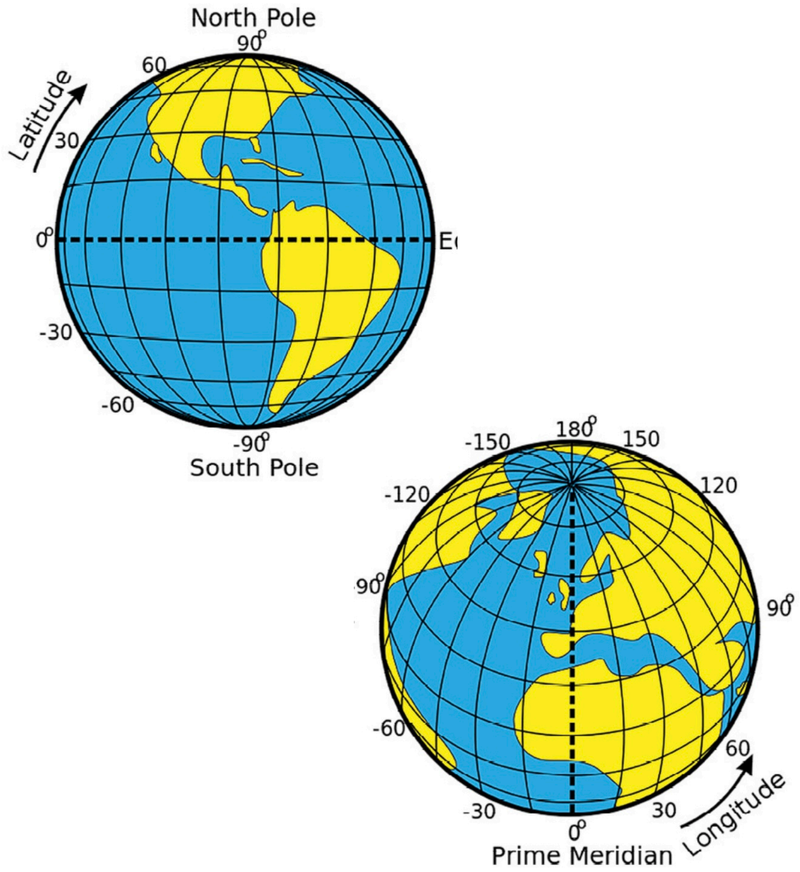


Blois et al.: Visualising Knowledge. Lessons from 25 years of policy-related data visualisation, PBL Netherlands Environmental Assessment Agency, 2022

(GEO-)SPATIAL DATA

- data can be available in X & Y (and z) positions, for geospatial data in latitude & longitude (and elevation)





- data can also be provided as regions (polygons)

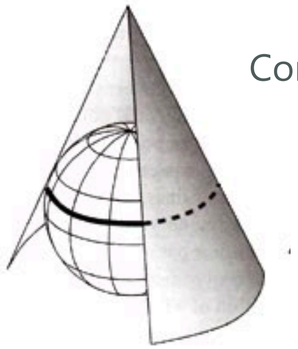
MAP PROJECTIONS

- one challenge with mapping data is the map projection the creator chooses to transform the world's sphere into a two-dimensional plane —> all map projections involve transformations that result in distortions



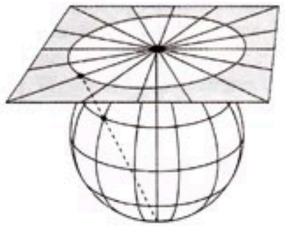
Cylinder

best for countries around equator



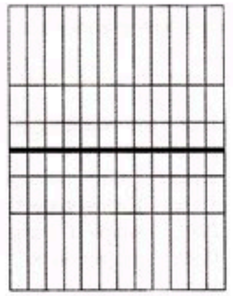
Conic

best for middle-latitude

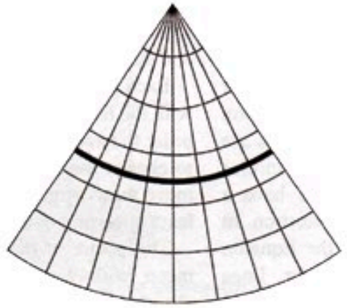


Planar

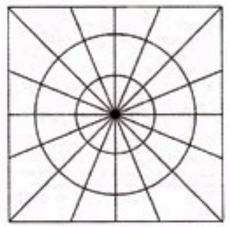
best for polar regions



Cylindrical



Conical



Azimuthal

PROJECTIONS

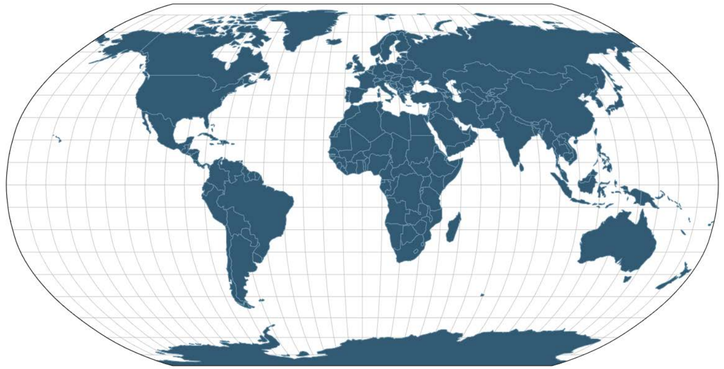
- distortion increased from point for contact: larger areas more prone to distortion

Equal-area projections



- preserve all relative areas and are are useful for visualizations in which the comparison of areas on the map is crucial
- e.g., Albers, Lambert, conic equal-area projection

„Compromise“ projection



- some projections offer acceptable compromises between conserving area and conserving angles
- e.g. Robinson projection

Conformal projections



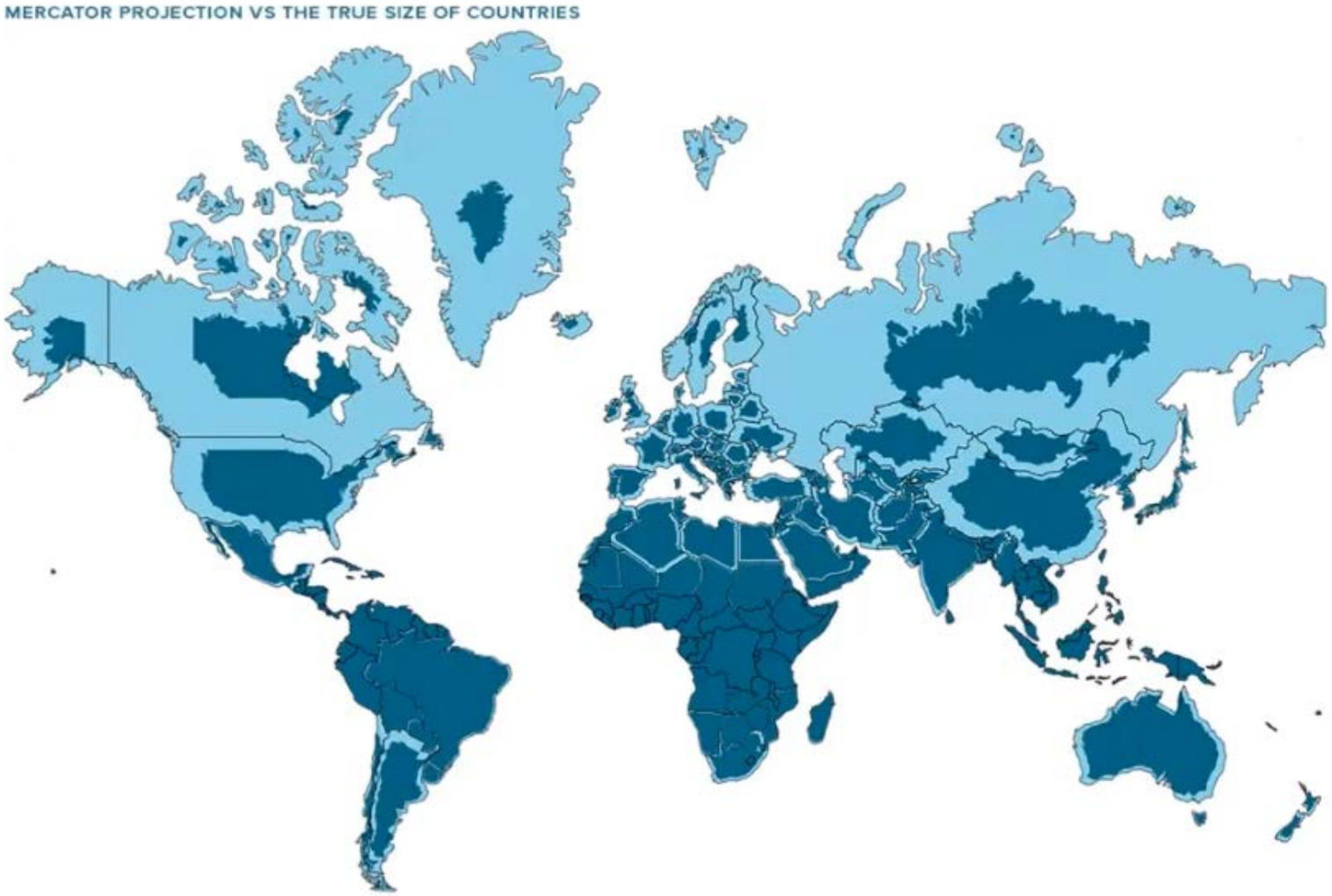
- conserve angular relationships: angle between any two intersecting lines will be same on the flat map as on the globe
- e.g. Mercator projection

https://observablehq.com/@etheralan/map-projections

MAP PROJECTIONS

Every map projection introduces distortion, and each has its own set of problems.

One of the most common criticisms of the **Mercator map** is that it exaggerates the size of countries nearer the poles (US, Russia, Europe), while downplaying the size of those near the equator (the African Continent).



https://thetruesize.com/

MAP PROJECTIONS FOR EUROPE



The Eqidistant conic projection is known since Classical times. It has the useful property that distances along the meridians are proportionately correct, and distances are also correct along two standard parallels that the mapmaker has chosen. The two standard parallels are also free of distortion.



The Albers equal-area conic projection is a conic, equal area map projection that uses two standard parallels. Although scale and shape are not preserved, distortion is minimal between the standard parallels.



The Lambert conformal conic projection is a conic map projection used for many national and regional mapping systems. It is one of seven projections introduced by Johann Heinrich Lambert in 1772.

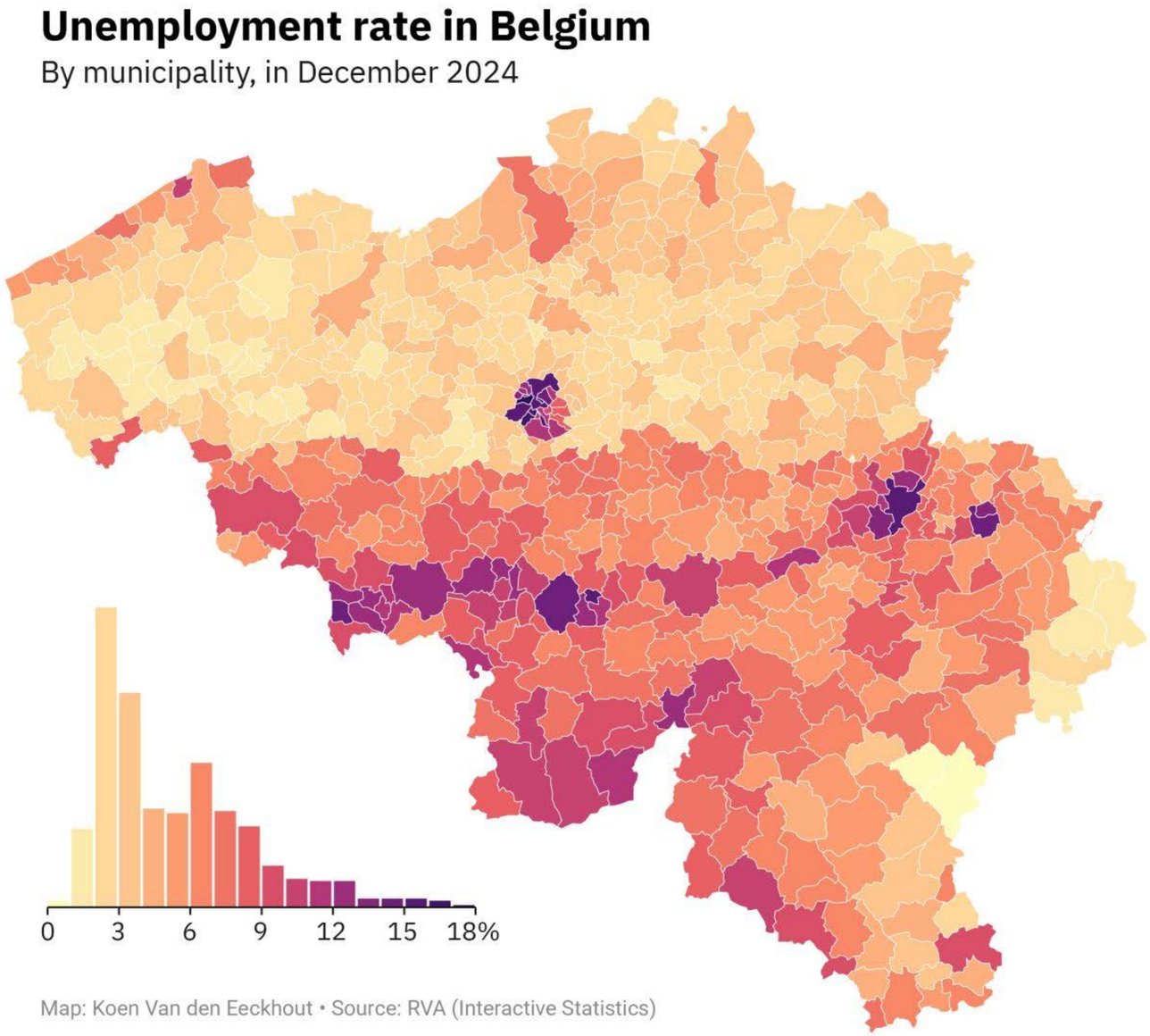


The Lambert azimuthal equal-area projection is a particular mapping from a sphere to a disk (that is, a region bounded by a circle). It accurately represents area in all regions of the sphere, but it does not accurately represent angles.

https://observablehq.com/@toja/five-map-projections-for-europe

CHOROPLETH MAP

- uses colors, shades, or patterns on geographic units to show proportionate quantities
- regions are predefined (specific spacial regions like states or countries)
- **important decision:** which level of detail should be used in the map? different patterns emerge from the map depending on the level of detail
- **second decision:** color map should be carefully chosen

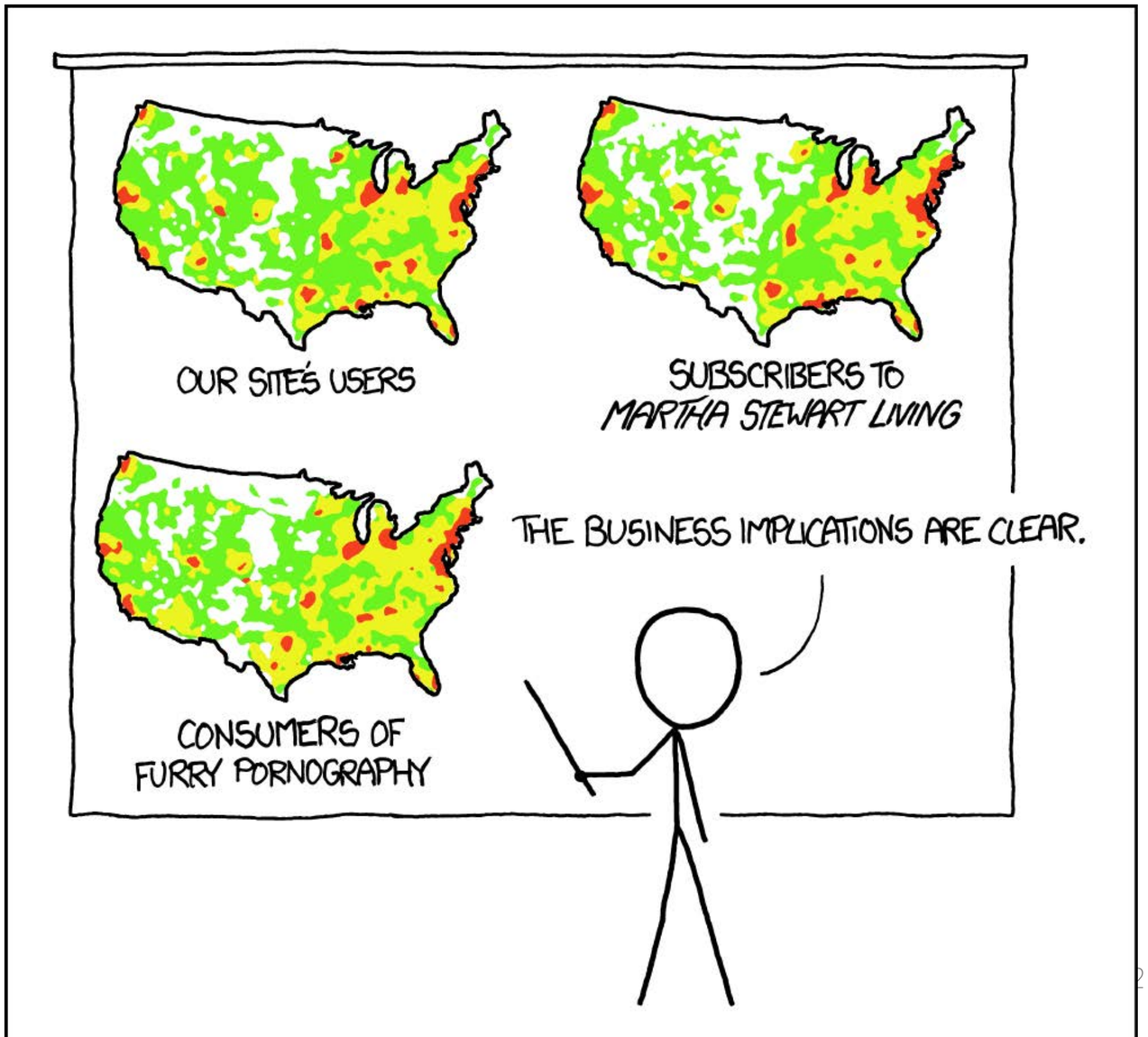


https://observablehq.com/@d3/choropleth

BEWARE: POPULATION MAPS TRICKINESS!

- **spurious correlations:** most attributes just show where people live
- beware of using **absolute numbers** as this can be misleading because the figures **don't weight in factors like population** or land extension
- consider when to normalize by population density (e.g. unemployed people per 100 citizens, mean family income)

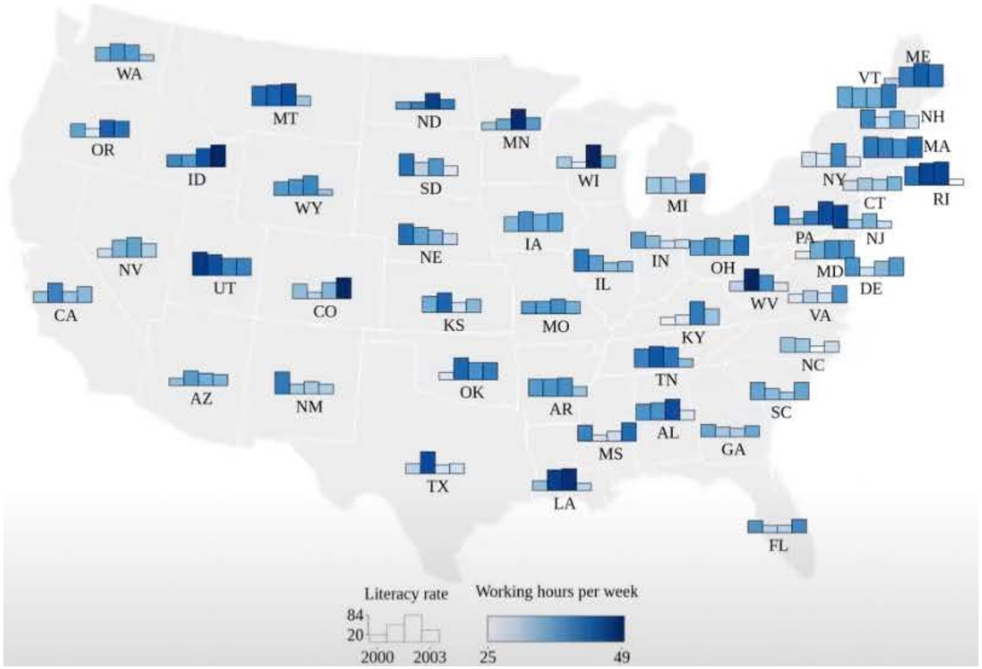
Geographic profile maps which are basically just population maps

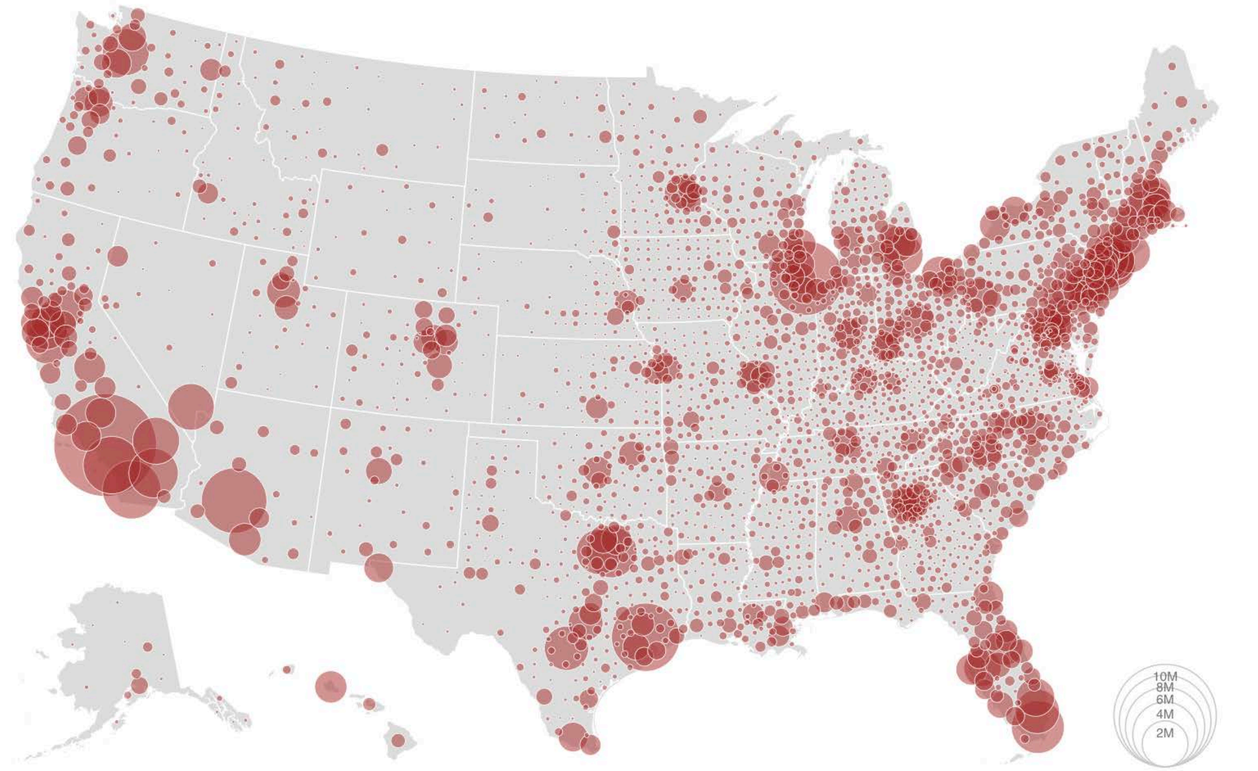


https://xkcd.com/1138

SYMBOL MAP » BUBBLE MAP, PROPORTIONAL SYMBOL MAP

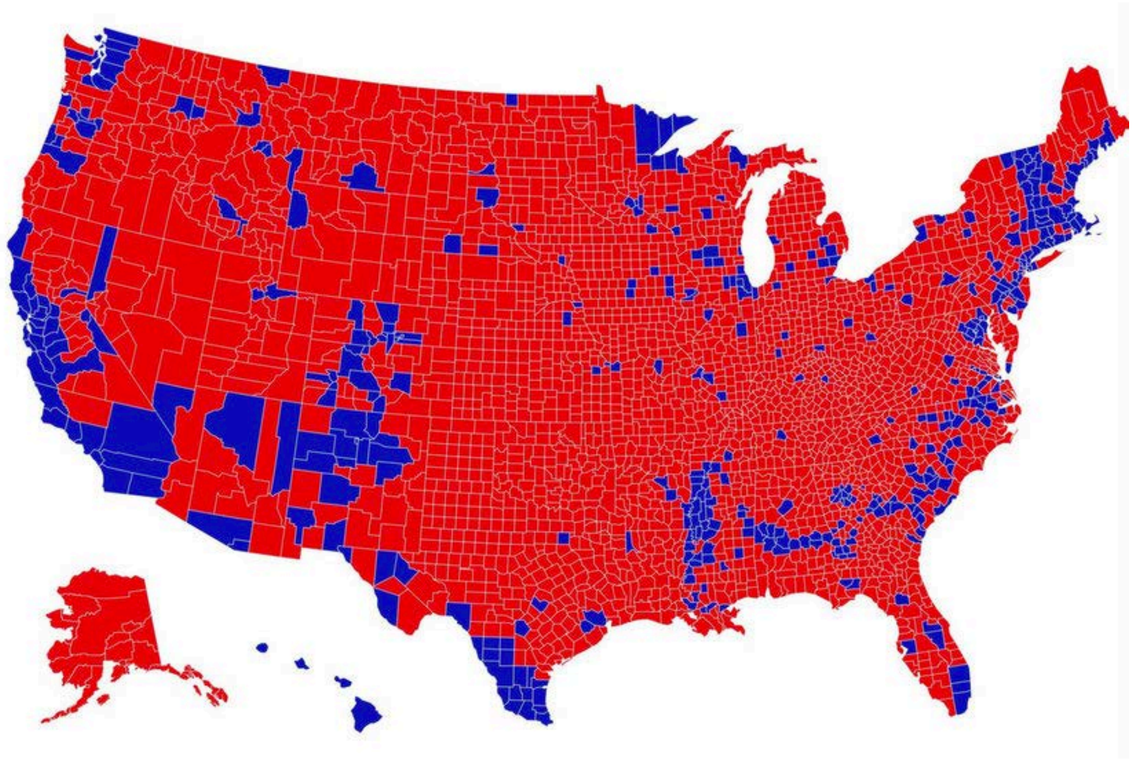
- different shapes and objects (lines, arrows, points, circles, icons, even glyphs can all be placed on the map)
- also known as proportional symbol maps, because symbols are sized proportionate to the data

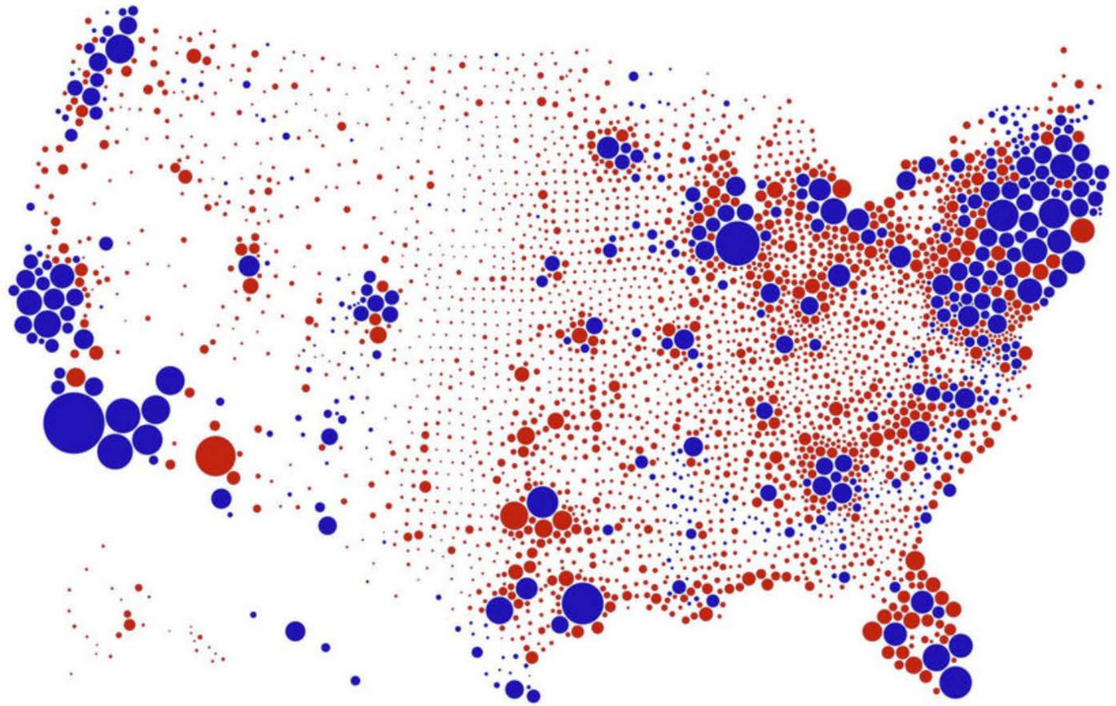




https://observablehq.com/@d3/bubble-map

"LAND DOESN'T VOTE. PEOPLE DO"

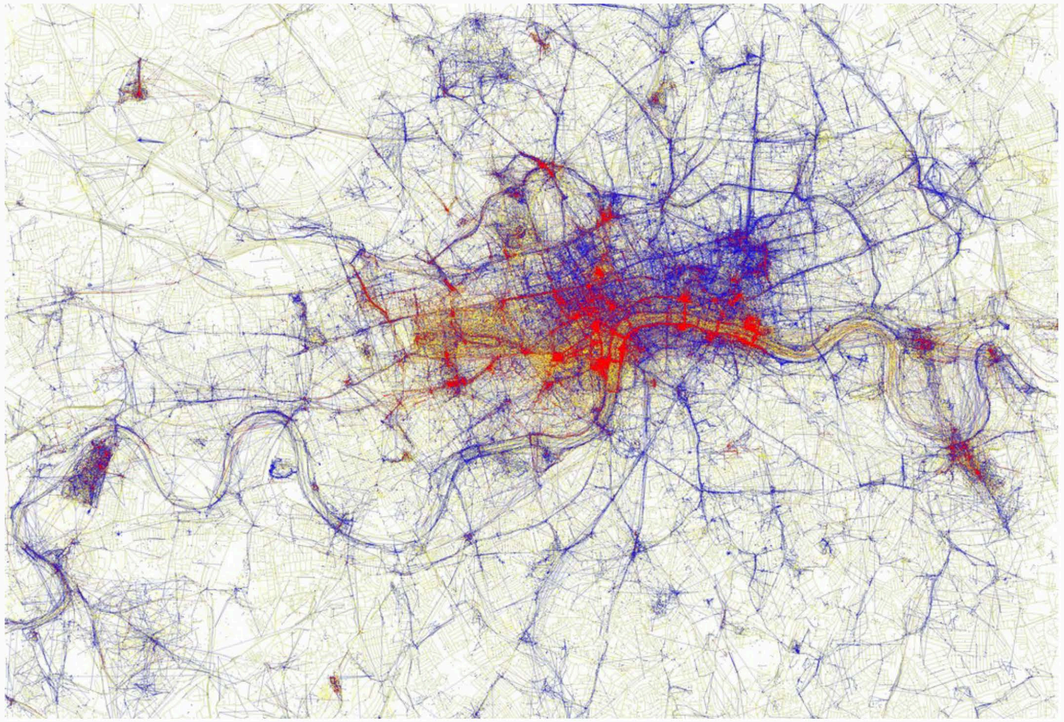




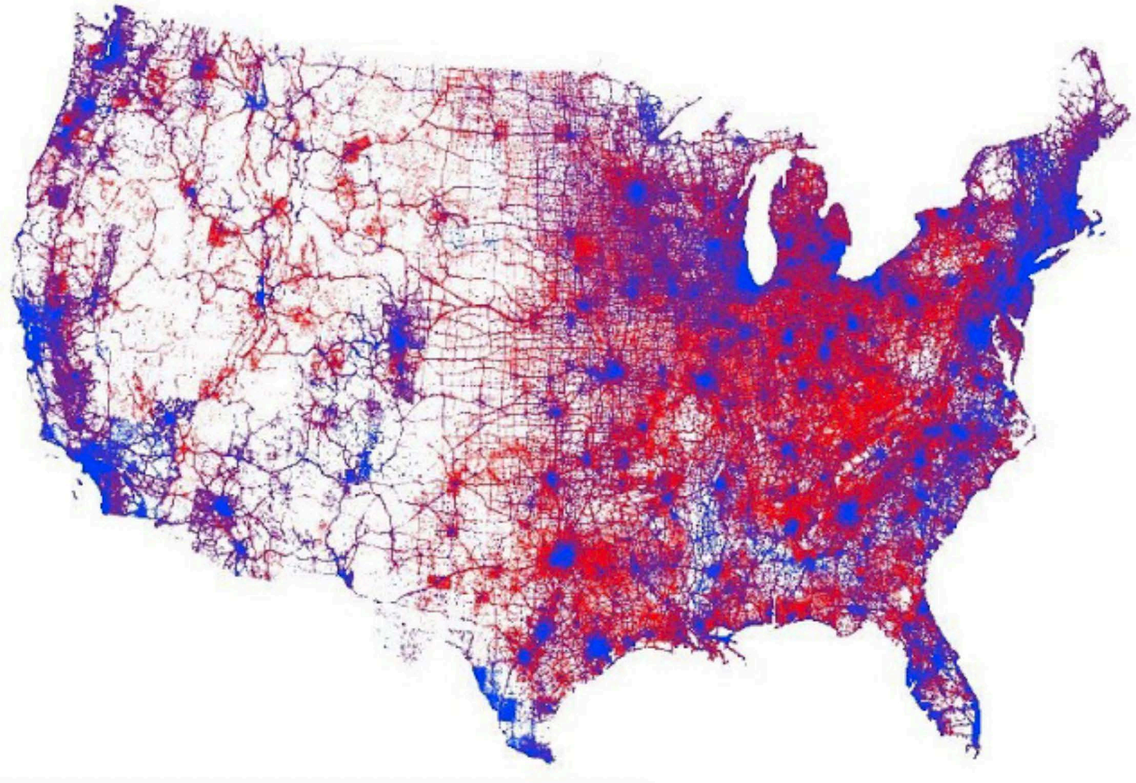
https://www.core77.com/posts/90771/A-Great-Example-of-Better-Data-Visualization-This-Voting-Map-GIF

DOT DENSITY MAP » DOT DISTRIBUTION MAP, DOT MAP, LOCATION MAP

- visualize distribution of a phenomenon by placing dots
- one symbol represents a constant number of items
- allows use of color channel; dots have uniform size



A. Cairo: How Charts Lie, 2019 / https://brilliantmaps.com/tourists-vs-locals/



Eric Fischer: U.S.: „Locals and Tourists“, 2010

- dots depict the location of photos that were geo-tagged and uploaded to the photo server Flickr
- red dots represent tourists, blue dots locals

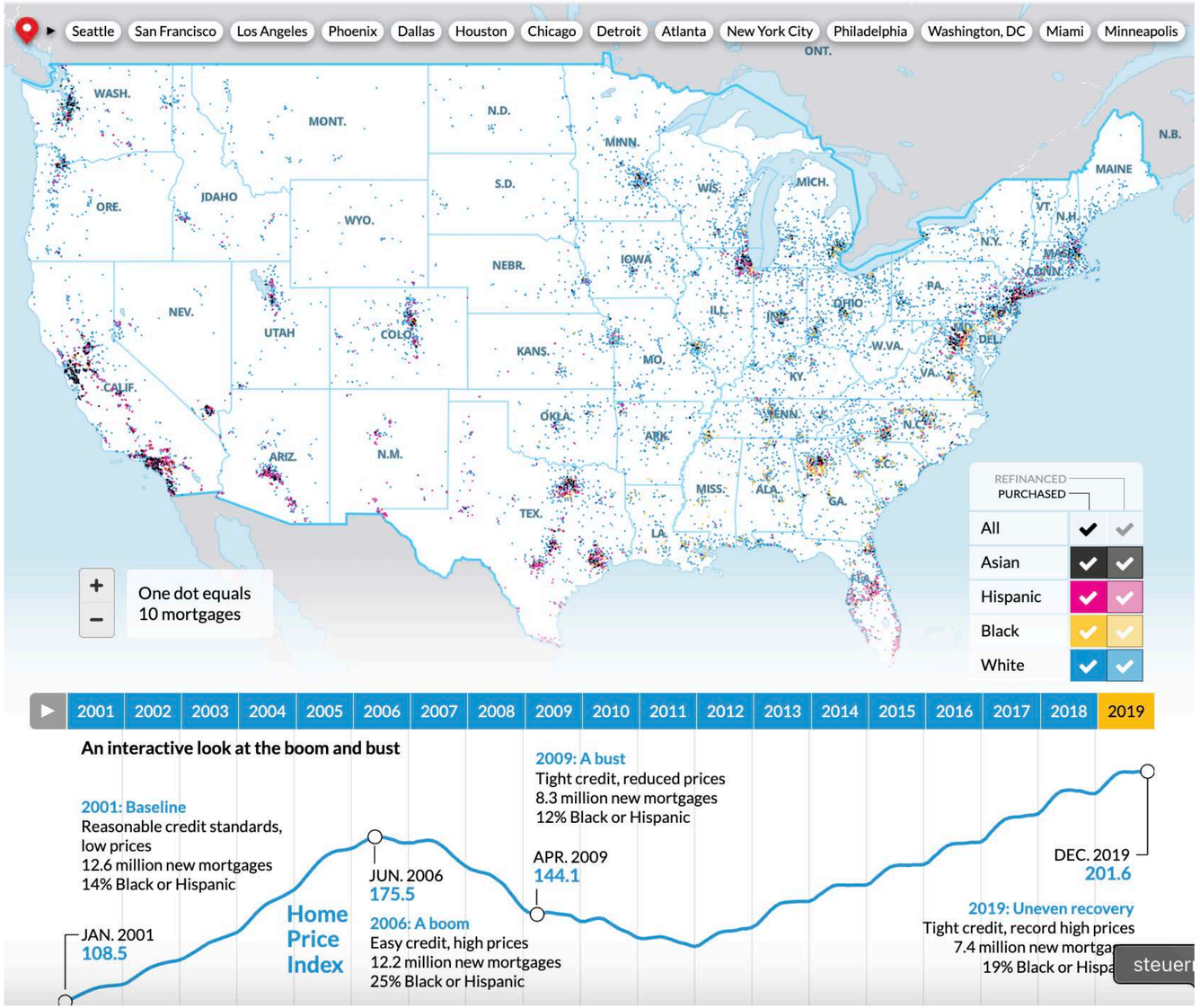
DOT DENSITY MAP

Advantages:

- straightforward to understand
- avoids choropleth non-uniform region size problems

Disadvantages:

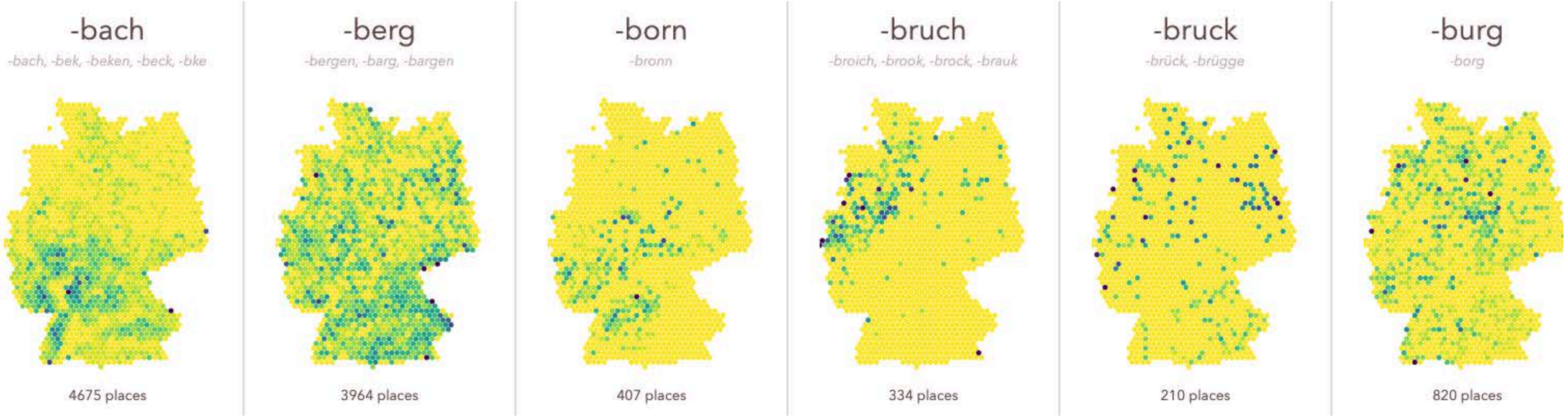
- challenge: normalization, just like choropleths
  - show population density (correlated with attribute), not effect of interest
- performance: rendering many dots can be slow



https://apps.urban.org/features/mortgages-by-race/, Munster 2014

BINNED MAP » HEX MAP

- organizes the areas as specific geometries such as hexagons
- visualize density without dividing countries in predefined regions



- a visual exploration of the spatial patterns in the endings of german town and village names

https://truth-and-beauty.net/experiments/ach-ingen-zell/

CONTINUOUS CARTOGRAMS

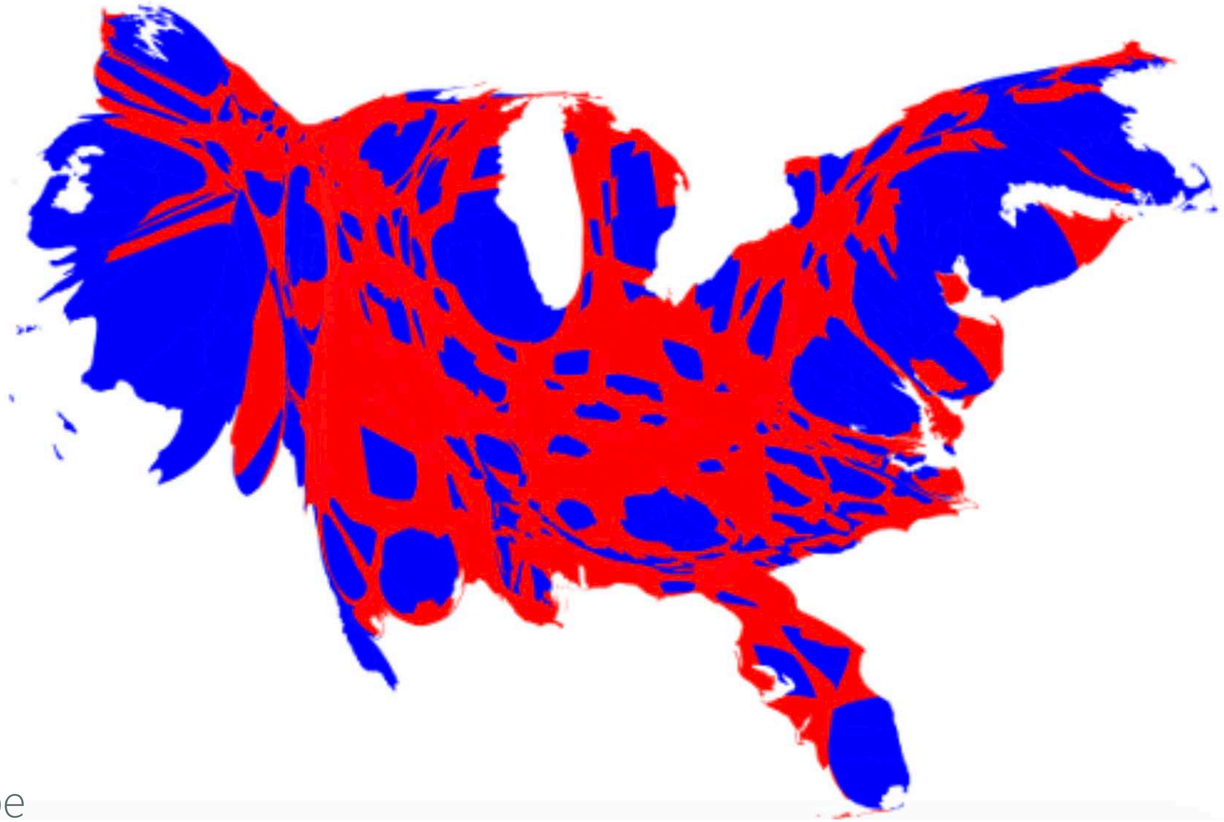
- stretches or shrinks the size of each county to accurately reflect the proportion of the total population in that area
- because it significantly compromises on its shape, it is unfamiliar to the public and confusing for the average citizen to view

Advantages:

- good at showing numbers (such as a population)
- Best case: strong and surprising size disparities

Disadvantages:

- since the geographic areas are distorted, areas might not be recognizable
- intended for map-literate audiences only

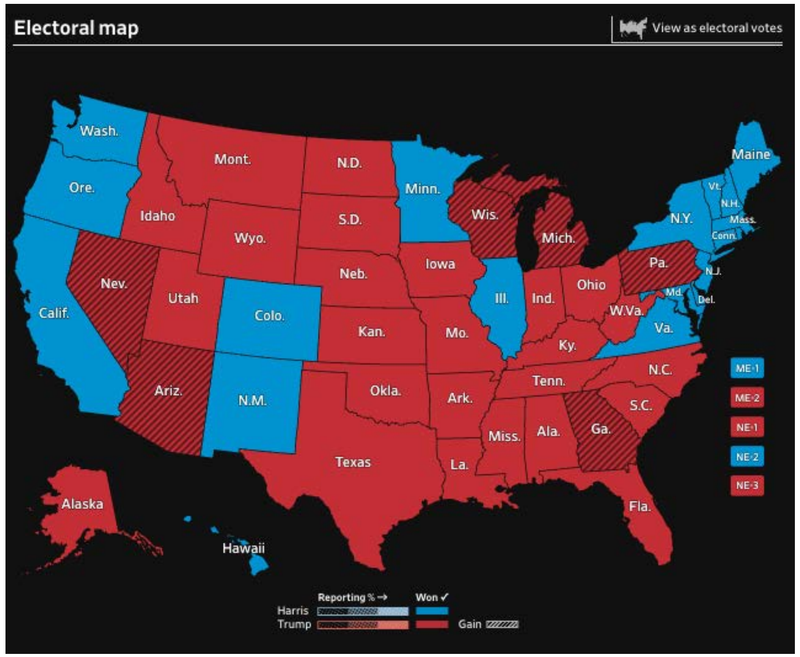


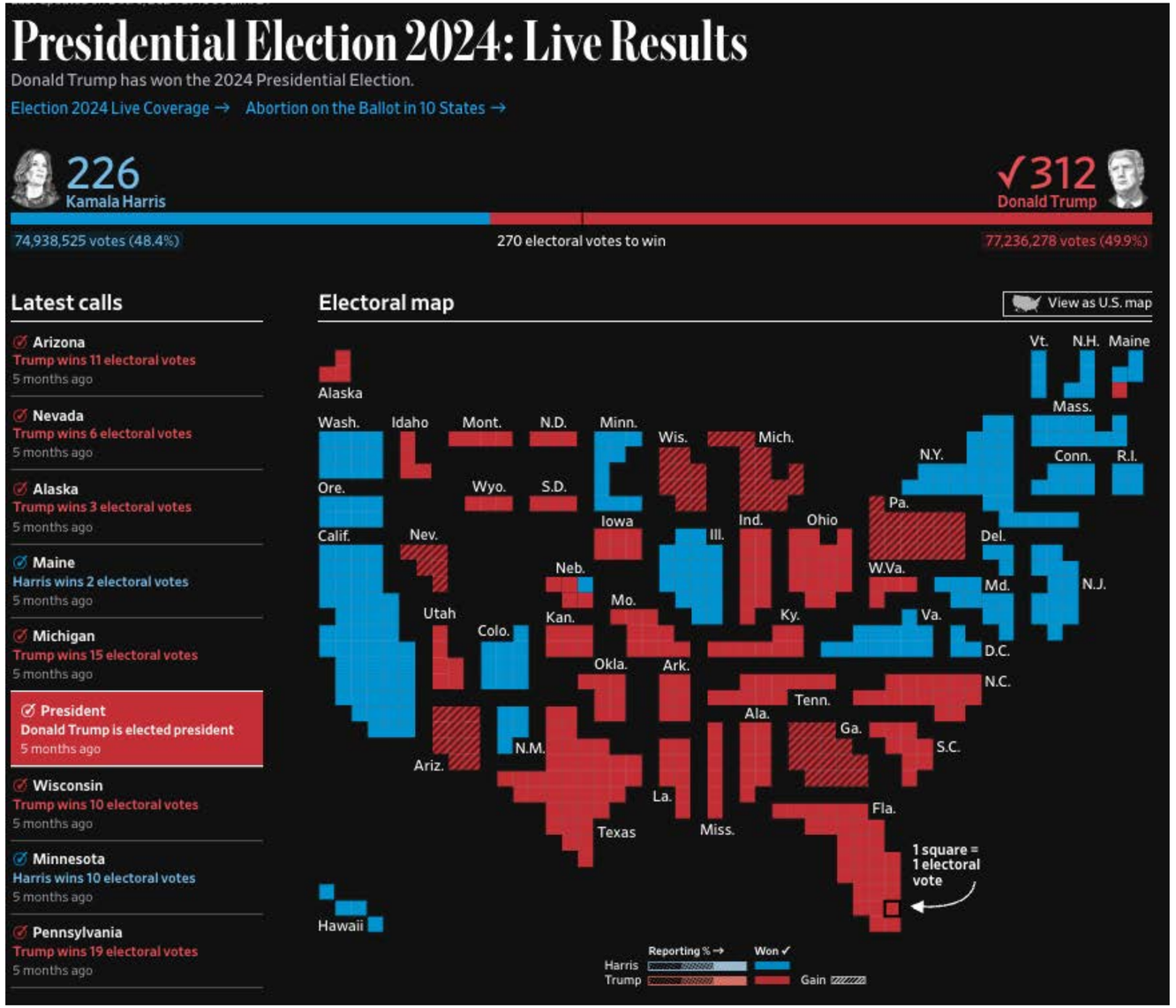
https://storymaps.arcgis.com/stories/0e636a652d44484b9457f953994b212b

GRID CARTOGRAMS

- uniformed-sized shapes arranged in rectilinear grid
- maintain approximate spatial position and arrangement

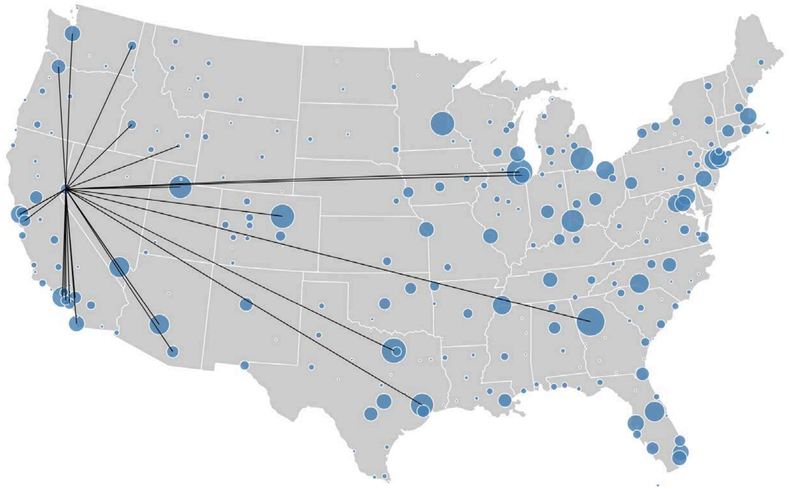
Comparison to Choropleth Map

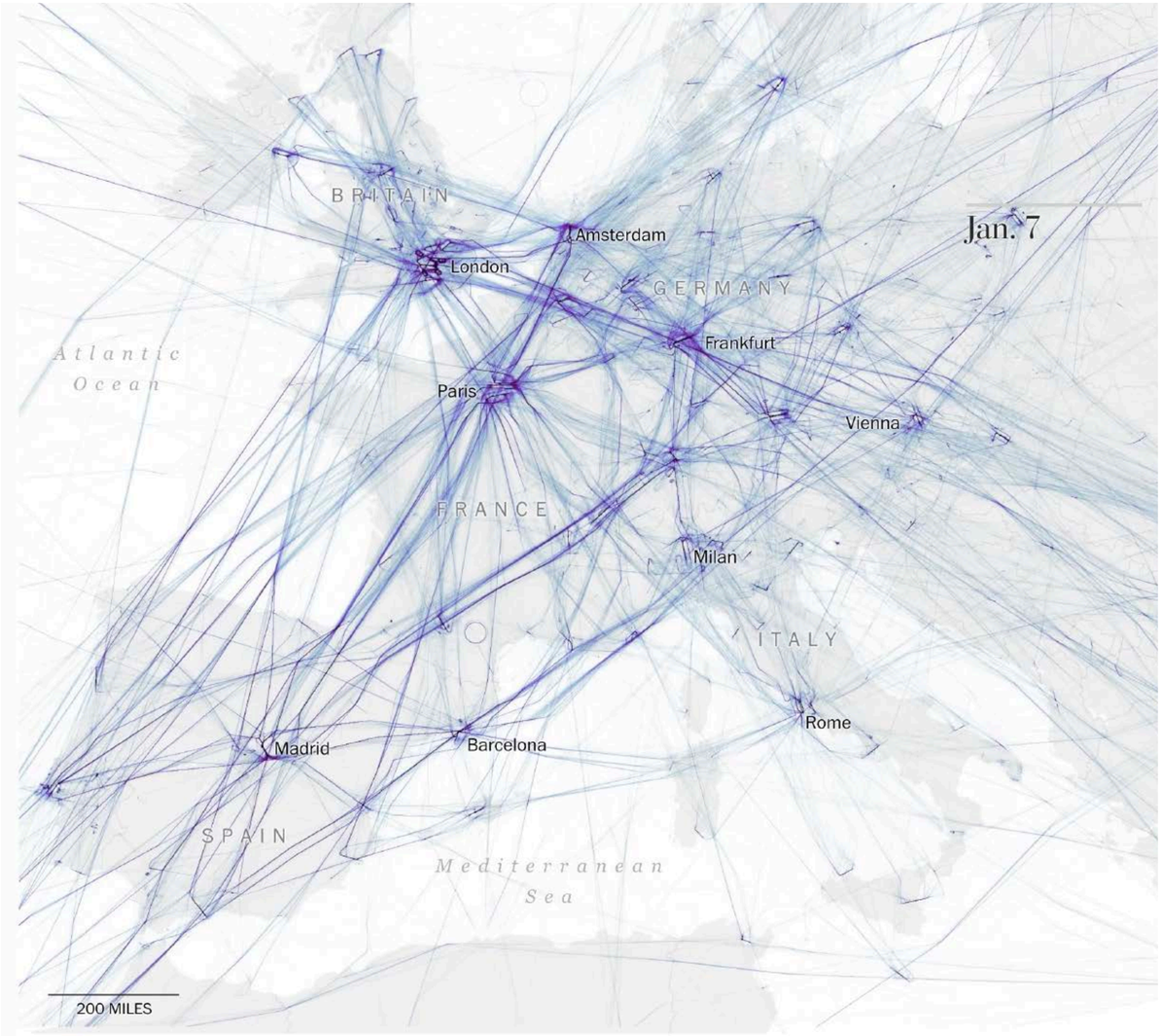




FLOW MAP » CONNECTION MAP, ROUTE MAP

- shows movement between places
- arrows and lines denote the direction of the flow, and the width of the line can correspond to the data values
- **radial flow maps** (also called origin-destination maps) show flows from a single source to many destinations





https://mbostock.github.io/d3/talk/20111116/airports.html

MAPS WITH GLYPHS

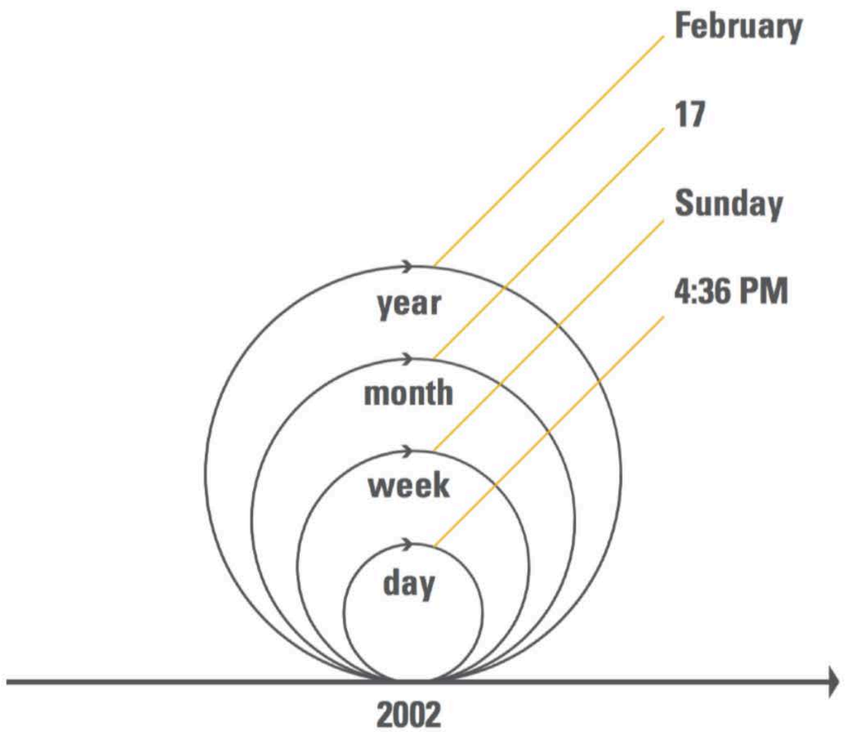




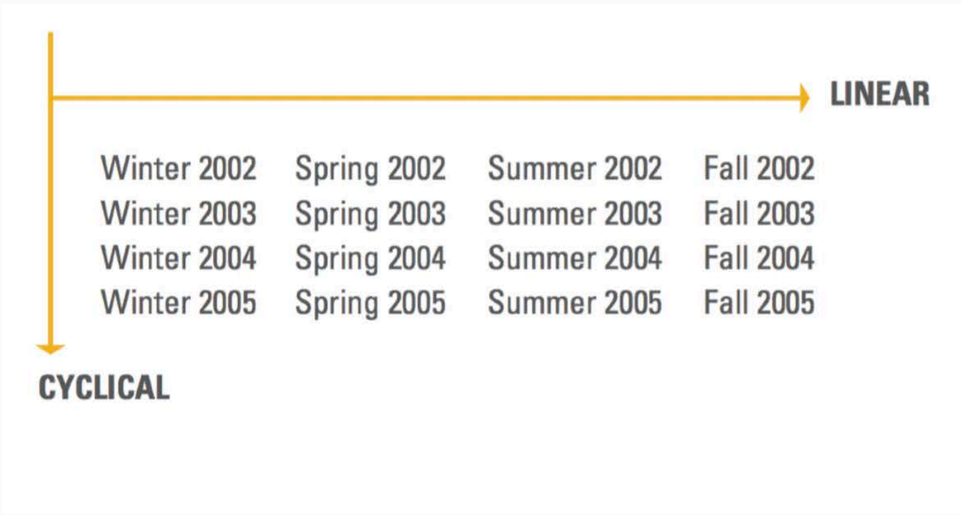
TEMPORAL DATA

Visualization Techniques

STRUCTURE OF TIME



hierarchical



linear & cyclic

I. Meirelles: Design for Information, Rockport Publishers, 2013

LINE CHART

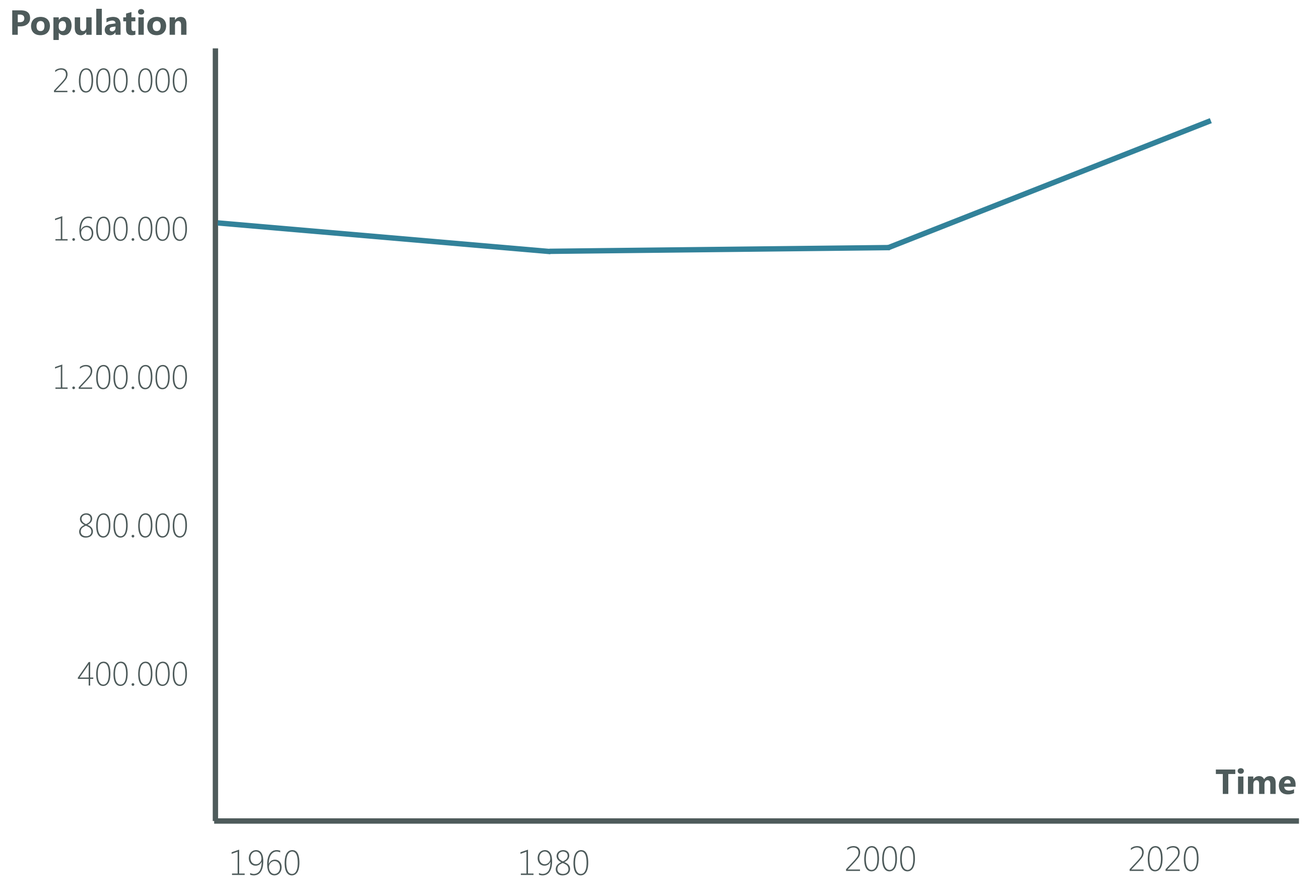
- data values are connected by lines to show values over a continuous period, tracking trends and patterns

WHAT ordinal (mostly temporal) + quantitative

TIME	POPULATION
1960	1.627.000
1980	1.531.000
2020	1.549.000
2022	1.900.000

WHY recognize trends

HOW



HANDLING MISSING VALUES IN LINE CHARTS







- unfortunately, some dataviz software still treats missing or null values as zero, but zero isn't the same thing as „missing" or „null"



HANDLING MISSING VALUES IN LINE CHARTS

So how can we avoid misleading the audience when there are missing values in a line chart?



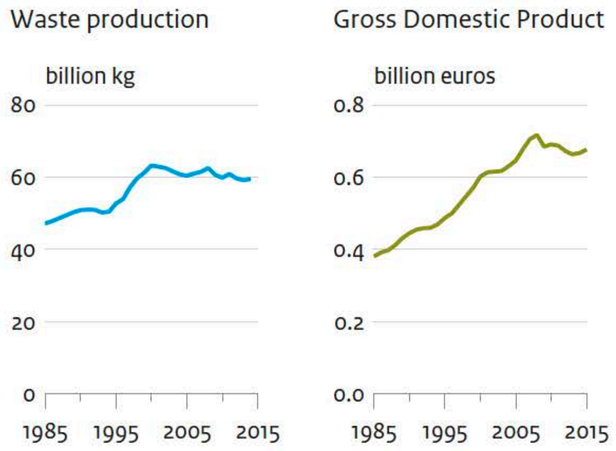


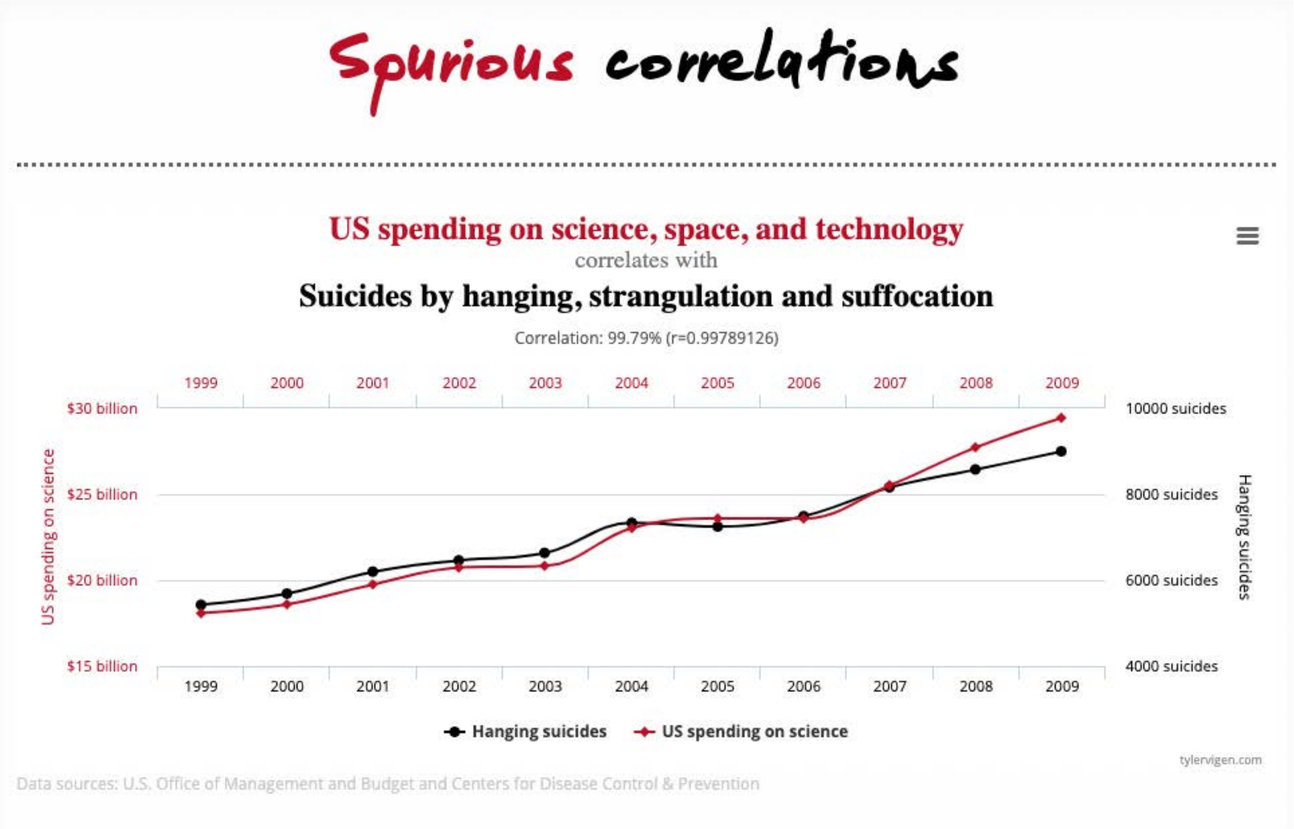


- Don't connect lines through missing values, show missing values as gaps in lines instead
- Reserve dashed or dotted lines for values of lower certainty or importance such as estimates and targets

THE PROBLEM WITH A 2ND Y-AXIS

- by arbitrarily choosing the axis range, we can make different data series look correlated, even if they are not
- **Alternative:** use side-by-side visualizations to make it clear that two different data sets are involved or normalize value range





On this website, „Spurious correlations“, Tyler vine shows all kinds of dual-axis charts in which arbitrary vertical axis scales created erroneous - and humorous - correlations

LINE CHART

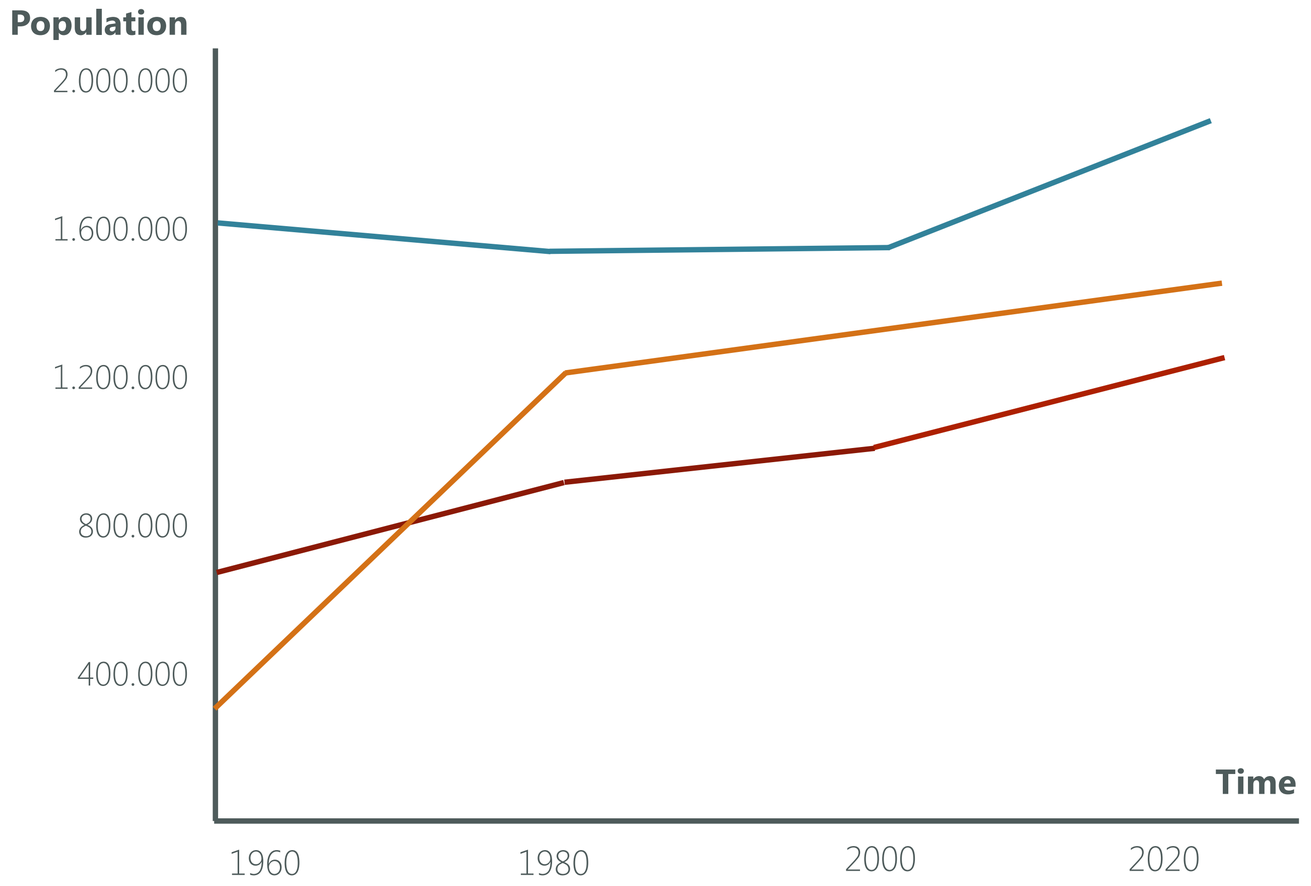
HOW

- visualize how a **quantity** changes in relation to another **quantity** (typically **time**)

WHAT

Multiple Lines =

TIME + QUANTITY + CATEGORIES



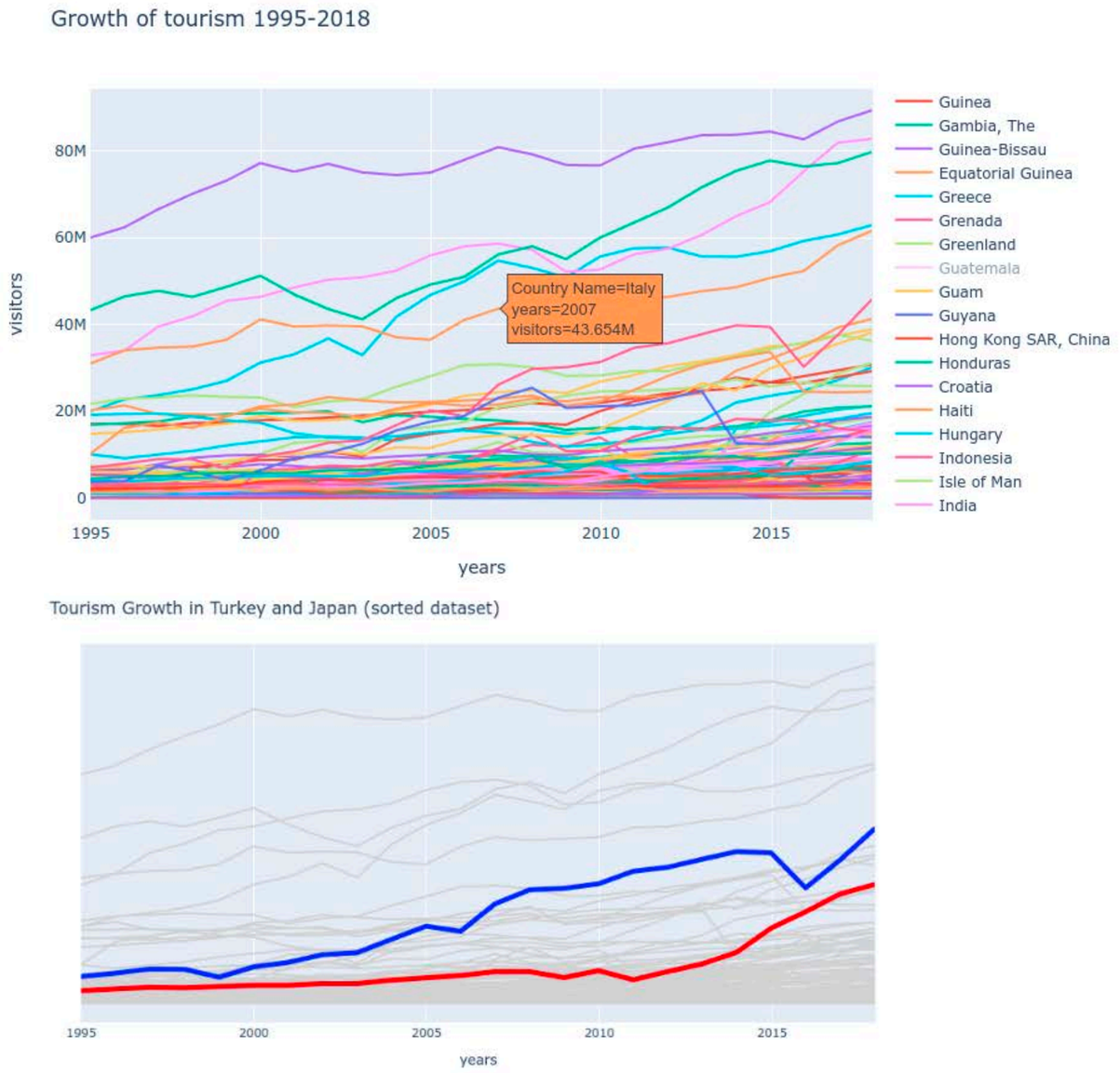
WHY compare values and trends

LINE CHART

- **Disadvantage:** it is getting cluttered with many lines and
  color is not suited for too many categories

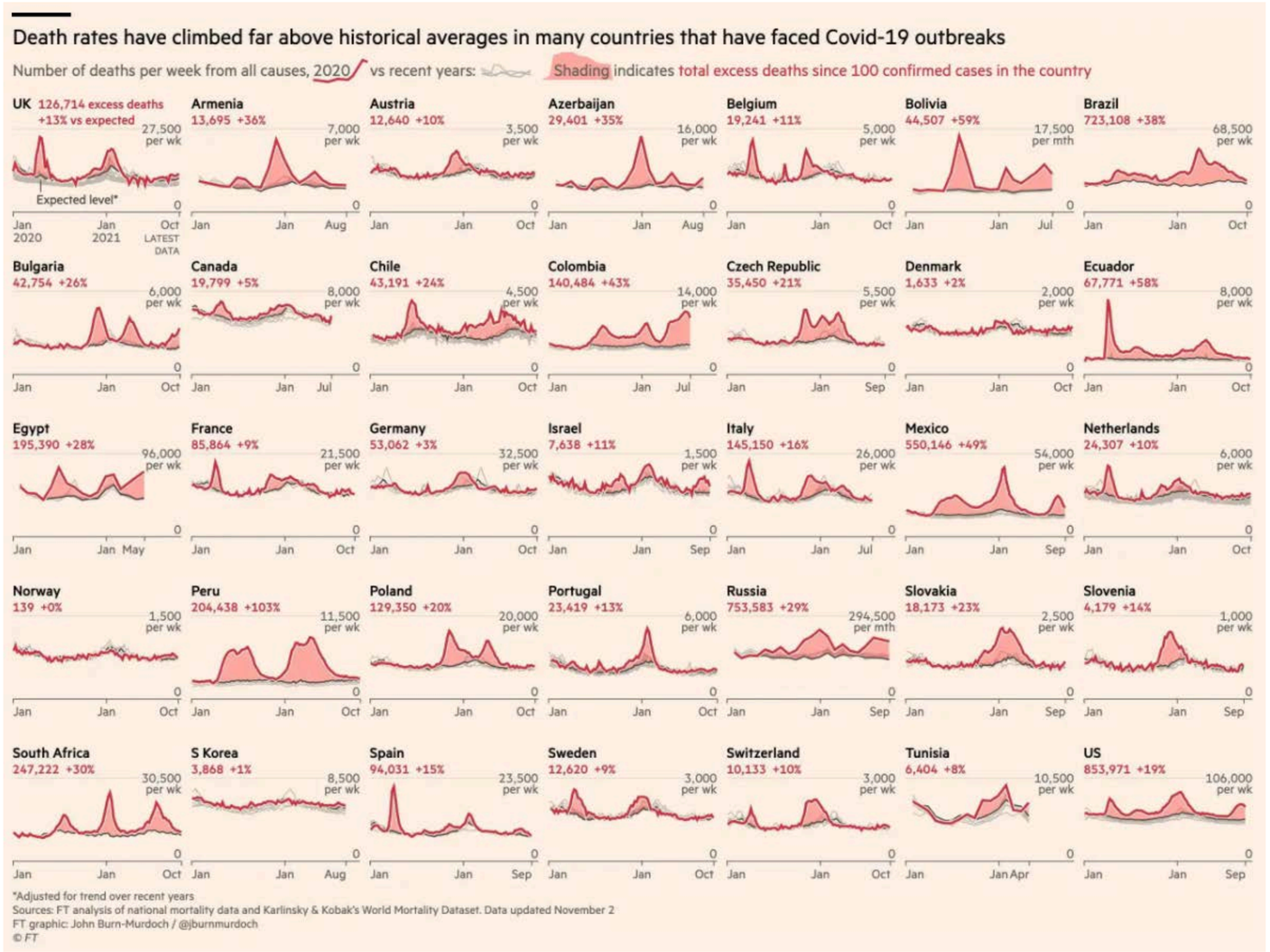
Possible Solutions:

- **Grouping:** group objects into categories
- **Filtering:** just communicate the important subsets
- **Highlighting:** selected lines are colored and the other lines in grey shade to reduce clutter
- **Small Multiples:** juxtaposition of line charts



SMALL MULTIPLES

- show each time series in its own chart
- alternative approach to reduce clutter and compare trends
- **disadvantage:** not appropriate for the identification and comparison of individual values

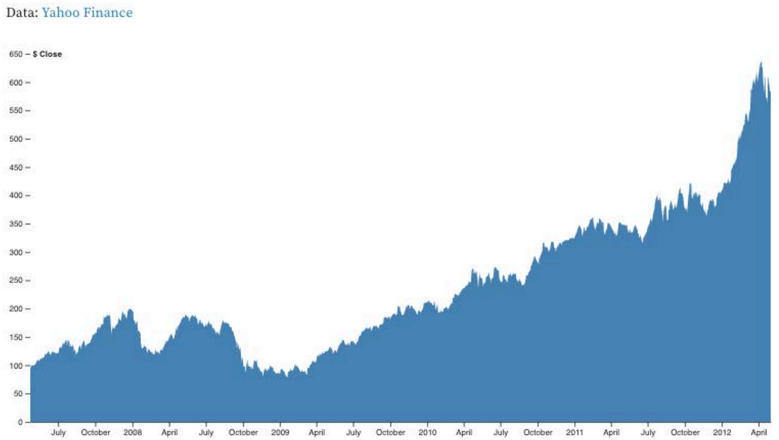


https://www.ft.com/content/a2901ce8-5eb7-4633-b89c-cbdf5b386938

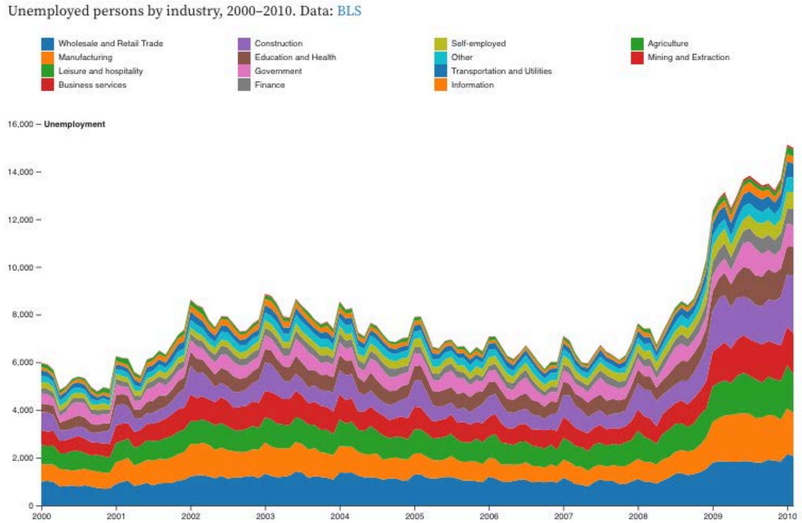
AREA CHART

- show Line charts with the area below the line filled in to give the series more visual weight
- **Stacked area charts:** build on typical area chart by showing multiple data series simultaneously with data sum to a total or a percentage
- they do not support negative numbers and are meaningless for data that should not be summed (e.g. temperatures)

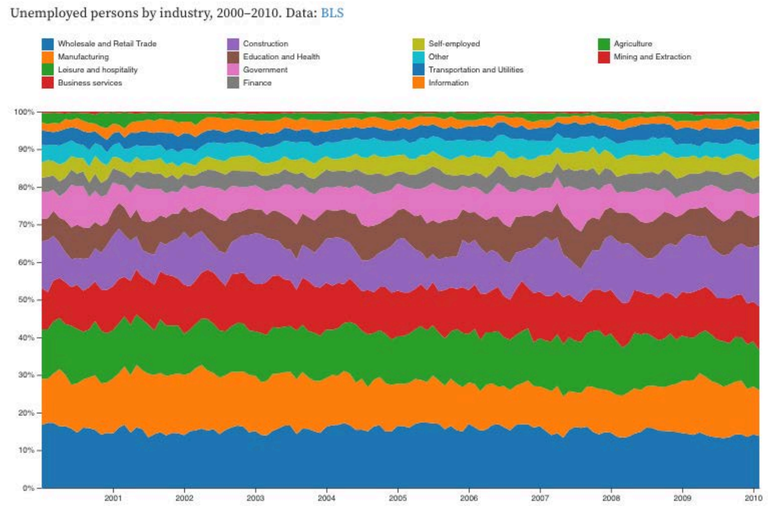
Area chart



Stacked area chart



Normalized stacked area chart

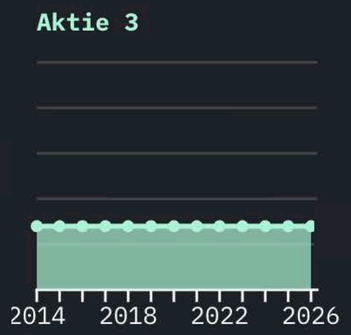


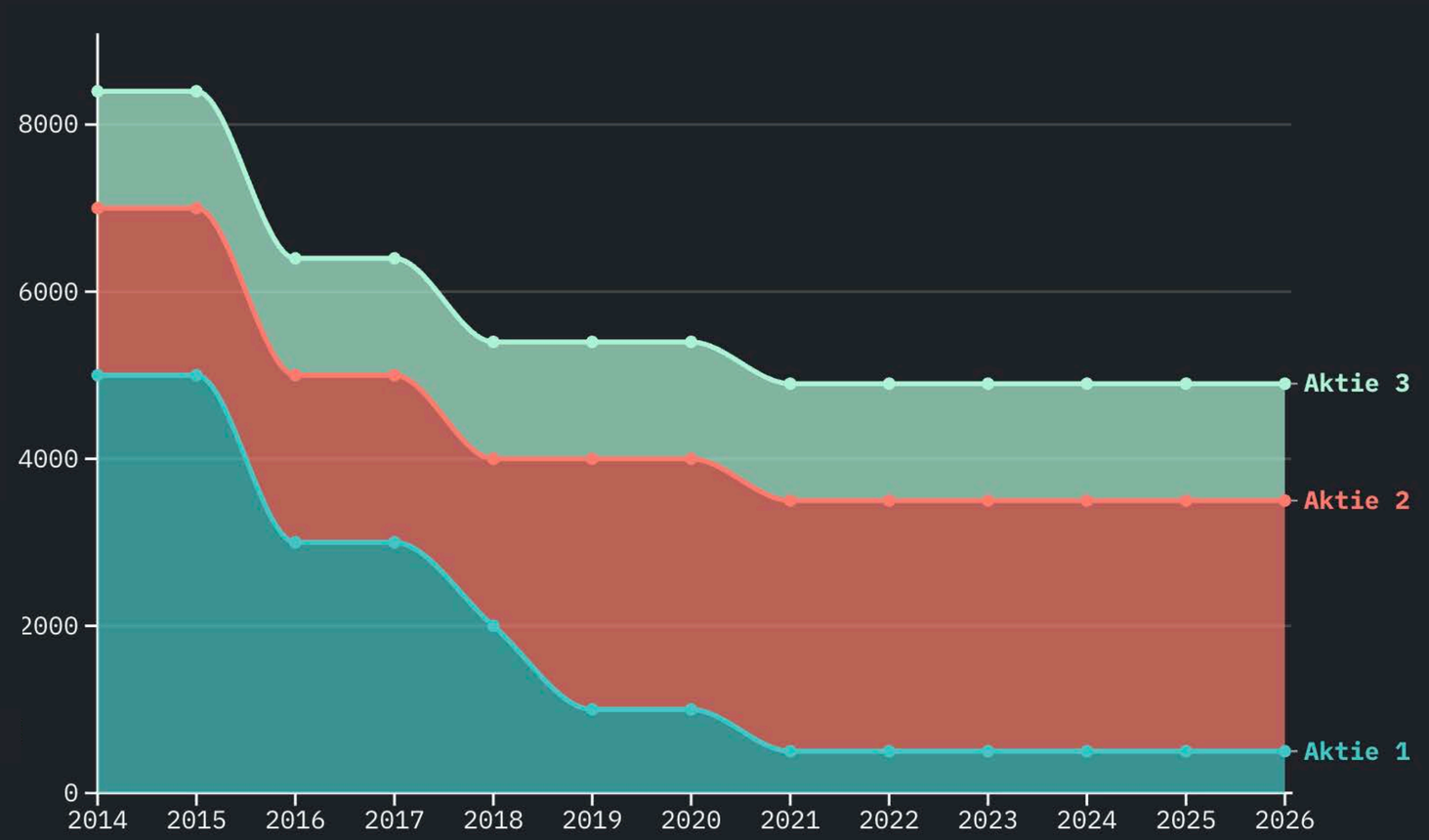
STACKED AREA CHART

Stock Portfolio



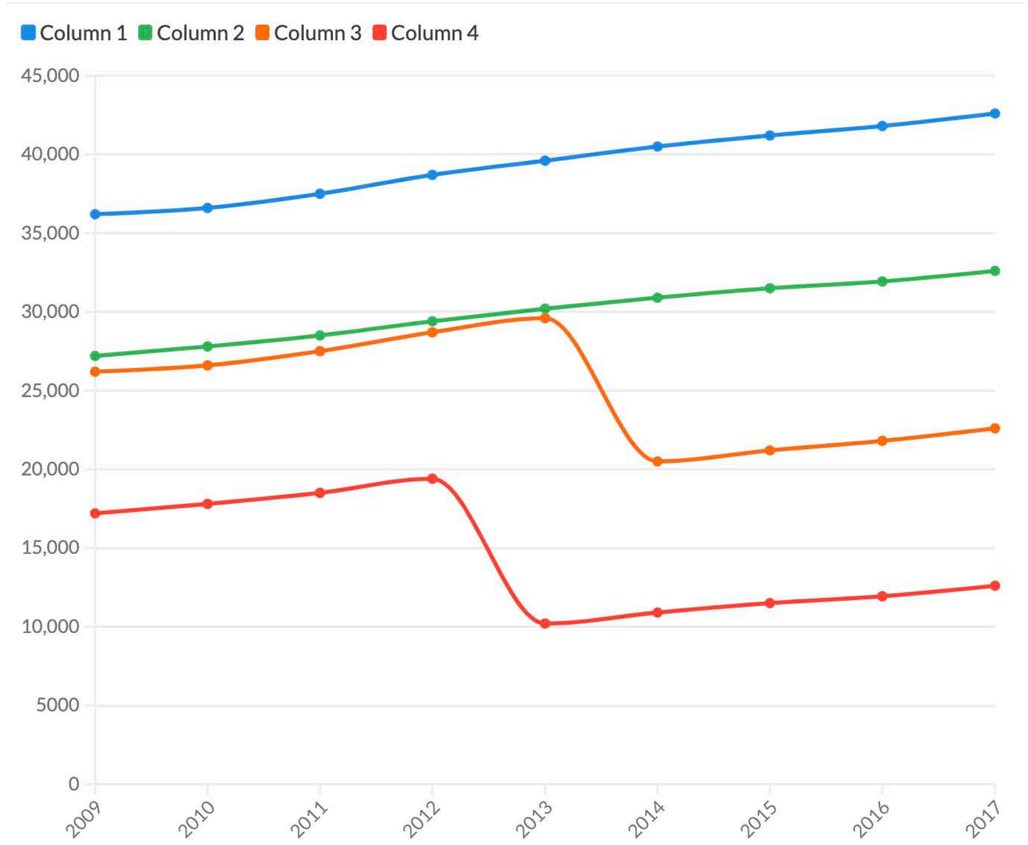




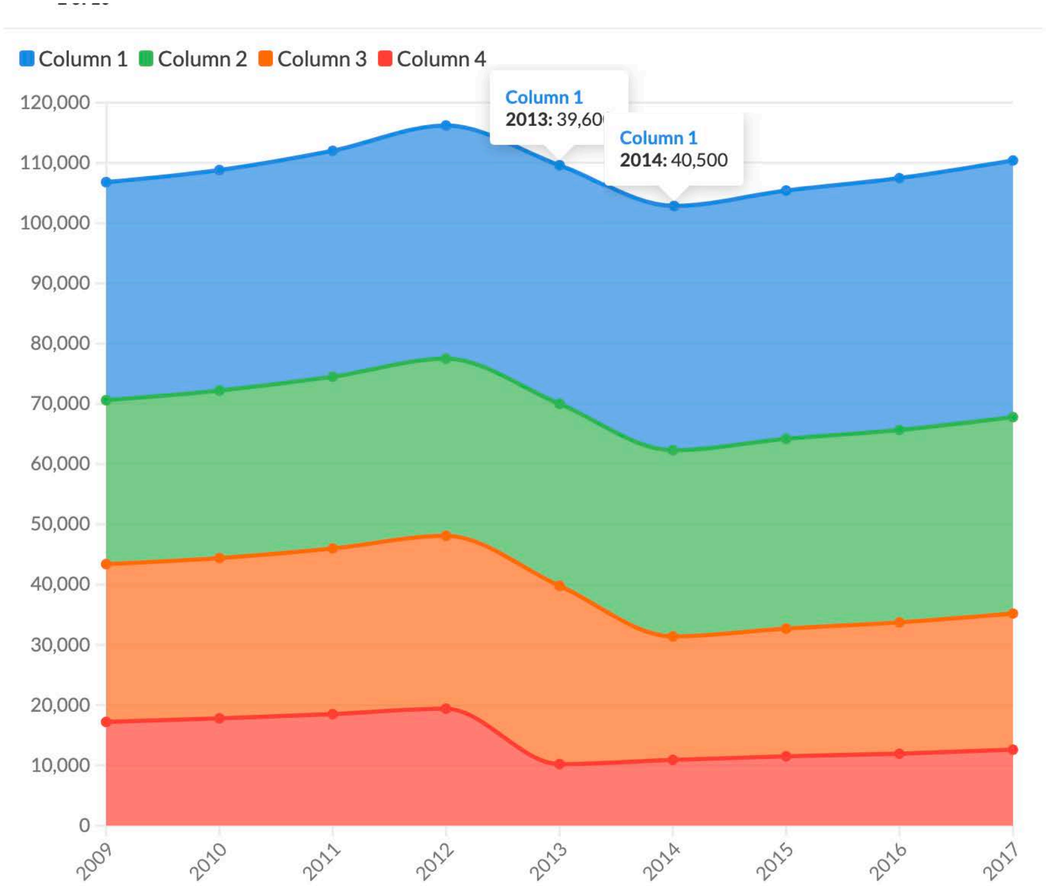


LINE CHART VS. STACKED AREA CHART

Line chart



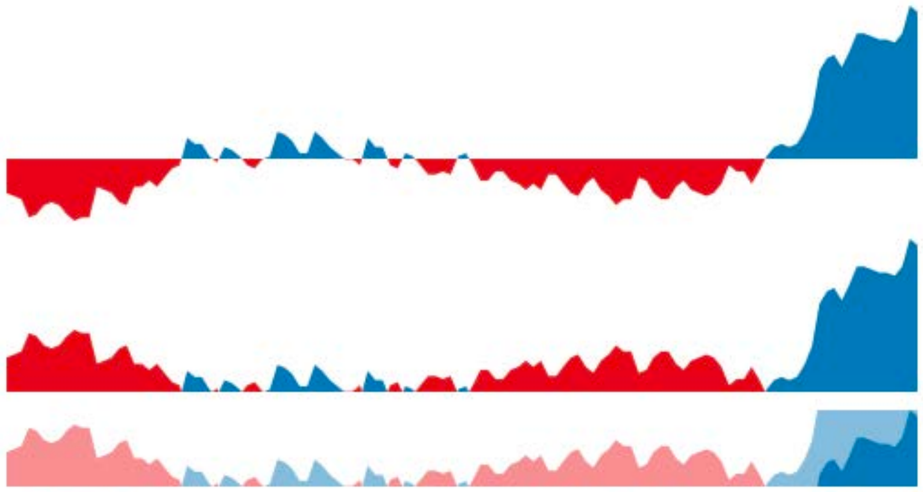
Stacked area chart

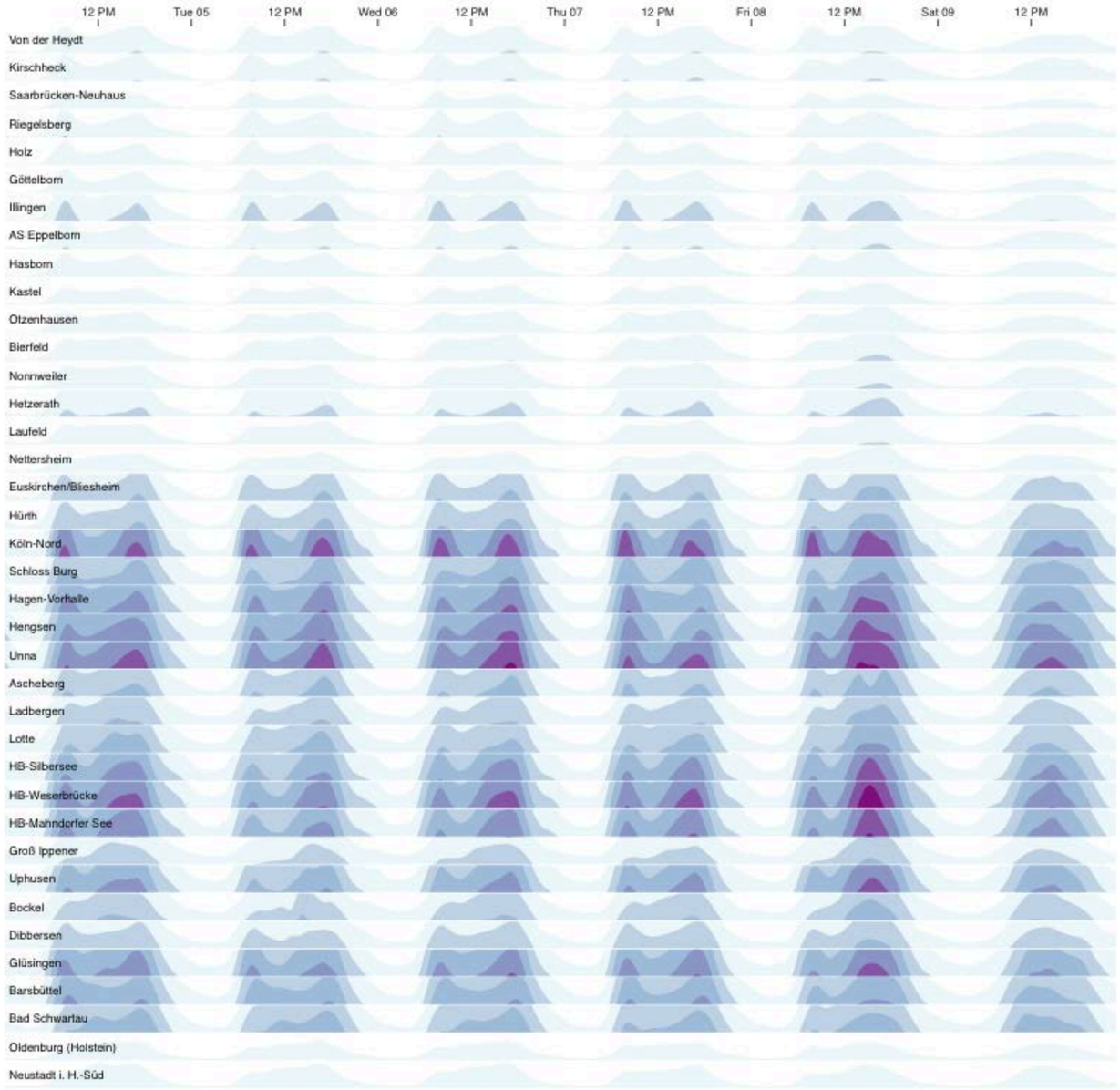


https://help.flourish.studio/article/25-line-bar-and-pie-charts

HORIZON CHARTS

- like an area chart that is sliced into equal horizontal intervals and collapsed down into single bands, which make the chart more compact
- are useful when time series data are so close in value so that the data mark in - e.g. a line chart - would lie on top of each other

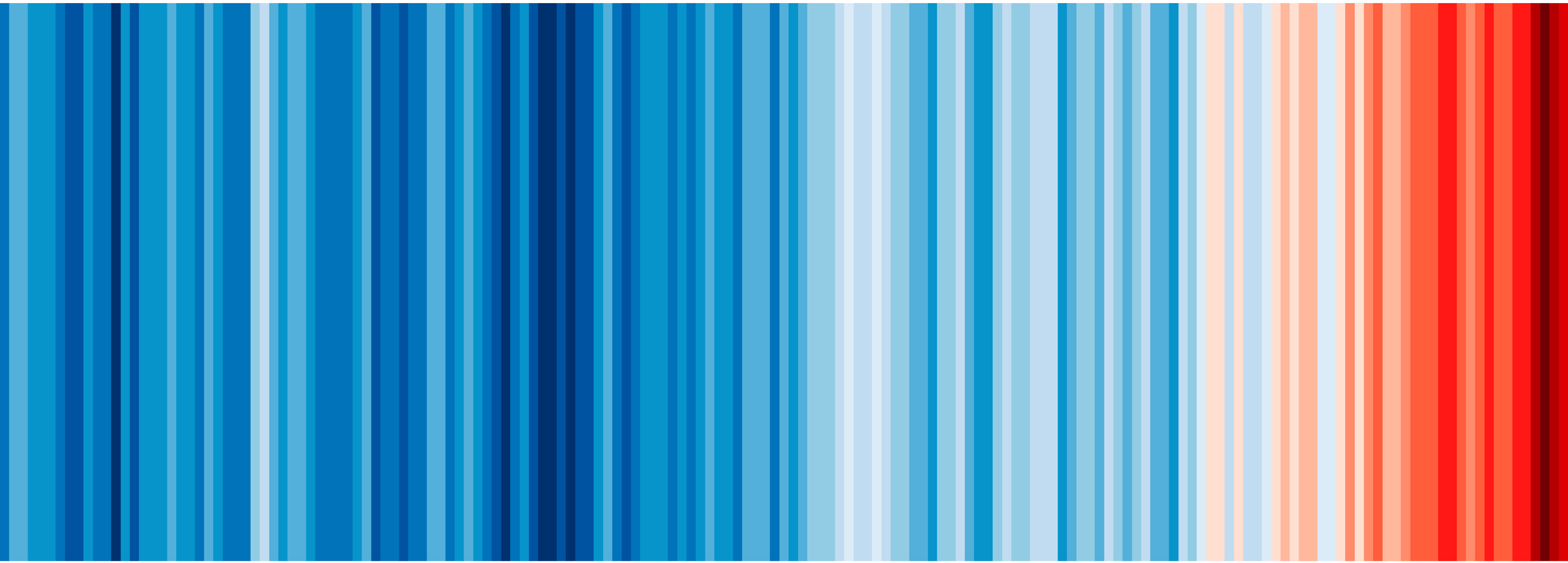




https://observablehq.com/@d3/horizon-chart - https://homes.cs.washington.edu/~jheer/files/zoo/ex/time/horizon.html

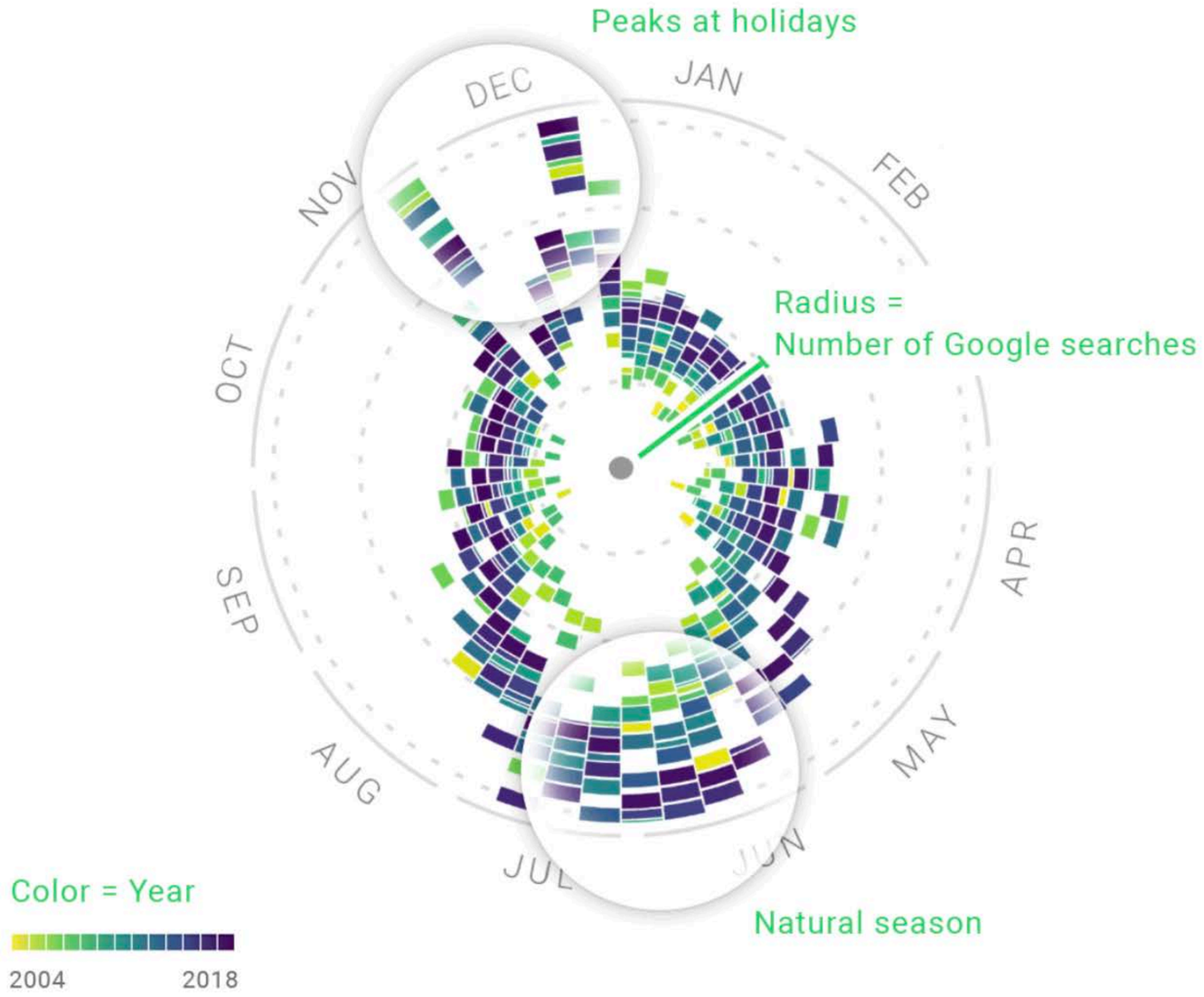
HEATMAP FOR TEMPORAL DATA

Warming Stripes: use a series of colored stripes chronologically ordered to visually portray long-term temperature trends



VISUALIZING PERIODIC PHENOMENA

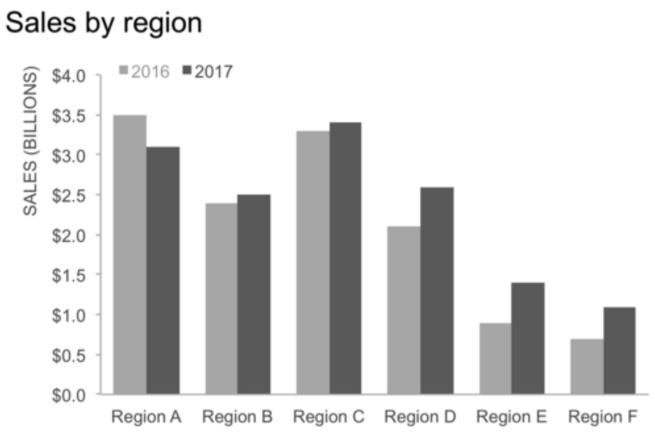
- time can also be cyclic
- radial layout can be used to detect periodic patterns

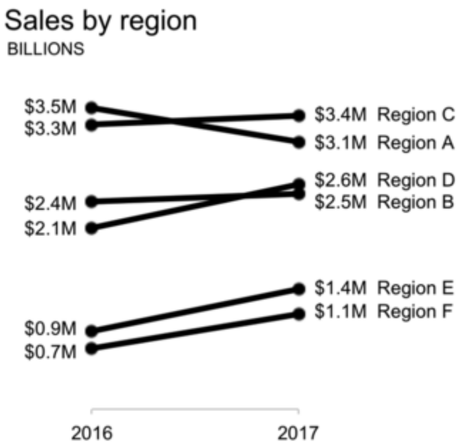


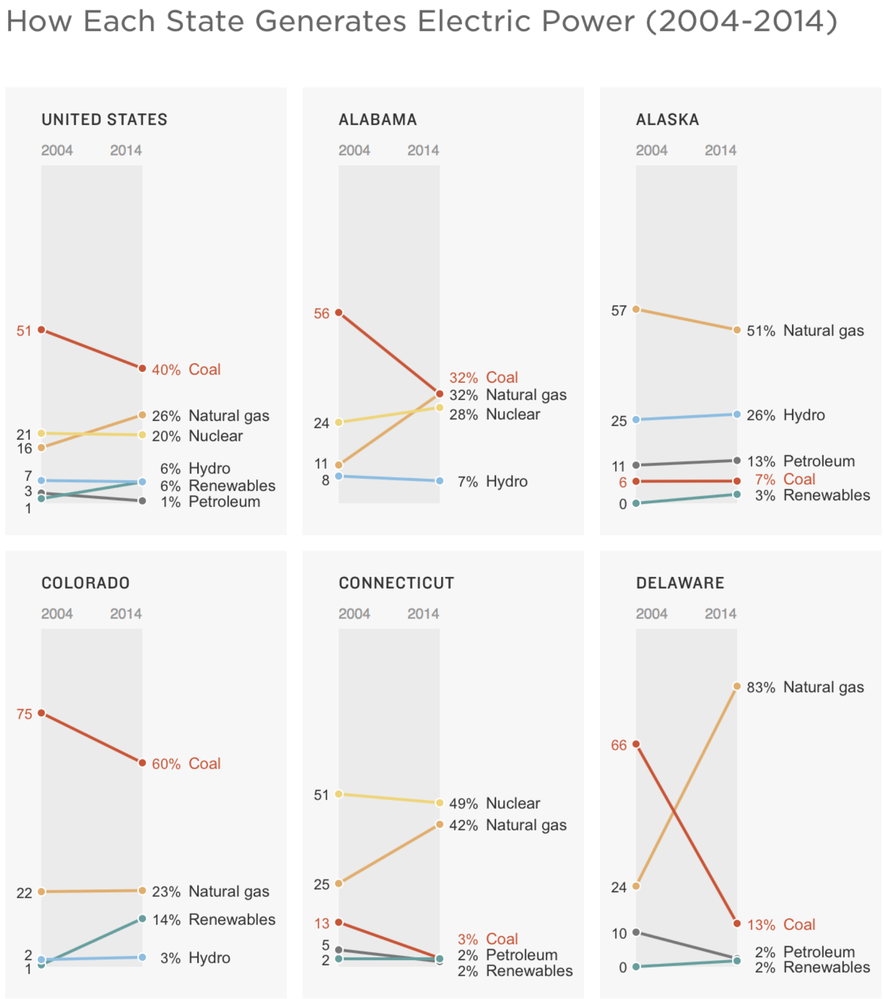
https://rhythm-of-food.net/

SLOPE CHART

- plots each data point on a separate vertical axis and connects the lines
- shows the level and the change of two different states in time
- alternative to Grouped Bar Chart if there are two observations at different points in time
- Grouped Bar chart quickly becomes very complex, so that the trends alone can be better recognized in the slope chart



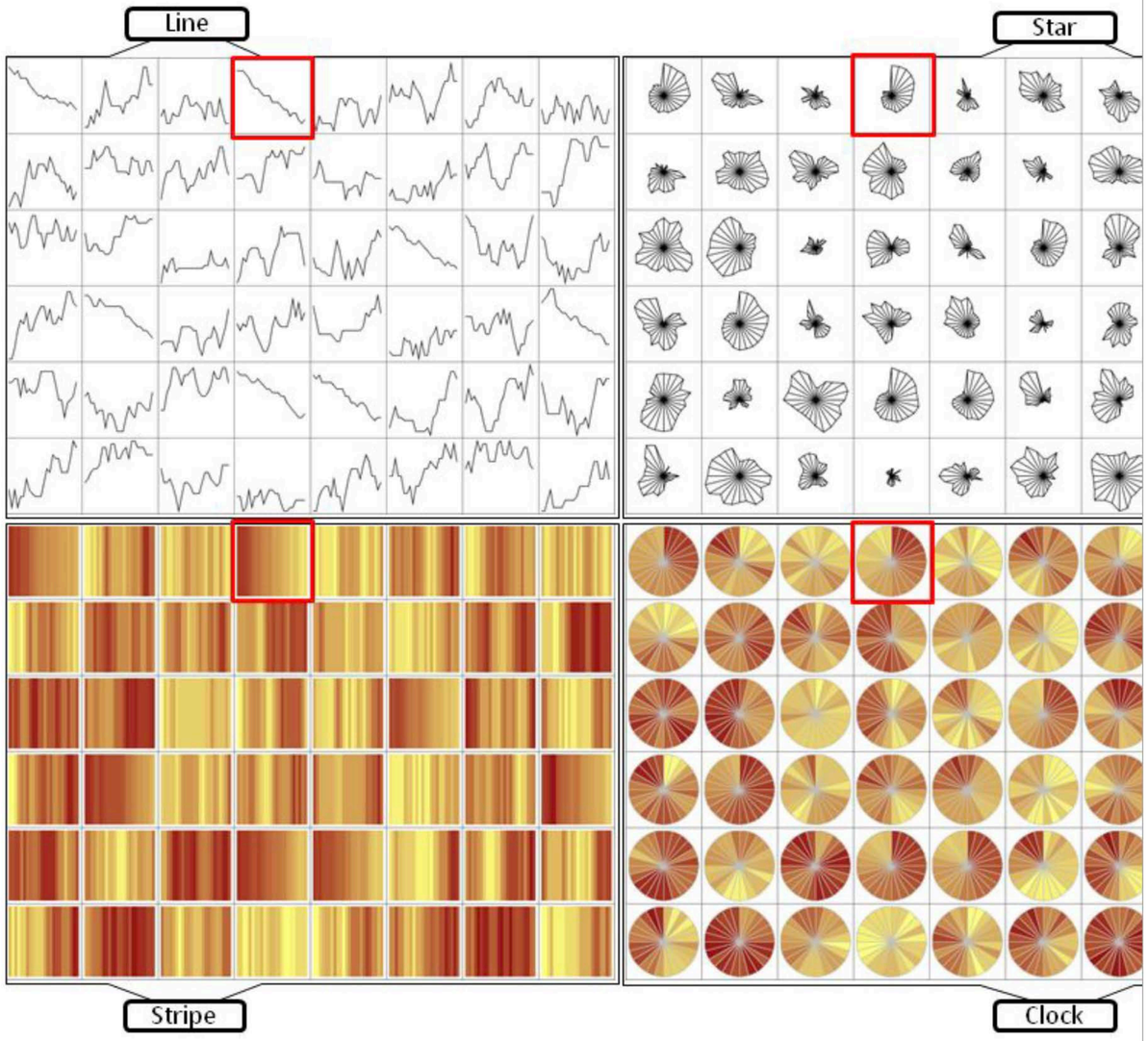




https://datavizproject.com/data-type/slope-chart/

GLYPHS FOR TEMPORAL DATA

- Line glyhs for trends and peaks
- Radial for value look-up



Fuchs, Johannes, et al. "Evaluation of alternative glyph designs for time series data in a small multiple setting." Proceedings of the SIGCHI 2013